

Use addition to check: is $6,000 - 2,348 = 3,652$ correct?

Correct? _____

Check: _____

True or false?

Because $4,519 + 2,786 = 7,305$, you know that $7,305 - 2,786 = 4,519$.

A. True

B. False

Which addition checks $9,104 - 5,677 = 3,427$?

A. $3,427 + 9,104 = 5,677$

B. $3,427 + 5,677 = 9,104$

C. $5,677 + 9,104 = 3,427$

D. $9,104 + 5,677 = 3,427$

Find $8,000 - 3,995$ by counting up from 3,995.

Difference: _____

True or false?

Because $12,450 + 6,780 = 19,230$, you know that $19,230 - 12,450 = 6,870$.

A. True

B. False

SET 26 • 3.NBT.A.2

FORWARD 2

Check decimal subtraction with addition • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Use addition to check: is $5.2 - 1.75 = 3.45$ correct?

Correct? _____

Check: _____

True or false?

Because $2.6 + 3.85 = 6.45$, you know that $6.45 - 3.85 = 2.6$.

A. True

B. False

Which addition checks $9.3 - 4.56 = 4.74$?

A. $9.3 + 4.56 = 4.74$

B. $4.74 + 4.56 = 9.3$

C. $4.74 + 9.3 = 4.56$

D. $4.56 + 9.3 = 4.74$

Find $4 - 2.35$ by counting up from 2.35.

Difference: _____

True or false?

Because $1.4 + 2.75 = 4.15$, you know that $4.15 - 1.4 = 2.65$.

- A. True
- B. False

Multiply one-digit whole numbers by multiples of 10

MAIN	Grade 3 main question	3.NBT.A.3
BACKWARD 1	A hundred is ten tens	2.NBT.A.1.a
BACKWARD 2	Skip-count by 10s	2.NBT.A.2
FORWARD 1	Multiply by multiples of 10, 100, and 1,000	4.NBT.B.5
FORWARD 2	Multiply by powers of 10 and explain the pattern in the zeros	5.NBT.A.2

SET 27 • 3.NBT.A.3

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.3

Multiply.

$$4 \times 70$$

Product: _____

True or false?

$$9 \times 20 = 160$$

A. True

B. False

$$6 \times 90 = ?$$

A. 5,400

B. 540

C. 54

D. 96

True or false?

$$7 \times 40 = 280$$

A. True

B. False

Explain how you can use $3 \times 8 = 24$ to find 3×80 .

Product: _____

Explanation: _____

SET 27 • 3.NBT.A.3

BACKWARD 1

A hundred is ten tens • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1.a

How many tens make 1 hundred?

Tens: _____

True or false?

10 tens and 1 hundred are the same amount.

A. True

B. False

Which is the same amount as 1 hundred?

- A.** 10 tens
- B.** 10 ones
- C.** 100 tens
- D.** 1 ten

Fill in the blank.

10 tens = ____

Blank: _____

True or false?

A bundle of 10 tens is the same amount as 10.

A. True

B. False

SET 27 • 3.NBT.A.3

BACKWARD 2

Skip-count by 10s • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 10s from 10.

What is the 6th number?

Number: _____

True or false?

Counting by 10s from 0, 8 jumps of 10 land on 80.

A. True

B. False

Count by 10s.

30, 40, 50, ____

A. 60

B. 70

C. 51

D. 500

Skip-count by 10s.

150, 160, 170, __, __

Next two: _____

True or false?

Counting by 10s from 0, 5 jumps of 10 land on 15.

A. True

B. False

SET 27 • 3.NBT.A.3

FORWARD 1

Multiply by multiples of 10, 100, and 1,000 • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$6 \times 400$$

Product: _____

True or false?

$$8 \times 3,000 = 24,000$$

A. True

B. False

$$5 \times 800 = ?$$

- A.** 4,000
- B.** 40,000
- C.** 805
- D.** 400

Multiply.

$$30 \times 40$$

Product: _____

True or false?

$$7 \times 600 = 420$$

A. True

B. False

SET 27 • 3.NBT.A.3

FORWARD 2

Multiply by powers of 10 and explain the pattern in the zeros • Grade 5 •
Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

$$36 \times 10^3$$

Product: _____

True or false?

$$4.2 \times 100 = 420$$

A. True

B. False

Which is equal to 5×10^2 ?

A. 5,000

B. 52

C. 500

D. 50

Explain why $7 \times 1,000$ ends in three zeros.

Explanation: _____

True or false?

$$0.6 \times 10 = 0.06$$

A. True

B. False

Understand a unit fraction as one part of a whole split into equal parts

MAIN	Grade 3 main question	3.NF.A.1
BACKWARD 1	Split shapes into halves, thirds, and fourths	2.G.A.3
BACKWARD 2	Halves and quarters of circles and rectangles	1.G.A.3
FORWARD 1	Write a fraction as a sum of unit fractions	4.NF.B.3.b
FORWARD 2	Find a unit fraction of a unit fraction	5.NF.B.4.a

SET 28 • 3.NF.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

The bar is split into 6 equal parts. One part is shaded.
What fraction of the bar is shaded?



Fraction: _____

A pizza is cut into 8 equal slices.

What fraction of the pizza is one slice?

A. $\frac{8}{1}$

B. $\frac{1}{7}$

C. $\frac{1}{8}$

D. $\frac{7}{8}$

True or false?

$\frac{1}{3}$ is one part when a whole is split into 3 equal parts.

- A.** True
- B.** False

Shade $\frac{1}{4}$ of the bar. Then explain why the part you shaded is $\frac{1}{4}$ of the bar.



Explanation: _____

True or false?

If a whole is cut into 2 parts of different sizes, each part is $\frac{1}{2}$ of the whole.

A. True

B. False

SET 28 • 3.NF.A.1

BACKWARD 1

Split shapes into halves, thirds, and fourths • Grade 2 • Geometry • 2.G.A.3

A rectangle is split into 3 equal shares.

What is each share called?

Name: _____

True or false?

Four fourths make one whole.

A. True

B. False

A circle is split into 2 equal shares.

How many halves make the whole circle?

A. 4

B. 2

C. 1

D. 3

A square is split into 4 equal shares.

What is each share called?

Name: _____

True or false?

A rectangle split into 3 equal shares is split into fourths.

- A.** True
- B.** False

SET 28 • 3.NF.A.1

BACKWARD 2

Halves and quarters of circles and rectangles • Grade 1 • Geometry • 1.G.A.3

True or false?

Cutting a circle into 4 equal parts makes smaller parts than cutting it into 2 equal parts.

A. True

B. False

A sandwich is cut into 2 equal parts.

What is each part called?

Name: _____

A pie is cut into 4 equal parts.

What is each part called?

- A.** A quarter
- B.** A whole
- C.** A half
- D.** A third

How many quarters make a whole circle?

Quarters: _____

True or false?

Halves of a rectangle are smaller than quarters of the same rectangle.

A. True

B. False

SET 28 • 3.NF.A.1

FORWARD 1

Write a fraction as a sum of unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $\frac{5}{6}$ as a sum of unit fractions.

Sum: _____

True or false?

$$\frac{3}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

A. True

B. False

Which sum is equal to $\frac{4}{5}$?

A. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

B. $\frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5}$

C. $\frac{4}{5} + \frac{1}{5}$

D. $\frac{2}{5} + \frac{1}{5}$

Write $\frac{7}{8}$ as a sum of two fractions with a denominator of 8.

Sum: _____

True or false?

$$\frac{2}{3} = \frac{1}{3} + \frac{1}{2}$$

A. True

B. False

SET 28 • 3.NF.A.1

FORWARD 2

Find a unit fraction of a unit fraction • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

What is $\frac{1}{2}$ of $\frac{1}{4}$?

Answer: _____

True or false?

$$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

A. True

B. False

Multiply.

$$\frac{1}{4} \times \frac{1}{3}$$

A. $\frac{1}{7}$

B. $\frac{1}{12}$

C. $\frac{4}{3}$

D. $\frac{2}{7}$

A ribbon is $\frac{1}{2}$ yard long. Ana uses $\frac{1}{5}$ of it.

What fraction of a yard does she use?

Answer: _____

True or false?

$\frac{1}{2}$ of $\frac{1}{2}$ is 1.

A. True

B. False

Understand a fraction a/b as a parts of size $1/b$

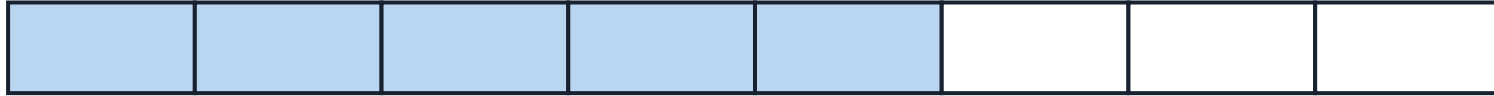
MAIN	Grade 3 main question	3.NF.A.1
BACKWARD 1	Count halves, thirds, and fourths of a shape	2.G.A.3
BACKWARD 2	Count halves and quarters of a shape	1.G.A.3
FORWARD 1	Understand a fraction a/b as a multiple of $1/b$	4.NF.B.4.a
FORWARD 2	Find a fraction of a quantity by first finding the unit fraction of it	5.NF.B.4.a

SET 29 • 3.NF.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

What fraction of the bar is shaded?



Fraction: _____

Which fraction is 3 parts of size $\frac{1}{4}$?

A. $\frac{1}{3}$

B. $\frac{4}{3}$

C. $\frac{3}{8}$

D. $\frac{3}{4}$

True or false?

$\frac{4}{6}$ is 4 parts of size $\frac{1}{6}$.

A. True

B. False

Shade $\frac{3}{8}$ of the bar.

How many parts of size $\frac{1}{8}$ did you shade?



Parts: _____

True or false?

$\frac{5}{6}$ is 6 parts of size $\frac{1}{5}$.

A. True

B. False

SET 29 • 3.NF.A.1

BACKWARD 1

Count halves, thirds, and fourths of a shape • Grade 2 • Geometry • 2.G.A.3

A rectangle is cut into 3 equal shares, and 2 shares are shaded.

How many thirds are shaded?

Thirds: _____

True or false?

A whole is 3 thirds.

A. True

B. False

A circle is cut into 4 equal shares.

How many fourths make the whole circle?

A. 2

B. 3

C. 4

D. 1

A rectangle is cut into 2 equal shares, and both are shaded.

How many halves are shaded?

Halves: _____

True or false?

A whole is 3 halves.

A. True

B. False

SET 29 • 3.NF.A.1

BACKWARD 2

Count halves and quarters of a shape • Grade 1 • Geometry • 1.G.A.3

A pizza is cut into 4 equal parts, and 3 parts are eaten.
How many quarters are eaten?

Quarters: _____

True or false?

Two halves make one whole.

A. True

B. False

A rectangle is cut into 4 equal parts.

How many parts make the whole?

A. 8

B. 2

C. 1

D. 4

A pie is cut into 2 equal parts, and 1 part is eaten.
What is the part that was eaten called?

Name: _____

True or false?

Four quarters make two wholes.

A. True

B. False

SET 29 • 3.NF.A.1

FORWARD 1

Understand a fraction a/b as a multiple of $1/b$ • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{5}{6}$ as a whole number times $\frac{1}{6}$.

Expression: _____

True or false?

$$\frac{7}{4} = 7 \times \frac{1}{4}$$

A. True

B. False

Which equation shows $\frac{3}{8}$ as a multiple of $\frac{1}{8}$?

A. $\frac{3}{8} = \frac{1}{8} \div 3$

B. $\frac{3}{8} = 3 \times \frac{1}{8}$

C. $\frac{3}{8} = 8 \times \frac{1}{3}$

D. $\frac{3}{8} = 3 + \frac{1}{8}$

Fill in the blank.

$$\frac{9}{10} = \underline{\hspace{1cm}} \times \frac{1}{10}$$

Blank: _____

True or false?

$$\frac{4}{5} = 5 \times \frac{1}{4}$$

A. True

B. False

SET 29 • 3.NF.A.1

FORWARD 2

Find a fraction of a quantity by first finding the unit fraction of it • Grade 5 •
Number & Operations–Fractions • 5.NF.B.4.a

Find $\frac{3}{5}$ of 20 by first finding $\frac{1}{5}$ of 20.

$\frac{1}{5}$ of 20: _____

$\frac{3}{5}$ of 20: _____

True or false?

$\frac{2}{3}$ of 9 is 2 parts of size 3, which is 6.

A. True

B. False

What is $\frac{5}{8}$ of 16?

A. 80

B. 8

C. 10

D. 2

Multiply.

$$\frac{4}{7} \times 14$$

Product: _____

True or false?

$\frac{3}{4}$ of 20 is 3 parts of size 4, which is 12.

A. True

B. False

Represent a unit fraction on a number line

MAIN	Grade 3 main question	3.NF.A.2.a
BACKWARD 1	Whole numbers as lengths from 0 on a number line	2.MD.B.6
BACKWARD 2	Measure lengths in whole inches with a ruler	2.MD.A.1
FORWARD 1	Write tenths and hundredths as decimals and locate them on a n...	4.NF.C.6
FORWARD 2	Divide a unit fraction by a whole number	5.NF.B.7.a

SET 30 • 3.NF.A.2.a

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.2.a

The space from 0 to 1 is split into 4 equal parts.

What fraction is at point A?



A = _____

Mark and label $\frac{1}{3}$ on the number line.



Into how many equal parts must the space from 0 to 1 be split to show $\frac{1}{6}$?

A. 1

B. 5

C. 6

D. 7

True or false?

On a number line, $\frac{1}{2}$ is halfway between 0 and 1.

A. True

B. False

True or false?

On a number line from 0 to 1 split into 8 equal parts, the first tick mark after 0 is $\frac{1}{7}$.

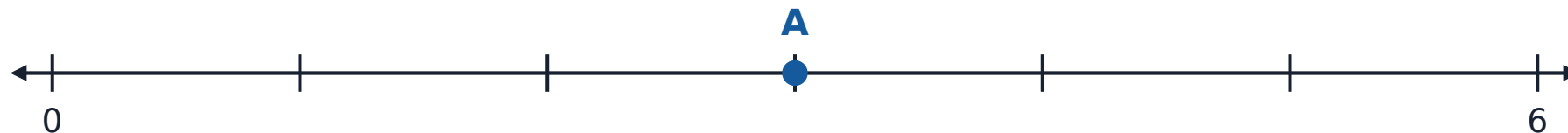
- A. True
- B. False

SET 30 • 3.NF.A.2.a

BACKWARD 1

Whole numbers as lengths from 0 on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

True or false?

On a number line, 5 is 5 equal lengths of 1 from 0.

A. True

B. False

A jump from 0 to 4 on a number line is how many lengths of 1?

A. 1

B. 4

C. 3

D. 5

What number is at point B?



B = _____

True or false?

On a number line, the distance from 0 to 7 is 6 lengths of 1.

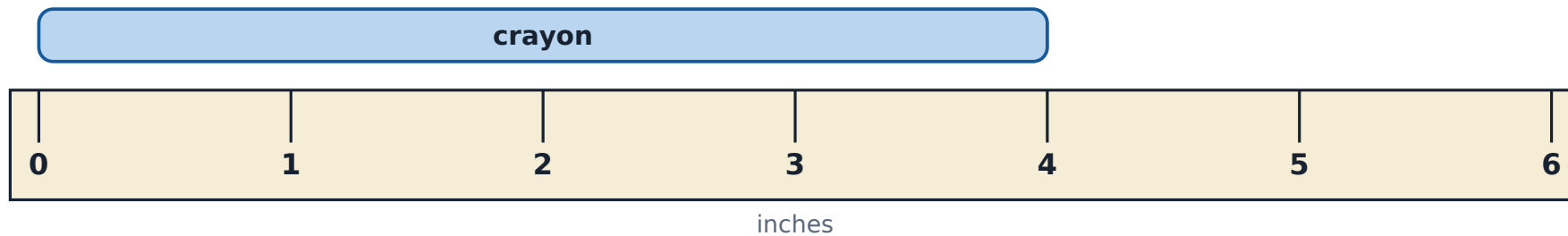
- A.** True
- B.** False

SET 30 • 3.NF.A.2.a

BACKWARD 2

Measure lengths in whole inches with a ruler • Grade 2 • Measurement & Data
• 2.MD.A.1

How long is the crayon?



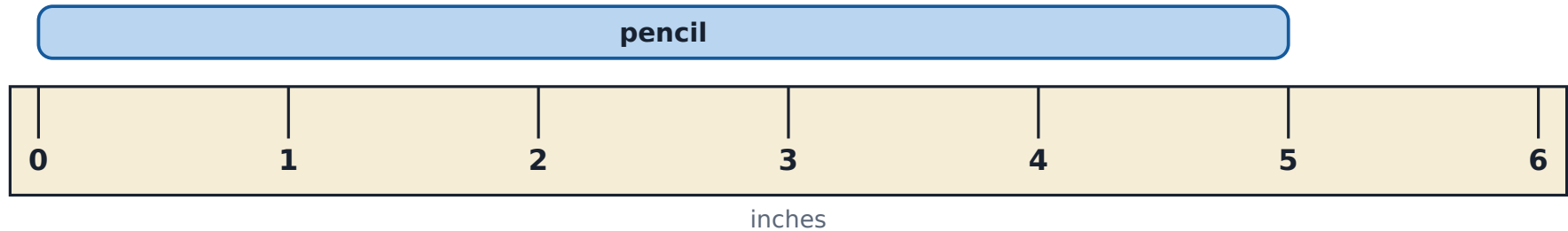
Length: _____

True or false?

To measure with a ruler, line up one end of the object with 0.

- A. True
- B. False

How long is the pencil?



A. 6 inches

B. 5 inches

C. 1 inch

D. 4 inches

A leaf reaches from 0 to 3 on an inch ruler.
How long is the leaf?

Length: _____

True or false?

A stick that reaches from 0 to 6 on an inch ruler is 7 inches long.

- A. True
- B. False

SET 30 • 3.NF.A.2.a

FORWARD 1

Write tenths and hundredths as decimals and locate them on a number line •
Grade 4 • Number & Operations–Fractions • 4.NF.C.6

The space from 0 to 1 is split into 10 equal parts.

Write the decimal at point A.



A = _____

True or false?

$$\frac{1}{10} = 0.1$$

A. True

B. False

Which decimal is equal to $\frac{1}{100}$?

A. 0.1

B. 0.01

C. 10.0

D. 1.01

Write $\frac{1}{100}$ as a decimal.

Decimal: _____

True or false?

On a number line, 0.1 is between 0 and $\frac{1}{100}$.

A. True

B. False

SET 30 • 3.NF.A.2.a

FORWARD 2

Divide a unit fraction by a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a

$\frac{1}{2}$ is split into 3 equal parts.

What fraction is each part?

Fraction: _____

True or false?

$$\frac{1}{4} \div 2 = \frac{1}{8}$$

A. True

B. False

$$\frac{1}{3} \div 2 = ?$$

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{5}$

D. 6

Divide.

$$\frac{1}{2} \div 4$$

Quotient: _____

True or false?

$$\frac{1}{5} \div 3 = \frac{3}{5}$$

A. True

B. False

Represent a fraction a/b on a number line by marking off a lengths of $1/b$

MAIN	Grade 3 main question	3.NF.A.2.b
BACKWARD 1	Mark whole numbers as lengths from 0 on a number line	2.MD.B.6
BACKWARD 2	Show sums as jumps on a number line	2.MD.B.6
FORWARD 1	Multiply a fraction by a whole number with jumps on a number li...	4.NF.B.4.b
FORWARD 2	Add fractions with unlike denominators	5.NF.A.1

SET 31 • 3.NF.A.2.b

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.2.b

What fraction is at point B?



B = _____

Start at 0 and mark off 5 lengths of $\frac{1}{6}$. Mark and label the point $\frac{5}{6}$.



A point is 3 lengths of $\frac{1}{8}$ from 0.

What fraction is at the point?

A. $\frac{1}{3}$

B. $\frac{3}{1}$

C. $\frac{3}{8}$

D. $\frac{8}{3}$

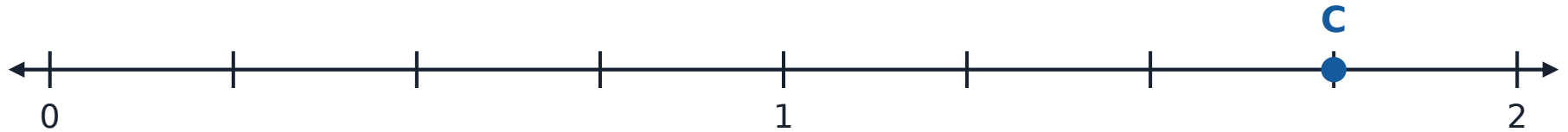
True or false?

The point $\frac{2}{3}$ is 2 lengths of $\frac{1}{3}$ from 0.

A. True

B. False

What fraction is at point C? Write it as a fraction with a denominator of 4.



C = _____

SET 31 • 3.NF.A.2.b

BACKWARD 1

Mark whole numbers as lengths from 0 on a number line • Grade 2 •
Measurement & Data • 2.MD.B.6

Mark and label 6 on the number line.



True or false?

The point 3 is 3 lengths of 1 from 0.

A. True

B. False

A point is 7 lengths of 1 from 0.

What number is it?

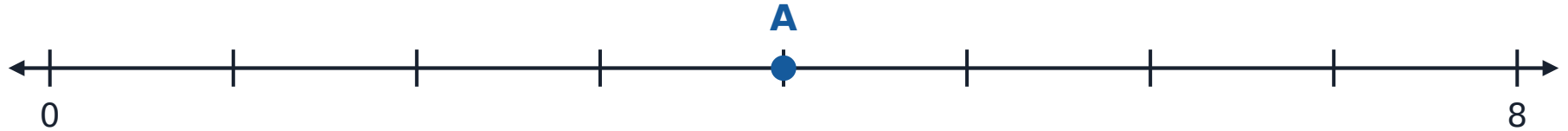
A. 8

B. 6

C. 1

D. 7

What number is at point A?



A = _____

True or false?

The point 9 is 10 lengths of 1 from 0.

A. True

B. False

SET 31 • 3.NF.A.2.b

BACKWARD 2

Show sums as jumps on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

Start at 20 on a number line and jump 15 more.

Where do you land?

Number: _____

True or false?

Starting at 0, a jump of 30 and then a jump of 20 lands on 50.

- A. True
- B. False

Start at 10 and make three jumps of 10.

Where do you land?

A. 40

B. 13

C. 30

D. 50

Start at 25 and jump 40 more.

Where do you land?

Number: _____

True or false?

Starting at 40, a jump of 15 lands on 65.

A. True

B. False

SET 31 • 3.NF.A.2.b

FORWARD 1

Multiply a fraction by a whole number with jumps on a number line • Grade 4 •
Number & Operations–Fractions • 4.NF.B.4.b

Start at 0 and make 4 jumps of $\frac{2}{3}$ on a number line.

Where do you land?

Number: _____

True or false?

3 jumps of $\frac{3}{4}$ from 0 land on $\frac{9}{4}$.

A. True

B. False

$$5 \times \frac{2}{6} = ?$$

A. $\frac{7}{6}$

B. $\frac{10}{30}$

C. $\frac{2}{30}$

D. $\frac{10}{6}$

Multiply.

$$6 \times \frac{3}{8}$$

Product: _____

True or false?

2 jumps of $\frac{5}{6}$ from 0 land on $\frac{10}{12}$.

A. True

B. False

SET 31 • 3.NF.A.2.b

FORWARD 2

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{2} + \frac{1}{3}$$

Sum: _____

True or false?

$$\frac{3}{4} + \frac{1}{8} = \frac{7}{8}$$

A. True

B. False

Add.

$$\frac{2}{3} + \frac{1}{6}$$

A. $\frac{5}{6}$

B. $\frac{3}{6}$

C. $\frac{3}{9}$

D. $\frac{2}{18}$

Add.

$$\frac{3}{8} + \frac{1}{4}$$

Sum: _____

True or false?

$$\frac{1}{4} + \frac{1}{3} = \frac{2}{7}$$

A. True

B. False

Understand equivalent fractions as the same size or the same point on a number line

MAIN	Grade 3 main question	3.NF.A.3.a
BACKWARD 1	Measure one length with two different units	2.MD.A.2
BACKWARD 2	Equal shares of the same whole	2.G.A.3
FORWARD 1	Explain equivalent fractions with visual models	4.NF.A.1
FORWARD 2	Use equivalent fractions to add fractions with unlike denominators	5.NF.A.1

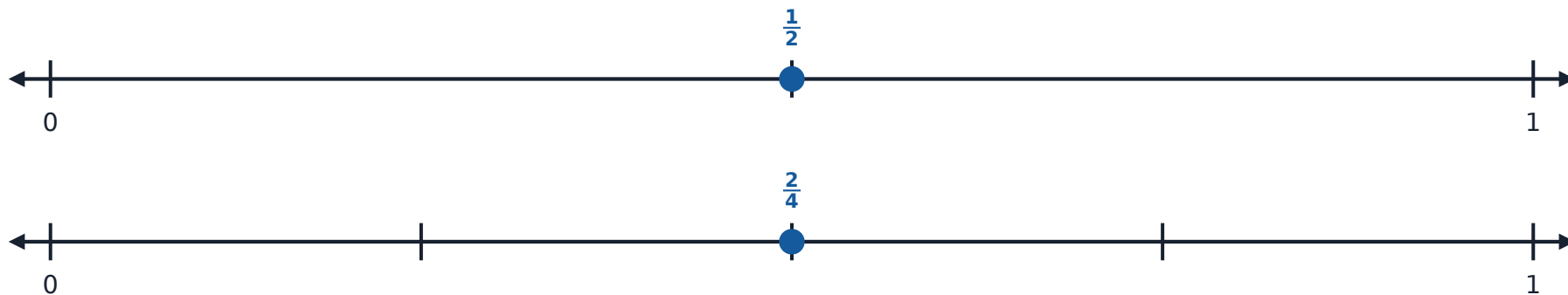
SET 32 • 3.NF.A.3.a

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.a

True or false?

The number lines are the same length. $\frac{1}{2}$ and $\frac{2}{4}$ are at the same point.

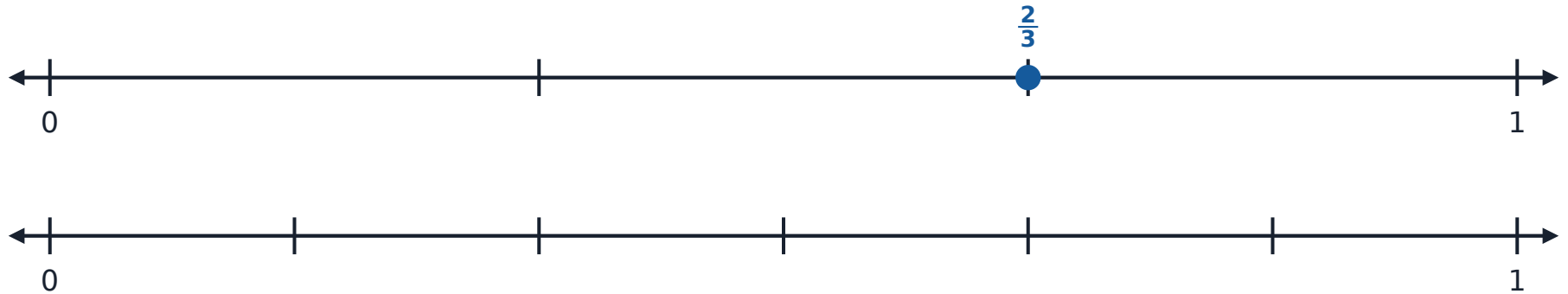


A. True

B. False

The number lines are the same length.

Which fraction is at the same point as $\frac{2}{3}$?



A. $\frac{2}{6}$

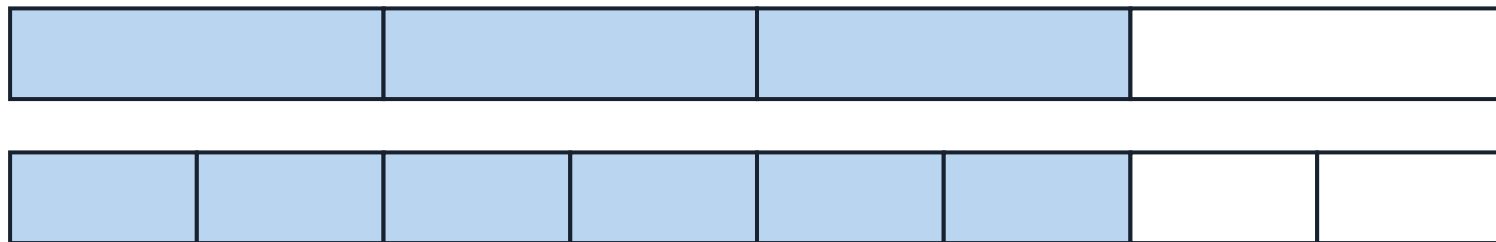
B. $\frac{3}{6}$

C. $\frac{5}{6}$

D. $\frac{4}{6}$

The two bars are the same size. The top bar shows $\frac{3}{4}$ shaded and the bottom bar shows $\frac{6}{8}$ shaded.

Are the fractions equivalent? Explain.



Answer: _____

Explanation: _____

True or false?

$\frac{1}{3}$ and $\frac{2}{6}$ are at different points on a number line from 0 to 1.

- A.** True
- B.** False

Name a fraction in eighths that is at the same point as $\frac{1}{4}$ on a number line.

Fraction: _____

SET 32 • 3.NF.A.3.a

BACKWARD 1

Measure one length with two different units • Grade 2 • Measurement & Data •
2.MD.A.2

True or false?

A table measured in centimeters gives a larger number than the same table measured in meters.

A. True

B. False

A pencil is 6 paper clips long and 3 crayons long.

Which unit is longer?

- A.** You cannot tell
- B.** The crayon
- C.** The paper clip
- D.** They are the same length

A shelf is 10 large blocks long and 20 small blocks long.
Did the shelf's length change?

Answer: _____

A ribbon is 4 feet long. Is its length in inches a number greater than or less than 4?

Answer: _____

True or false?

When you measure with a bigger unit, you get a bigger number.

- A. True
- B. False

SET 32 • 3.NF.A.3.a

BACKWARD 2

Equal shares of the same whole • Grade 2 • Geometry • 2.G.A.3

True or false?

Two halves of the same rectangle can have different shapes and still be equal shares.

- A. True
- B. False

A square is cut into 2 equal triangles. Another square of the same size is cut into 2 equal rectangles.

Is a triangle the same size as a rectangle?

Answer: _____

A circle is cut into 4 equal shares.

What is each share called?

- A.** A third
- B.** A fourth
- C.** A whole
- D.** A half

How many halves make one whole?

Halves: _____

True or false?

A third of a rectangle is larger than a half of the same rectangle.

A. True

B. False

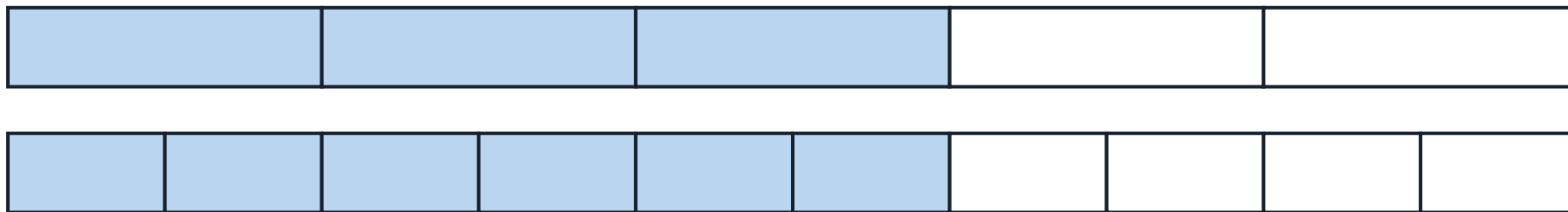
SET 32 • 3.NF.A.3.a

FORWARD 1

Explain equivalent fractions with visual models • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

The bars show $\frac{3}{5}$ and $\frac{6}{10}$ of the same whole.

Explain why the fractions are equal.



Explanation: _____

True or false?

$\frac{2}{5} = \frac{4}{10}$ because the numerator and the denominator are both multiplied by 2.

A. True

B. False

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{9}{12}$

B. $\frac{3}{12}$

C. $\frac{6}{4}$

D. $\frac{4}{3}$

Find the missing number.

$$\frac{4}{5} = \frac{?}{10}$$

Missing number: _____

True or false?

$$\frac{1}{3} = \frac{3}{6}$$

A. True

B. False

SET 32 • 3.NF.A.3.a

FORWARD 2

Use equivalent fractions to add fractions with unlike denominators • Grade 5 •
Number & Operations–Fractions • 5.NF.A.1

Rewrite $\frac{1}{2}$ and $\frac{1}{5}$ with a common denominator. Then add.

Rewritten: _____

Sum: _____

True or false?

$$\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12}$$

A. True

B. False

Add.

$$\frac{1}{6} + \frac{3}{4}$$

A. $\frac{3}{24}$

B. $\frac{4}{12}$

C. $\frac{4}{10}$

D. $\frac{11}{12}$

Add.

$$\frac{2}{5} + \frac{1}{10}$$

Sum: _____

True or false?

$$\frac{1}{3} + \frac{1}{2} = \frac{2}{5}$$

A. True

B. False

Recognize and generate simple equivalent fractions, and explain why they are equivalent

MAIN	Grade 3 main question	3.NF.A.3.b
BACKWARD 1	Halves, thirds, and fourths that make a whole	2.G.A.3
BACKWARD 2	Write even numbers as doubles	2.OA.C.3
FORWARD 1	Generate equivalent fractions by multiplying the numerator and th...	4.NF.A.1
FORWARD 2	Explain equivalent fractions as multiplying by a fraction equal to 1	5.NF.B.5.b

SET 33 • 3.NF.A.3.b

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

Find the missing number.

$$\frac{1}{3} = \frac{?}{6}$$

Missing number: _____

Which fraction is equivalent to $\frac{3}{4}$?

A. $\frac{4}{3}$

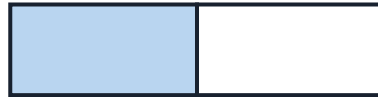
B. $\frac{4}{6}$

C. $\frac{6}{8}$

D. $\frac{3}{8}$

The bar shows $\frac{1}{2}$ shaded. Draw lines to split each half into 4 equal parts.

Then write the equivalent fraction in eighths.



Fraction: _____

True or false?

$$\frac{2}{6} = \frac{1}{3}$$

A. True

B. False

Explain why $\frac{2}{8} = \frac{1}{4}$. You may use a model.

Explanation: _____

SET 33 • 3.NF.A.3.b

BACKWARD 1

Halves, thirds, and fourths that make a whole • Grade 2 • Geometry • 2.G.A.3

How many fourths make a whole?

Fourths: _____

True or false?

Three thirds make one whole.

A. True

B. False

Which describes one whole?

A. 3 fourths

B. 3 halves

C. 2 thirds

D. 2 halves

A rectangle is cut into 3 equal shares.

How many of the shares make the whole rectangle?

Shares: _____

True or false?

Four halves make one whole.

A. True

B. False

SET 33 • 3.NF.A.3.b

BACKWARD 2

Write even numbers as doubles • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.3

Write 8 as the sum of two equal addends.

Sum: _____

True or false?

$$6 = 3 + 3$$

A. True

B. False

Which number is the double of 4?

- A.** 8
- B.** 6
- C.** 4
- D.** 16

What is the double of 3?

Double: _____

True or false?

$10 = 6 + 4$ shows 10 as the sum of two equal addends.

A. True

B. False

FORWARD 1

Generate equivalent fractions by multiplying the numerator and the denominator by the same number • Grade 4 • Number & Operations–Fractions
• 4.NF.A.1

Find the missing number.

$$\frac{3}{5} = \frac{?}{15}$$

Missing number: _____

True or false?

$$\frac{4}{6} = \frac{8}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{2}{5}$?

A. $\frac{3}{6}$

B. $\frac{5}{2}$

C. $\frac{4}{10}$

D. $\frac{2}{10}$

Write two fractions that are equivalent to $\frac{3}{4}$.

Fractions: _____

True or false?

$$\frac{5}{8} = \frac{10}{24}$$

A. True

B. False

SET 33 • 3.NF.A.3.b

FORWARD 2

Explain equivalent fractions as multiplying by a fraction equal to 1 • Grade 5 •
Number & Operations–Fractions • 5.NF.B.5.b

True or false?

$$\frac{2}{3} \times \frac{4}{4} = \frac{8}{12}, \text{ and } \frac{8}{12} = \frac{2}{3} \text{ because } \frac{4}{4} = 1.$$

A. True

B. False

To rewrite $\frac{3}{5}$ as $\frac{9}{15}$, by which whole number must you multiply BOTH its numerator and its denominator?

Which fraction equal to 1 is that the same as multiplying by?

Whole number: _____

Fraction equal to 1: _____

Which equation shows that $\frac{5}{6} = \frac{10}{12}$?

A. $\frac{5}{6} \times \frac{2}{1} = \frac{10}{12}$

B. $\frac{5}{6} + \frac{5}{6} = \frac{10}{12}$

C. $\frac{5}{6} \times \frac{2}{2} = \frac{10}{12}$

D. $\frac{5}{6} \times 2 = \frac{10}{12}$

Explain why multiplying $\frac{1}{4}$ by $\frac{3}{3}$ does not change its value.

Explanation: _____

True or false?

$\frac{1}{2} \times \frac{2}{3} = \frac{2}{6}$ shows that $\frac{1}{2} = \frac{2}{6}$.

A. True

B. False

Express whole numbers as fractions and recognize fractions that are equal to whole numbers

MAIN	Grade 3 main question	3.NF.A.3.c
BACKWARD 1	Two halves or four quarters make a whole	1.G.A.3
BACKWARD 2	Count lengths of 1 on a number line	2.MD.B.6
FORWARD 1	Write mixed numbers as fractions and fractions as mixed numbers	4.NF.B.3.b
FORWARD 2	Interpret a fraction as the numerator divided by the denominator	5.NF.B.3

SET 34 • 3.NF.A.3.c

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.c

Write 5 as a fraction with a denominator of 1.

Fraction: _____

Which fraction is equal to 2?

A. $\frac{4}{8}$

B. $\frac{2}{4}$

C. $\frac{2}{8}$

D. $\frac{8}{4}$

True or false?

$$\frac{6}{6} = 6$$

A. True

B. False

The number line from 0 to 3 is marked in thirds.

What fraction in thirds names the point 2?



Fraction: _____

True or false?

$$\frac{3}{3} = 3$$

A. True

B. False

SET 34 • 3.NF.A.3.c

BACKWARD 1

Two halves or four quarters make a whole • Grade 1 • Geometry • 1.G.A.3

How many halves make one whole pizza?

Halves: _____

True or false?

4 quarters of a sandwich make the whole sandwich.

A. True

B. False

A pie is cut into 4 equal parts. All 4 parts are eaten.

How much of the pie is eaten?

- A.** The whole pie
- B.** Half of the pie
- C.** A quarter of the pie
- D.** None of the pie

How many quarters make one whole?

Quarters: _____

True or false?

2 quarters make a whole circle.

A. True

B. False

SET 34 • 3.NF.A.3.c

BACKWARD 2

Count lengths of 1 on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

How many lengths of 1 are there from 0 to 3 on a number line?

Lengths: _____

True or false?

From 0 to 5 on a number line there are 5 lengths of 1.

A. True

B. False

Which number is 2 lengths of 1 from 0?

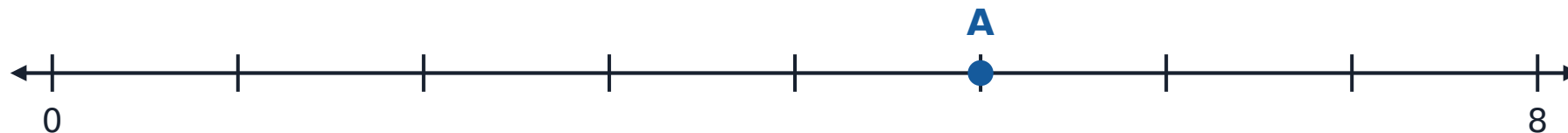
A. 2

B. 0

C. 1

D. 3

What number is at point A?



A = _____

True or false?

From 0 to 4 on a number line there are 3 lengths of 1.

A. True

B. False

SET 34 • 3.NF.A.3.c

FORWARD 1

Write mixed numbers as fractions and fractions as mixed numbers • Grade 4 •
Number & Operations–Fractions • 4.NF.B.3.b

Write $2\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$1\frac{3}{4} = \frac{4}{4} + \frac{3}{4}$$

A. True

B. False

Which mixed number is equal to $\frac{11}{4}$?

A. $2\frac{3}{4}$

B. $1\frac{3}{4}$

C. $2\frac{1}{4}$

D. $3\frac{1}{4}$

Write $\frac{9}{2}$ as a mixed number.

Mixed number: _____

True or false?

$$3\frac{2}{5} = \frac{32}{5}$$

A. True

B. False

SET 34 • 3.NF.A.3.c

FORWARD 2

Interpret a fraction as the numerator divided by the denominator • Grade 5 •
Number & Operations–Fractions • 5.NF.B.3

Write $\frac{20}{4}$ as a division. Then find its value.

Division: _____

Value: _____

True or false?

$$\frac{15}{3} = 15 \div 3 = 5$$

A. True

B. False

Which division is equal to $\frac{7}{2}$?

A. $7 - 2$

B. 7×2

C. $7 \div 2$

D. $2 \div 7$

Find the value of $\frac{42}{6}$.

Value: _____

True or false?

$$\frac{5}{10} = 10 \div 5$$

A. True

B. False

Compare two fractions with the same denominator

MAIN	Grade 3 main question	3.NF.A.3.d
BACKWARD 1	Compare two- and three-digit numbers	2.NBT.A.4
BACKWARD 2	Order objects by length	1.MD.A.1
FORWARD 1	Compare fractions with different numerators and different denomi...	4.NF.A.2
FORWARD 2	Use benchmark fractions to estimate sums of fractions	5.NF.A.2

SET 35 • 3.NF.A.3.d

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{3}{8} \text{ — } \frac{5}{8}$$

Symbol: _____

True or false?

$$\frac{5}{6} > \frac{1}{6}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{1}{4}$

B. $\frac{3}{4}$

C. $\frac{0}{4}$

D. $\frac{2}{4}$

Compare $\frac{2}{3}$ and $\frac{1}{3}$. Write $>$, $=$, or $<$, and explain how you know.

Comparison: _____

Explanation: _____

True or false?

$$\frac{4}{8} < \frac{3}{8}$$

A. True

B. False

SET 35 • 3.NF.A.3.d

BACKWARD 1

Compare two- and three-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $=$, or $<$.

57 75

Symbol: _____

True or false?

$$130 > 103$$

A. True

B. False

Which number is least?

A. 902

B. 209

C. 920

D. 290

Write $>$, $=$, or $<$.

408 480

Symbol: _____

True or false?

$$615 < 561$$

A. True

B. False

SET 35 • 3.NF.A.3.d

BACKWARD 2

Order objects by length • Grade 1 • Measurement & Data • 1.MD.A.1

A red pencil is longer than a blue pencil. The blue pencil is longer than a green pencil.

Which pencil is longest?

Answer: _____

True or false?

If rope A is longer than rope B, then rope B is shorter than rope A.

A. True

B. False

Ty's string is shorter than Ann's string. Ann's string is shorter than Bo's string.

Whose string is shortest?

- A.** Bo's
- B.** Ann's
- C.** Ty's
- D.** They are the same length.

A bat is longer than a stick.

Is the stick longer or shorter than the bat?

Answer: _____

True or false?

If a crayon is shorter than a marker, then the marker is shorter than the crayon.

- A.** True
- B.** False

SET 35 • 3.NF.A.3.d

FORWARD 1

Compare fractions with different numerators and different denominators •
Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Write $>$, $=$, or $<$.

$$\frac{2}{3} \text{ — } \frac{5}{8}$$

Symbol: _____

True or false?

$$\frac{3}{5} > \frac{1}{2}$$

A. True

B. False

Which comparison is true?

A. $\frac{3}{4} < \frac{7}{10}$

B. $\frac{3}{4} = \frac{7}{10}$

C. $\frac{3}{4} > \frac{7}{10}$

D. $\frac{7}{10} > \frac{3}{4}$

Compare $\frac{5}{6}$ and $\frac{7}{9}$. Write $>$, $=$, or $<$.

Comparison: _____

True or false?

$$\frac{2}{5} > \frac{1}{2}$$

A. True

B. False

SET 35 • 3.NF.A.3.d

FORWARD 2

Use benchmark fractions to estimate sums of fractions • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Is $\frac{2}{5} + \frac{3}{8}$ more or less than 1? Explain using benchmark fractions.

Answer: _____

Reason: _____

True or false?

$\frac{4}{7} + \frac{3}{5} > 1$, because each fraction is greater than $\frac{1}{2}$.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{5}{6}$?

A. About 3

B. About $\frac{1}{2}$

C. About 1

D. About 2

Use benchmark fractions to estimate $\frac{1}{12} + \frac{6}{11}$.

Estimate: _____

True or false?

$\frac{1}{8} + \frac{1}{9}$ is about 1.

A. True

B. False

Compare two fractions with the same numerator

MAIN	Grade 3 main question	3.NF.A.3.d
BACKWARD 1	More equal shares make smaller shares	1.G.A.3
BACKWARD 2	Compare halves, thirds, and fourths of the same shape	2.G.A.3
FORWARD 1	Compare fractions by finding a common numerator or a common...	4.NF.A.2
FORWARD 2	Compare the size of a product with the size of one factor without...	5.NF.B.5.a

SET 36 • 3.NF.A.3.d

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{1}{3} \text{ — } \frac{1}{6}$$

Symbol: _____

True or false?

$$\frac{2}{8} < \frac{2}{4}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{1}{6}$

B. $\frac{1}{8}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

Compare $\frac{2}{3}$ and $\frac{2}{6}$ of the same whole. Write $>$, $=$, or $<$, and explain how you know.

Comparison: _____

Explanation: _____

True or false?

$$\frac{5}{8} > \frac{5}{6}$$

A. True

B. False

SET 36 • 3.NF.A.3.d

BACKWARD 1

More equal shares make smaller shares • Grade 1 • Geometry • 1.G.A.3

True or false?

A pizza cut into 4 equal parts has larger parts than the same pizza cut into 2 equal parts.

A. True

B. False

Which is bigger: half of a sandwich or a quarter of the same sandwich?

Answer: _____

The same cake is cut in two different ways.

Which way makes the smaller pieces?

- A.** 4 equal pieces
- B.** You cannot tell
- C.** Both make the same size
- D.** 2 equal pieces

A ribbon is cut into 2 equal parts. Another ribbon of the same length is cut into 4 equal parts.

Which ribbon has the longer parts?

Answer: _____

True or false?

Halves of a circle are larger than quarters of the same circle.

A. True

B. False

SET 36 • 3.NF.A.3.d

BACKWARD 2

Compare halves, thirds, and fourths of the same shape • Grade 2 • Geometry •
2.G.A.3

Which is larger: a third or a fourth of the same rectangle?

Answer: _____

True or false?

A half of a circle is larger than a third of the same circle.

A. True

B. False

Which share of the same square is smallest?

- A.** A fourth
- B.** The whole
- C.** A half
- D.** A third

Ann eats a half of a pie. Bo eats a third of a pie of the same size.

Who eats more?

Answer: _____

True or false?

A fourth of a rectangle is larger than a half of the same rectangle.

A. True

B. False

FORWARD 1

Compare fractions by finding a common numerator or a common denominator

• Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Rewrite $\frac{3}{5}$ with a numerator of 6. Then compare it with $\frac{6}{12}$ using $>$, $=$, or $<$.

Rewritten: _____

Comparison: _____

True or false?

$$\frac{4}{7} > \frac{4}{9}$$

A. True

B. False

Which comparison is true?

A. $\frac{2}{3} > \frac{4}{10}$

B. $\frac{2}{3} = \frac{4}{10}$

C. $\frac{4}{10} > \frac{2}{3}$

D. $\frac{2}{3} < \frac{4}{10}$

Write $>$, $=$, or $<$.

$$\frac{5}{12} \text{ — } \frac{5}{9}$$

Symbol: _____

True or false?

$$\frac{3}{10} > \frac{3}{5}$$

A. True

B. False

FORWARD 2

Compare the size of a product with the size of one factor without multiplying •
Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

True or false?

$\frac{3}{4} \times 10$ is less than 10.

A. True

B. False

Without multiplying, is $\frac{5}{3} \times 12$ greater than, less than, or equal to 12?

Answer: _____

Which product is less than 8?

A. $\frac{3}{2} \times 8$

B. $\frac{4}{4} \times 8$

C. $\frac{2}{3} \times 8$

D. 2×8

Without multiplying, which is greater: $\frac{1}{3} \times 9$ or $\frac{1}{6} \times 9$?

Answer: _____

True or false?

$\frac{7}{8} \times 5$ is greater than 5.

A. True

B. False

Compare fractions only when they refer to the same whole, and justify the comparison with a model

MAIN	Grade 3 main question	3.NF.A.3.d
BACKWARD 1	Find how much longer one object is than another	2.MD.A.4
BACKWARD 2	Equal shares of same-size wholes	2.G.A.3
FORWARD 1	Compare fractions with the benchmark $\frac{1}{2}$ and explain the comp...	4.NF.A.2
FORWARD 2	Explain why multiplying by a fraction greater or less than 1 makes...	5.NF.B.5.b

SET 37 • 3.NF.A.3.d

MAIN

Grade 3 main question • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.d

True or false?

Ana ate $\frac{1}{2}$ of a small pizza. Ben ate $\frac{1}{2}$ of a large pizza. They ate the same amount of pizza.

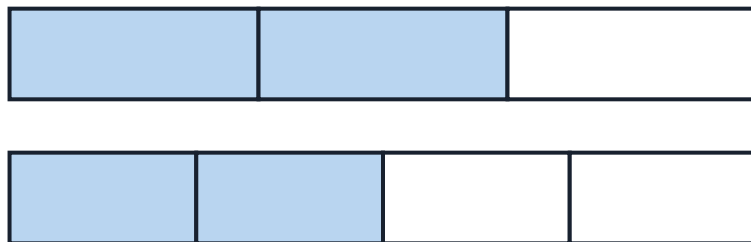
- A.** True
- B.** False

Explain why $\frac{1}{4}$ of a large cake can be more cake than $\frac{1}{2}$ of a small cake.

Explanation: _____

The two bars are the same size.

Which comparison do they show?



A. $\frac{2}{3} > \frac{2}{4}$

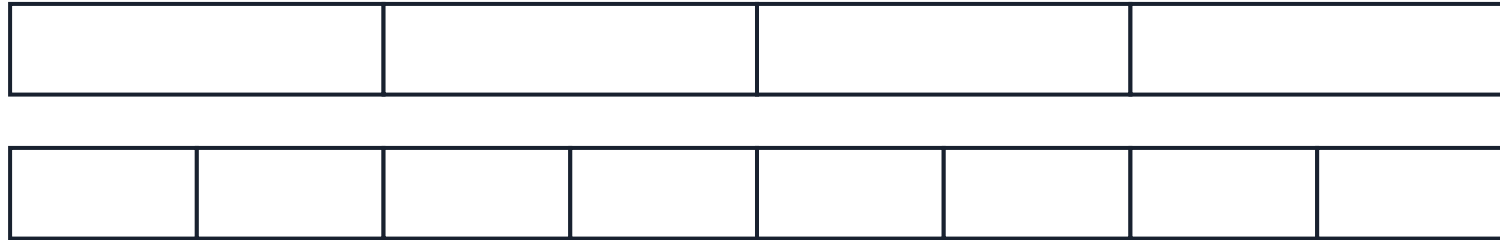
B. $\frac{2}{3} < \frac{2}{4}$

C. $\frac{2}{4} > \frac{2}{3}$

D. $\frac{2}{3} = \frac{2}{4}$

The bars are the same size. Shade $\frac{3}{4}$ of the top bar and $\frac{3}{8}$ of the bottom bar.

Then write $>$, $=$, or $<$ to compare $\frac{3}{4}$ and $\frac{3}{8}$.



Comparison: _____

True or false?

To compare $\frac{2}{6}$ and $\frac{5}{6}$ with models, both models must show the same whole.

- A. True
- B. False

SET 37 • 3.NF.A.3.d

BACKWARD 1

Find how much longer one object is than another • Grade 2 • Measurement & Data • 2.MD.A.4

A pencil is 7 inches long. A crayon is 4 inches long.
How much longer is the pencil?

Answer: _____

True or false?

A 10-cm stick is 6 cm longer than a 4-cm stick.

A. True

B. False

A book is 12 inches long and a card is 5 inches long.

How much longer is the book?

- A.** 17 inches
- B.** 12 inches
- C.** 5 inches
- D.** 7 inches

A rope is 9 feet long and a string is 6 feet long.
How much longer is the rope?

Answer: _____

True or false?

An 8-inch ribbon is 5 inches longer than a 2-inch ribbon.

- A. True
- B. False

SET 37 • 3.NF.A.3.d

BACKWARD 2

Equal shares of same-size wholes • Grade 2 • Geometry • 2.G.A.3

True or false?

Halves of a big circle and halves of a small circle are the same size.

A. True

B. False

Two same-size rectangles are each cut in half.
Are all four halves the same size?

Answer: _____

Which pair shows the same amount?

- A.** A third of a sheet and a half of the same sheet
- B.** A fourth of a sheet and a third of the same sheet
- C.** Half of a big sheet and half of a small sheet
- D.** Half of a sheet of paper and half of a same-size sheet

A small pizza and a large pizza are each cut into fourths.
Which pizza has the larger fourths?

Answer: _____

True or false?

Equal shares of the same rectangle are the same size.

- A.** True
- B.** False

SET 37 • 3.NF.A.3.d

FORWARD 1

Compare fractions with the benchmark $\frac{1}{2}$ and explain the comparison • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Compare $\frac{2}{5}$ and $\frac{5}{8}$ by comparing each one with $\frac{1}{2}$. Write $>$, $=$, or $<$.

Comparison: _____

Reason: _____

True or false?

$$\frac{7}{12} > \frac{1}{2}$$

A. True

B. False

Which fraction is less than $\frac{1}{2}$?

A. $\frac{3}{7}$

B. $\frac{5}{8}$

C. $\frac{6}{10}$

D. $\frac{4}{6}$

Explain why $\frac{3}{4}$ of a 1-meter rope is longer than $\frac{3}{4}$ of a 1-foot rope.

Explanation: _____

True or false?

$$\frac{5}{12} > \frac{1}{2}$$

A. True

B. False

FORWARD 2

Explain why multiplying by a fraction greater or less than 1 makes a product larger or smaller • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.b

True or false?

Multiplying 6 by $\frac{5}{4}$ gives a number greater than 6.

A. True

B. False

Explain why $\frac{2}{3} \times 15$ is less than 15.

Explanation: _____

Which product is greater than 20?

A. $\frac{1}{2} \times 20$

B. $\frac{5}{5} \times 20$

C. $\frac{6}{5} \times 20$

D. $\frac{4}{5} \times 20$

Explain why $\frac{4}{4} \times 7 = 7$.

Explanation: _____

True or false?

Multiplying 9 by $\frac{3}{8}$ gives a number greater than 9.

- A. True
- B. False

Tell and write time to the nearest minute

MAIN	Grade 3 main question	3.MD.A.1
BACKWARD 1	Tell time to the nearest five minutes	2.MD.C.7
BACKWARD 2	Skip-count by 5s	2.NBT.A.2
FORWARD 1	Express hours in minutes and minutes in seconds	4.MD.A.1
FORWARD 2	Convert between units of time, including fractions of an hour	5.MD.A.1

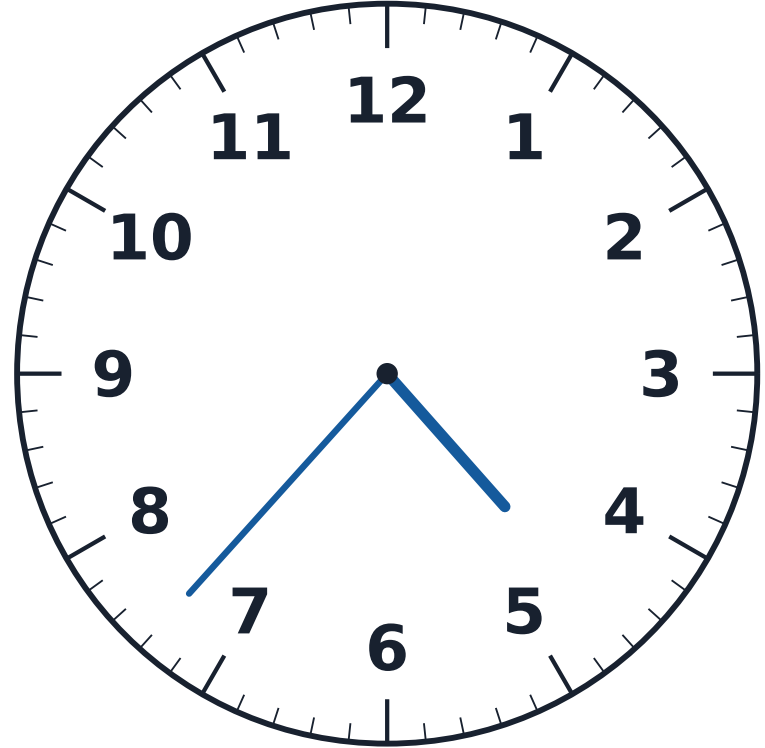
SET 38 • 3.MD.A.1

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.1

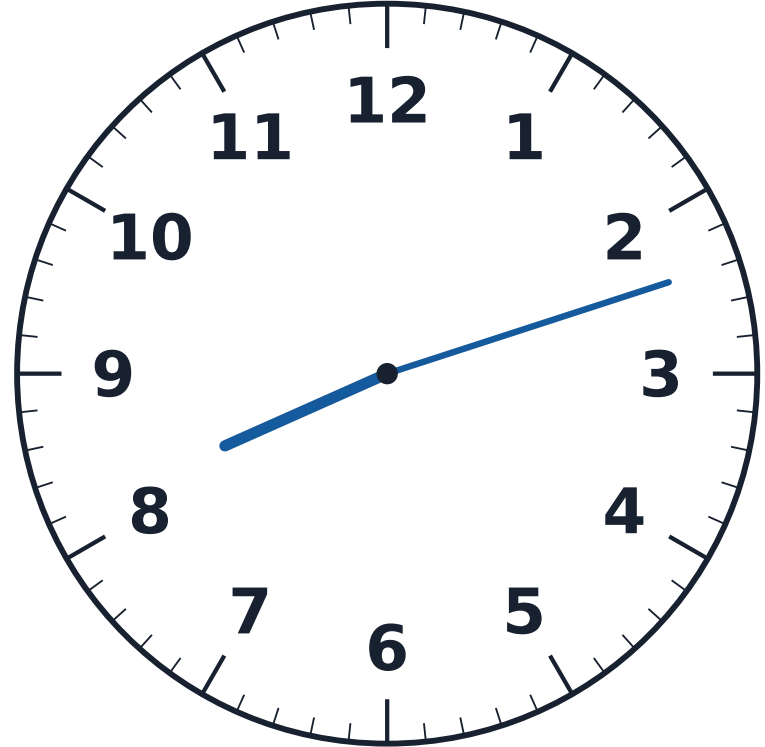
What time does the clock show?

Time: _____

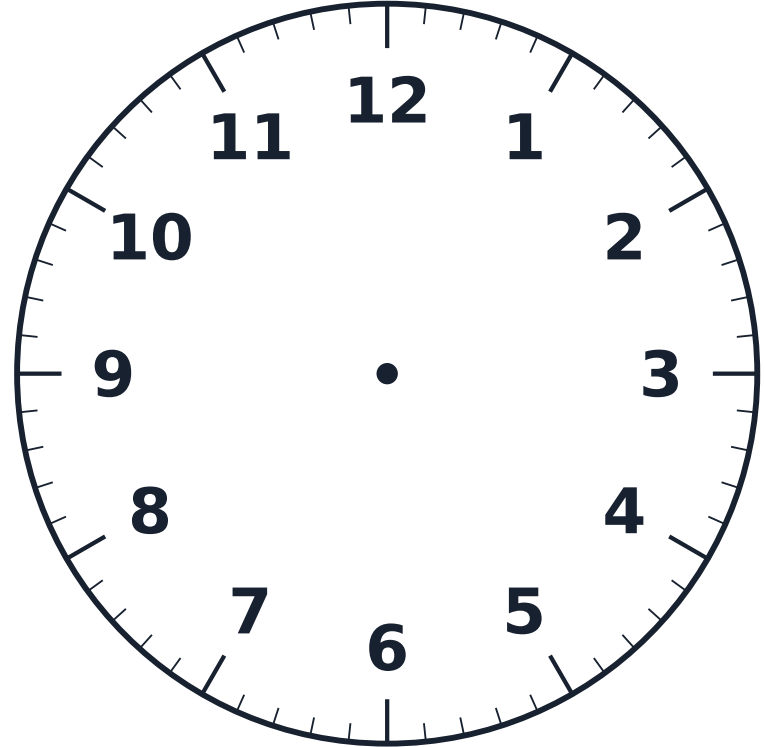


What time does the clock show?

- A.** 8:12
- B.** 8:14
- C.** 2:40
- D.** 3:40



Draw the hands on the clock to show 10:53.

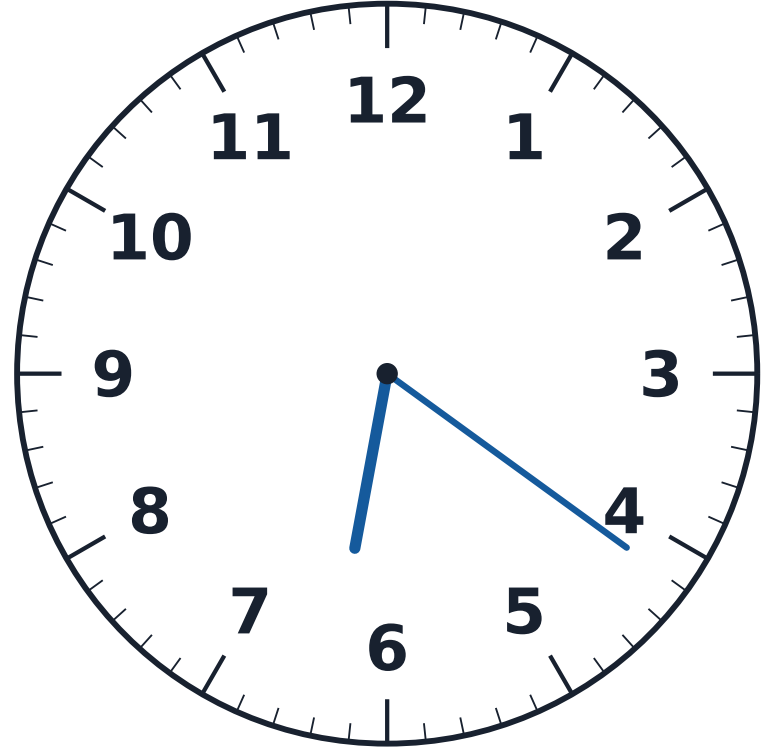


True or false?

The clock shows 6:21.

A. True

B. False



True or false?

The clock shows 1:48.

A. True

B. False



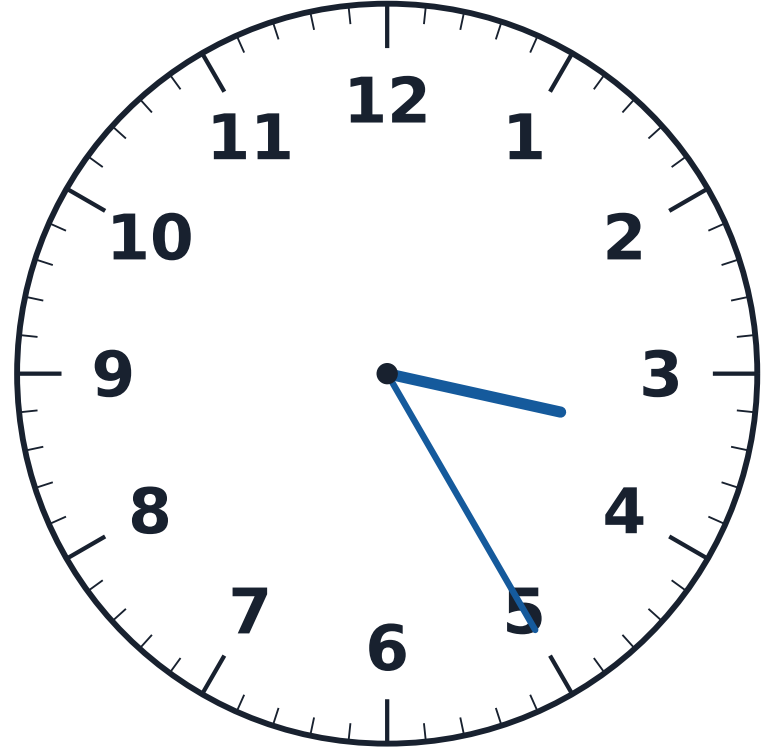
SET 38 • 3.MD.A.1

BACKWARD 1

Tell time to the nearest five minutes • Grade 2 • Measurement & Data •
2.MD.C.7

What time does the clock show?

Time: _____

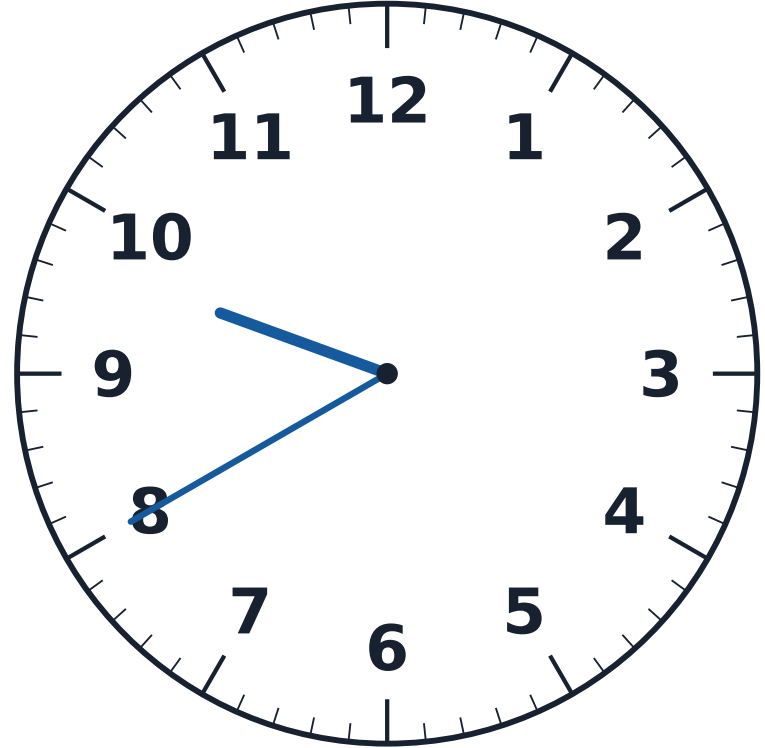


True or false?

The clock shows 9:40.

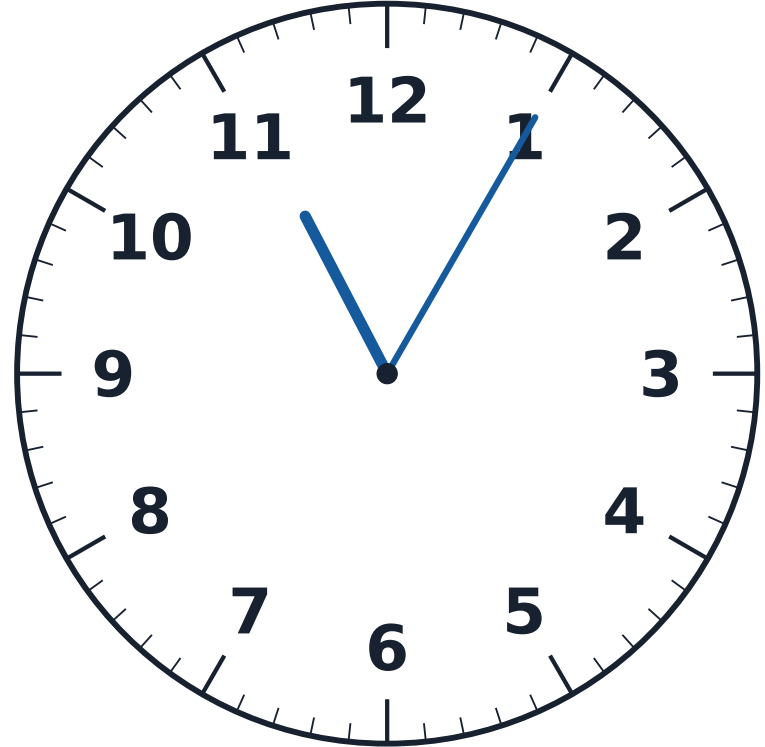
A. True

B. False



What time does the clock show?

- A.** 11:01
- B.** 12:05
- C.** 11:05
- D.** 1:55



When the minute hand points to the 6, how many minutes after the hour is it?

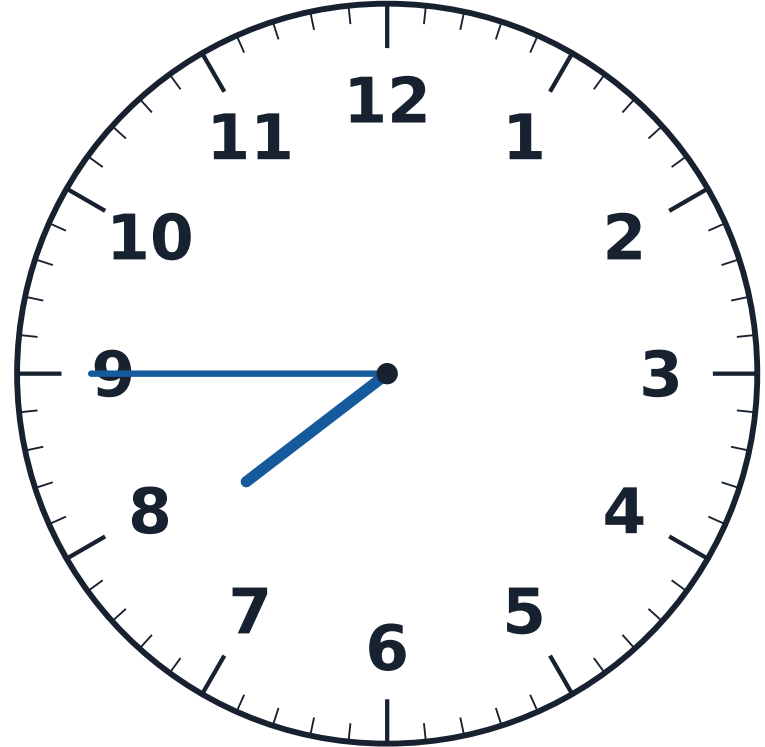
Minutes: _____

True or false?

The clock shows 7:15.

A. True

B. False



SET 38 • 3.MD.A.1

BACKWARD 2

Skip-count by 5s • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 5s from 0 to 35.

How many 5s did you count?

Fives: _____

True or false?

Counting by 5s: 40, 45, 50, 55.

A. True

B. False

Count by 5s.

15, 20, 25, ____

A. 30

B. 35

C. 50

D. 26

Skip-count by 5s.

35, 40, 45, __, __

Next two: _____

True or false?

Counting by 5s from 5, the 9th number is 40.

A. True

B. False

SET 38 • 3.MD.A.1

FORWARD 1

Express hours in minutes and minutes in seconds • Grade 4 • Measurement & Data • 4.MD.A.1

How many minutes are in 3 hours?

Minutes: _____

True or false?

5 minutes = 300 seconds

A. True

B. False

1 hour 20 minutes = ____ minutes

- A.** 140
- B.** 80
- C.** 70
- D.** 120

Complete the table.

Minutes	1	2	3	4
Seconds	60	?	180	?

Missing values: _____

True or false?

2 hours = 200 minutes

A. True

B. False

SET 38 • 3.MD.A.1

FORWARD 2

Convert between units of time, including fractions of an hour • Grade 5 •
Measurement & Data • 5.MD.A.1

How many hours is 150 minutes?

Hours: _____

True or false?

90 seconds = $1\frac{1}{2}$ minutes

A. True

B. False

A movie is 135 minutes long.

How long is it in hours and minutes?

- A.** 1 hour 35 minutes
- B.** 2 hours 15 minutes
- C.** 2 hours 35 minutes
- D.** 13 hours 5 minutes

How many minutes are in $\frac{3}{4}$ of an hour?

Minutes: _____

True or false?

$3\frac{1}{2}$ hours = 330 minutes

A. True

B. False

Solve word problems about time intervals in minutes

MAIN	Grade 3 main question	3.MD.A.1
BACKWARD 1	Tell time and use a.m. and p.m.	2.MD.C.7
BACKWARD 2	Add two-digit numbers within 100	2.NBT.B.5
FORWARD 1	Solve word problems about time intervals in hours and minutes	4.MD.A.2
FORWARD 2	Solve multistep time problems with conversions and fractions of a...	5.MD.A.1

SET 39 • 3.MD.A.1

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.1

A soccer game starts at 2:15 and ends at 3:05.
How many minutes long is the game?

Minutes: _____

Mia starts reading at 6:40. She reads for 35 minutes.
What time does she stop?

Time: _____

A bus ride takes 25 minutes and ends at 8:10.

What time did the ride start?

A. 7:55

B. 8:35

C. 7:35

D. 7:45

True or false?

Ben practices piano for 20 minutes and then guitar for 15 minutes. He practices for 35 minutes in all.

- A.** True
- B.** False

True or false?

A class starts at 10:50 and lasts 45 minutes. It ends at 11:25.

- A.** True
- B.** False

SET 39 • 3.MD.A.1

BACKWARD 1

Tell time and use a.m. and p.m. • Grade 2 • Measurement & Data • 2.MD.C.7

True or false?

Times from midnight until noon are a.m. times.

- A.** True
- B.** False

Which time is in the afternoon?

A. 11:00 a.m.

B. 3:00 a.m.

C. 6:00 a.m.

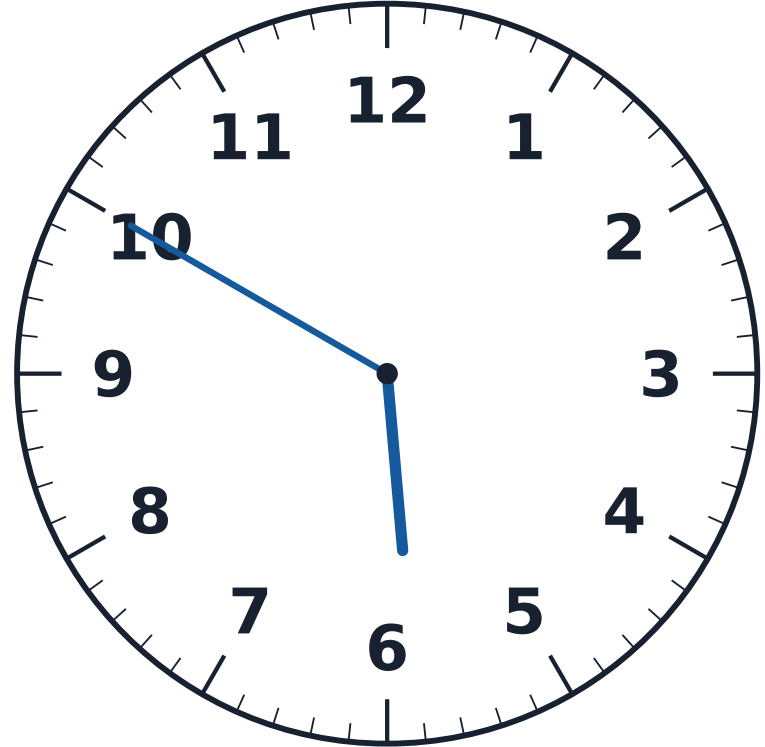
D. 3:00 p.m.

Is 9:00 at night an a.m. time or a p.m. time?

Answer: _____

What time does the clock show?

Time: _____



True or false?

Noon is written as 12:00 a.m.

A. True

B. False

SET 39 • 3.MD.A.1

BACKWARD 2

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$35 + 25$$

Sum: _____

True or false?

$$45 + 15 = 60$$

A. True

B. False

$$20 + 55 = ?$$

A. 35

B. 75

C. 85

D. 65

Add.

$$40 + 35$$

Sum: _____

True or false?

$$25 + 50 = 85$$

A. True

B. False

SET 39 • 3.MD.A.1

FORWARD 1

Solve word problems about time intervals in hours and minutes • Grade 4 •
Measurement & Data • 4.MD.A.2

A train leaves at 9:45 a.m. and arrives at 1:20 p.m.

How long is the trip?

Time: _____

True or false?

A movie starts at 6:50 p.m. and lasts 1 hour 45 minutes. It ends at 8:35 p.m.

- A.** True
- B.** False

A hike takes 2 hours 40 minutes and ends at 4:10 p.m.

When did it start?

- A.** 2:30 p.m.
- B.** 6:50 p.m.
- C.** 1:50 p.m.
- D.** 1:30 p.m.

Ty practices for 35 minutes each day for 4 days.

How long does he practice in all? Write the answer in hours and minutes.

Time: _____

True or false?

A 50-minute lesson that starts at 11:30 a.m. ends at 12:10 p.m.

A. True

B. False

SET 39 • 3.MD.A.1

FORWARD 2

Solve multistep time problems with conversions and fractions of a minute or hour • Grade 5 • Measurement & Data • 5.MD.A.1

A race has 3 laps. Each lap takes $1\frac{1}{2}$ minutes.

How many seconds does the race take?

Seconds: _____

True or false?

A trip of 2 hours 15 minutes is 2.25 hours long.

A. True

B. False

Ana runs for 0.75 hour and then walks for 20 minutes.

How many minutes is that in all?

- A.** 95
- B.** 65
- C.** 75
- D.** 55

A video is $2\frac{1}{4}$ minutes long.

How many seconds is that?

Seconds: _____

True or false?

$\frac{1}{3}$ of an hour is 30 minutes.

A. True

B. False

Represent time-interval problems on a number line diagram

MAIN	Grade 3 main question	3.MD.A.1
BACKWARD 1	Show sums and differences within 100 on a number line	2.MD.B.6
BACKWARD 2	Read clocks to the nearest five minutes	2.MD.C.7
FORWARD 1	Use number line diagrams for time problems in hours and minutes	4.MD.A.2
FORWARD 2	Use fractions and decimals of an hour in time problems	5.MD.A.1

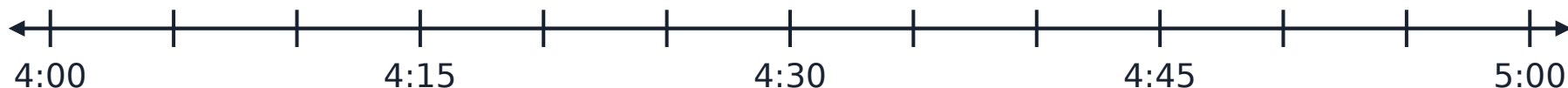
SET 40 • 3.MD.A.1

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.1

Lu starts her homework at 4:10 and works for 40 minutes.

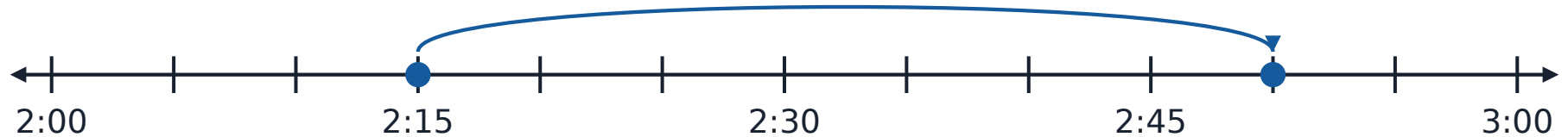
Show the time with jumps on the number line. What time does she finish?



Time: _____

The number line shows a trip.

How many minutes long is the trip?



Minutes: _____

A number line from 3:00 to 4:00 is marked every 5 minutes. Kim jumps from 3:15 to 3:40.

How many minutes is the jump?

A. 25

B. 55

C. 5

D. 40

True or false?

On a number line from 9:00 to 10:00, 9:30 is halfway between 9:00 and 10:00.

- A.** True
- B.** False

True or false?

On a number line, the jump from 7:20 to 8:05 is 35 minutes.

A. True

B. False

SET 40 • 3.MD.A.1

BACKWARD 1

Show sums and differences within 100 on a number line • Grade 2 •
Measurement & Data • 2.MD.B.6

Start at 30 on a number line and jump 25.

Where do you land?

Number: _____

True or false?

A jump from 15 to 45 on a number line is a jump of 30.

A. True

B. False

A jump from 40 to 75 on a number line is a jump of how much?

- A.** 115
- B.** 35
- C.** 25
- D.** 45

Show $20 + 15$ with jumps on the number line.



True or false?

Starting at 10 on a number line, a jump of 30 lands on 50.

- A.** True
- B.** False

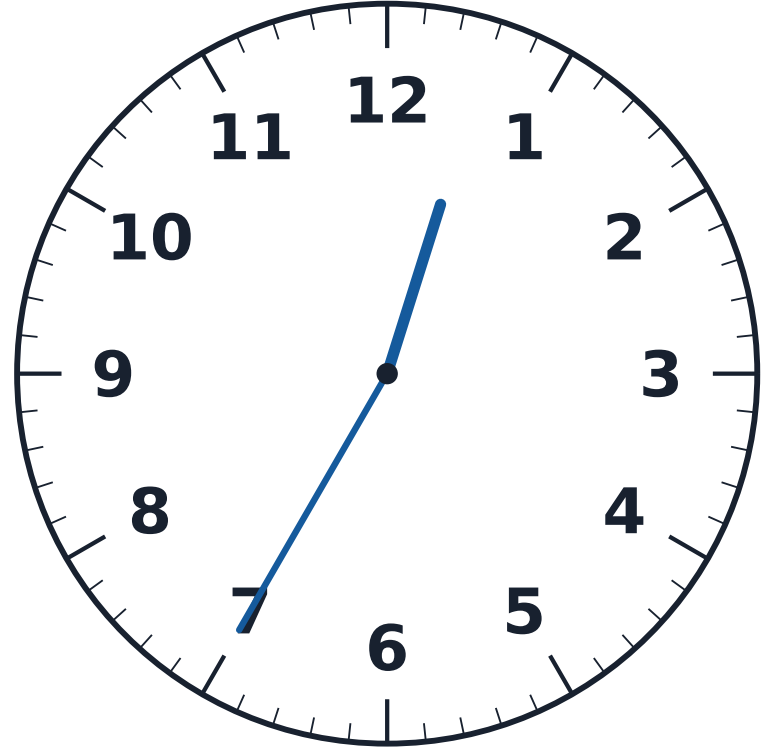
SET 40 • 3.MD.A.1

BACKWARD 2

Read clocks to the nearest five minutes • Grade 2 • Measurement & Data •
2.MD.C.7

What time does the clock show?

Time: _____

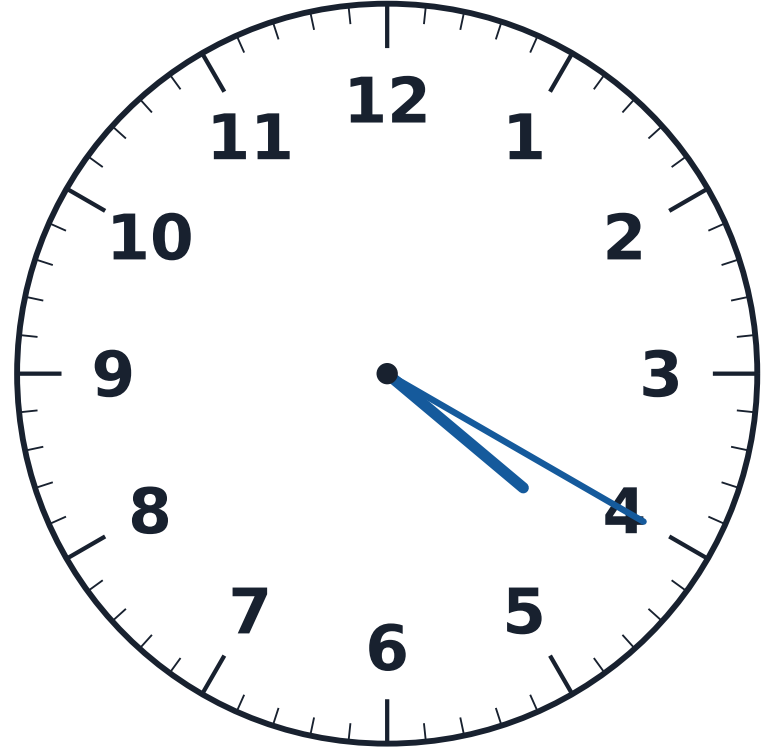


True or false?

The clock shows 4:20.

A. True

B. False



What time does the clock show?

- A.** 10:10
- B.** 11:10
- C.** 10:02
- D.** 2:50



At 8:45, which number does the minute hand point to?

Number: _____

True or false?

At 3:30, the minute hand points to the 3.

- A. True
- B. False

SET 40 • 3.MD.A.1

FORWARD 1

Use number line diagrams for time problems in hours and minutes • Grade 4 •
Measurement & Data • 4.MD.A.2

A field trip goes from 9:40 a.m. to 12:15 p.m.
Use a number line to find how long the trip is.

Time: _____

True or false?

On a number line, the jump from 10:45 a.m. to 12:30 p.m. is 1 hour 45 minutes.

- A.** True
- B.** False

A jump of 1 hour 20 minutes starts at 2:50 p.m. on a number line.

Where does it end?

A. 3:70 p.m.

B. 4:10 p.m.

C. 3:10 p.m.

D. 4:20 p.m.

A show starts at 7:35 p.m. and lasts 1 hour 40 minutes.

Show the jumps on the number line. When does the show end?



Ends at: _____

True or false?

On a number line, the jump from 11:50 a.m. to 1:05 p.m. is 1 hour 25 minutes.

A. True

B. False

SET 40 • 3.MD.A.1

FORWARD 2

Use fractions and decimals of an hour in time problems • Grade 5 •
Measurement & Data • 5.MD.A.1

How many minutes are in $\frac{1}{4}$ of an hour?

Minutes: _____

True or false?

$1\frac{1}{2}$ hours is 90 minutes.

A. True

B. False

A trip takes $2\frac{1}{4}$ hours.

How many minutes is that?

- A.** 135
- B.** 150
- C.** 225
- D.** 125

Mia reads for 0.5 hour and then for 0.75 hour.
How many minutes does she read in all?

Minutes: _____

True or false?

$\frac{2}{3}$ of an hour is 45 minutes.

A. True

B. False

Measure and estimate liquid volumes and masses in liters, grams, and kilograms

MAIN	Grade 3 main question	3.MD.A.2
BACKWARD 1	Estimate lengths in inches, feet, centimeters, and meters	2.MD.A.3
BACKWARD 2	Choose and use tools to measure length	2.MD.A.1
FORWARD 1	Know the relative sizes of grams and kilograms, and of milliliters...	4.MD.A.1
FORWARD 2	Convert metric units of mass and liquid volume with decimals	5.MD.A.1

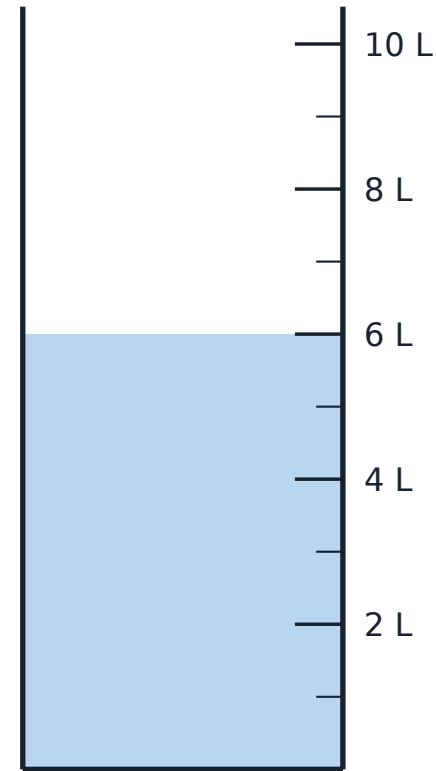
SET 41 • 3.MD.A.2

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.2

How many liters of water are
in the container?

Liters: _____



Which object has a mass of about 1 kilogram?

- A.** A grape
- B.** A paper clip
- C.** A pineapple
- D.** A car

True or false?

A bathtub holds more than 1 liter of water.

A. True

B. False

Would you measure the mass of a bean in grams or in kilograms?

Unit: _____

Shade the container to show
3 liters.



SET 41 • 3.MD.A.2

BACKWARD 1

Estimate lengths in inches, feet, centimeters, and meters • Grade 2 •
Measurement & Data • 2.MD.A.3

About how long is a new pencil: 7 inches or 7 feet?

Answer: _____

True or false?

A door is about 2 meters tall.

A. True

B. False

Which is the best estimate of the length of a car?

- A.** 40 meters
- B.** 4 inches
- C.** 4 centimeters
- D.** 4 meters

About how long is a paper clip: 3 centimeters or 3 meters?

Answer: _____

True or false?

A school bus is about 12 inches long.

A. True

B. False

SET 41 • 3.MD.A.2

BACKWARD 2

Choose and use tools to measure length • Grade 2 • Measurement & Data •
2.MD.A.1

True or false?

A meter stick is a good tool for measuring the length of a classroom.

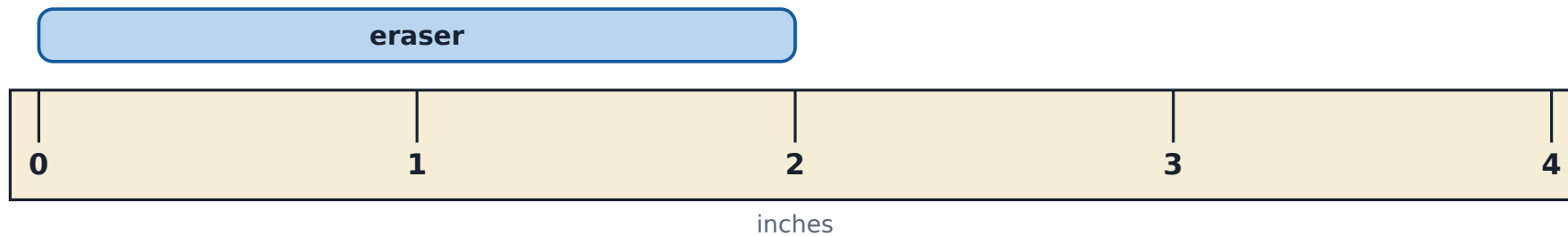
A. True

B. False

Which tool is best for measuring the length of a crayon?

- A.** A ruler
- B.** A clock
- C.** A scale
- D.** A measuring cup

How long is the eraser?



Length: _____

Name a tool you can use to measure the length of a hallway.

Tool: _____

True or false?

A clock is a tool for measuring length.

A. True

B. False

SET 41 • 3.MD.A.2

FORWARD 1

Know the relative sizes of grams and kilograms, and of milliliters and liters •

Grade 4 • Measurement & Data • 4.MD.A.1

How many grams are in 3 kilograms?

Grams: _____

True or false?

1 liter = 1,000 milliliters

A. True

B. False

Which is heaviest?

- A.** 200 grams
- B.** 2 kilograms
- C.** 1,500 grams
- D.** 20 grams

Complete the table.

Liters	1	2	3	4
Milliliters	1,000	?	3,000	?

Missing values: _____

True or false?

5 kilograms = 500 grams

A. True

B. False

SET 41 • 3.MD.A.2

FORWARD 2

Convert metric units of mass and liquid volume with decimals • Grade 5 •
Measurement & Data • 5.MD.A.1

Convert.

2.5 kilograms = ____ grams

Grams: _____

True or false?

750 milliliters = 0.75 liter

A. True

B. False

Convert.

3,200 grams = ____ kilograms

- A.** 32
- B.** 320
- C.** 3.2
- D.** 0.32

Convert.

1.25 liters = ____ milliliters

Milliliters: _____

True or false?

400 grams = 4 kilograms

A. True

B. False

Solve one-step word problems about masses or liquid volumes

MAIN	Grade 3 main question	3.MD.A.2
BACKWARD 1	Subtract lengths in one-step word problems within 100	2.MD.B.5
BACKWARD 2	Add two-digit numbers within 100	2.NBT.B.5
FORWARD 1	Solve liquid volume and mass problems that need a unit conversi...	4.MD.A.2
FORWARD 2	Solve multistep problems with metric conversions and decimals	5.MD.A.1

SET 42 • 3.MD.A.2

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.A.2

A pot holds 9 liters of soup. 4 liters are served.
How many liters are left?

Liters: _____

Each bag of rice has a mass of 5 kilograms.

What is the mass of 6 bags?

Mass: _____

A 36-liter tank is filled with a 4-liter bucket.
How many full buckets fill the tank?

- A.** 8
- B.** 9
- C.** 40
- D.** 32

True or false?

A 450-gram melon and a 300-gram apple have a total mass of 850 grams.

- A.** True
- B.** False

True or false?

24 liters of juice are shared equally among 3 jugs.
Each jug gets 6 liters.

- A. True
- B. False

SET 42 • 3.MD.A.2

BACKWARD 1

Subtract lengths in one-step word problems within 100 • Grade 2 •
Measurement & Data • 2.MD.B.5

A rope is 85 cm long. 38 cm is cut off.

How long is the rope now?

Length: _____

True or false?

An 80-inch shelf with 35 inches cut off is 45 inches long.

- A. True
- B. False

A path is 64 m long. Ana has walked 29 m.

How far does she have left?

A. 25 m

B. 93 m

C. 35 m

D. 45 m

A stick is 74 inches long. Lu cuts off 26 inches.
How long is the stick now?

Length: _____

True or false?

A 90-cm board with 45 cm cut off is 55 cm long.

A. True

B. False

SET 42 • 3.MD.A.2

BACKWARD 2

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$45 + 30$$

Sum: _____

True or false?

$$60 + 25 = 85$$

A. True

B. False

$$38 + 27 = ?$$

A. 11

B. 55

C. 75

D. 65

Add.

$$54 + 19$$

Sum: _____

True or false?

$$29 + 46 = 65$$

A. True

B. False

SET 42 • 3.MD.A.2

FORWARD 1

Solve liquid volume and mass problems that need a unit conversion • Grade 4 •
Measurement & Data • 4.MD.A.2

A jug holds 2 liters. Ella pours out 750 milliliters.
How many milliliters are left?

Milliliters: _____

True or false?

Four 250-gram bags of nuts have a total mass of 1 kilogram.

A. True

B. False

A 3-kilogram bag of flour is split into 6 equal bags.
What is the mass of each bag?

- A.** 2 grams
- B.** 50 grams
- C.** 500 grams
- D.** 5,000 grams

A box has a mass of 1 kg 200 g. Another box has a mass of 850 g.

What is their total mass in grams?

Grams: _____

True or false?

A 5-liter jug fills exactly 8 cups that each hold 500 milliliters.

A. True

B. False

SET 42 • 3.MD.A.2

FORWARD 2

Solve multistep problems with metric conversions and decimals • Grade 5 •
Measurement & Data • 5.MD.A.1

A 1.5-liter bottle fills 6 equal cups.

How many milliliters go in each cup?

Milliliters: _____

True or false?

3 bags that each have a mass of 0.4 kg have a total mass of 1,200 grams.

A. True

B. False

A 2.4-kg cheese is cut into 8 equal pieces.

What is the mass of each piece?

- A.** 0.03 kilogram
- B.** 3 kilograms
- C.** 300 grams
- D.** 30 grams

A tank has 4.5 liters of water. 1,750 mL leaks out.
How many liters are left?

Liters: _____

True or false?

Five 350-mL glasses hold 1.5 liters in all.

- A. True
- B. False

Draw a scaled picture graph to represent data

MAIN

Grade 3 main question

3.MD.B.3

BACKWARD 1

Draw and read a picture graph with a single-unit scale

2.MD.D.10

BACKWARD 2

Skip-count by 5s and 10s

2.NBT.A.2

FORWARD 1

Make a line plot of measurements in fractions of a unit

4.MD.B.4

FORWARD 2

Use operations on fractions to solve problems with line plot data

5.MD.B.2

SET 43 • 3.MD.B.3

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.3

Make a picture graph of the data. Each ● stands for 2 votes.

Apples: 6 votes, Bananas: 10 votes, Grapes: 4 votes

Favorite fruit	
Apples	
Bananas	
Grapes	

Key: ● = 2 votes

In a picture graph, each ● stands for 5 books. Ty read 20 books.

How many ● should be drawn for Ty?

A. 4

B. 20

C. 5

D. 15

A picture graph has the key $\bullet = 3$ cars. The row for red cars has 5 $\bullet$.

How many red cars are there?

Red cars: _____

True or false?

If ● = 4 students, then 7 ● stand for 28 students.

- A. True
- B. False

Data: Dogs 8, Cats 12, Fish 4. You will make a picture graph with ● = 4 pets.

How many ● go in each row?

Dogs: _____

Cats: _____

Fish: _____

SET 43 • 3.MD.B.3

BACKWARD 1

Draw and read a picture graph with a single-unit scale • Grade 2 •
Measurement & Data • 2.MD.D.10

Make a picture graph. Draw one ● for each pet.

Dogs: 4, Cats: 3, Birds: 2

Pets	
Dogs	
Cats	
Birds	

Key: ● = 1 pet

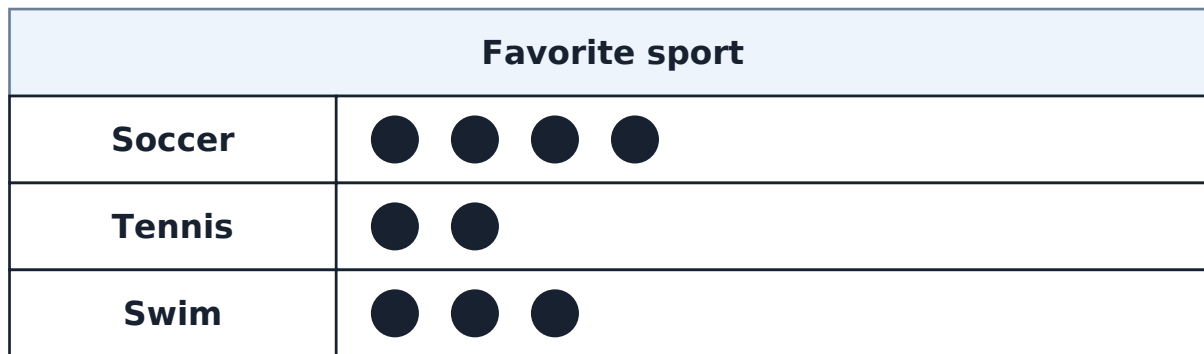
True or false?

In a picture graph where ● = 1 child, 5 ● stand for 5 children.

- A. True
- B. False

Use the picture graph.

How many children chose soccer?



Key: ● = 1 child

A. 5

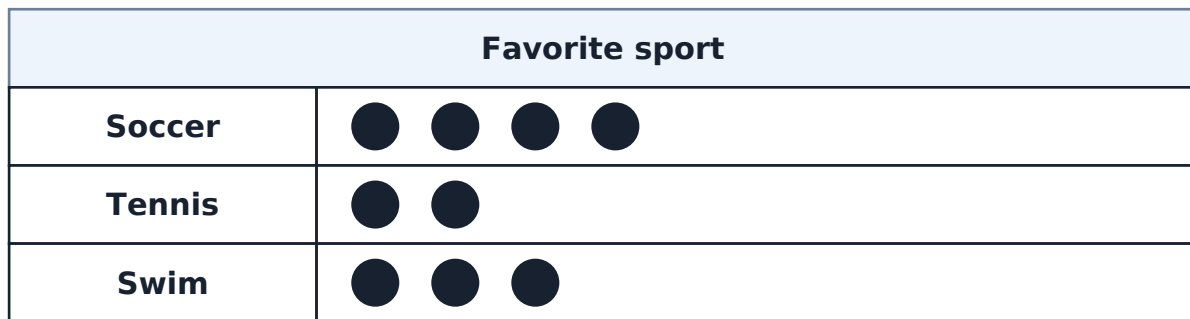
B. 4

C. 2

D. 3

Use the picture graph.

How many more children chose swim than tennis?

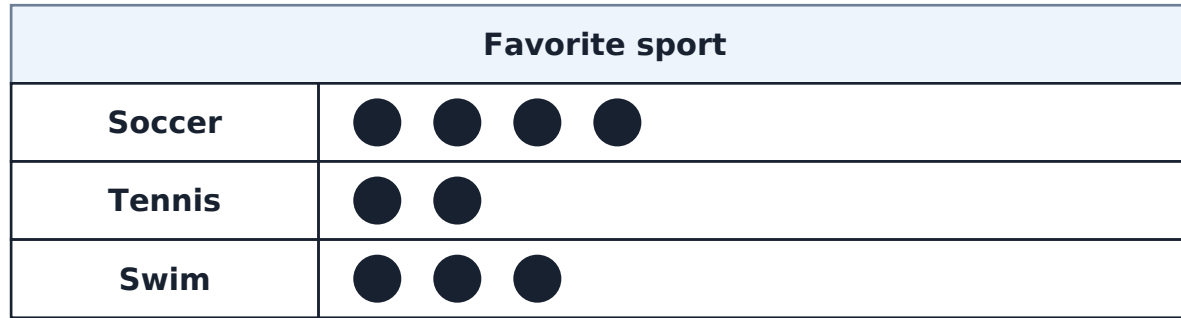


Key: ● = 1 child

More: _____

True or false?

In the picture graph, tennis has the most votes.



Key: ● = 1 child

A. True

B. False

SET 43 • 3.MD.B.3

BACKWARD 2

Skip-count by 5s and 10s • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.2

Skip-count by 5s from 5.

What is the 4th number?

Number: _____

True or false?

Skip-counting by 10s: 10, 20, 30, 40, 50.

A. True

B. False

Skip-count by 5s.

5, 10, 15, 20, ____

A. 21

B. 30

C. 35

D. 25

Skip-count by 10s from 10.

What is the 7th number?

Number: _____

True or false?

Skip-counting by 5s: 5, 10, 15, 25.

A. True

B. False

SET 43 • 3.MD.B.3

FORWARD 1

Make a line plot of measurements in fractions of a unit • Grade 4 •
Measurement & Data • 4.MD.B.4

Make a line plot of the data.

Bean lengths (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, 1



True or false?

A line plot shows each data value as an X above a number line.

- A.** True
- B.** False

Which number line works best for a line plot of $\frac{1}{8}$, $\frac{3}{8}$, and $\frac{5}{8}$ inch?

- A.** A line from 0 to 1 marked in halves
- B.** A line from 0 to 1 marked in eighths
- C.** A line from 0 to 8 marked in ones
- D.** A line from 1 to 2 marked in fourths

Lengths (inches): $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, $\frac{1}{2}$

In a line plot of the data, how many X's go above $\frac{1}{2}$?

X's: _____

True or false?

In a line plot of $\frac{1}{4}$, $\frac{1}{4}$, and $\frac{3}{4}$, there are 3 X's above $\frac{1}{4}$.

A. True

B. False

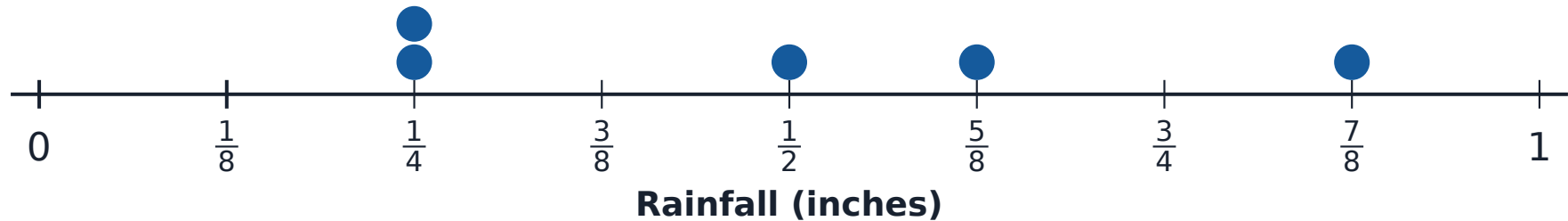
SET 43 • 3.MD.B.3

FORWARD 2

Use operations on fractions to solve problems with line plot data • Grade 5 •
Measurement & Data • 5.MD.B.2

The line plot shows rainfall on 5 days.

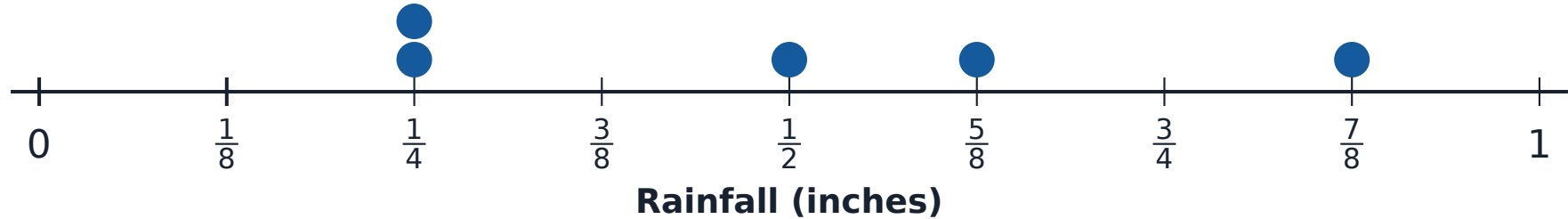
What is the total rainfall?



Total: _____

True or false?

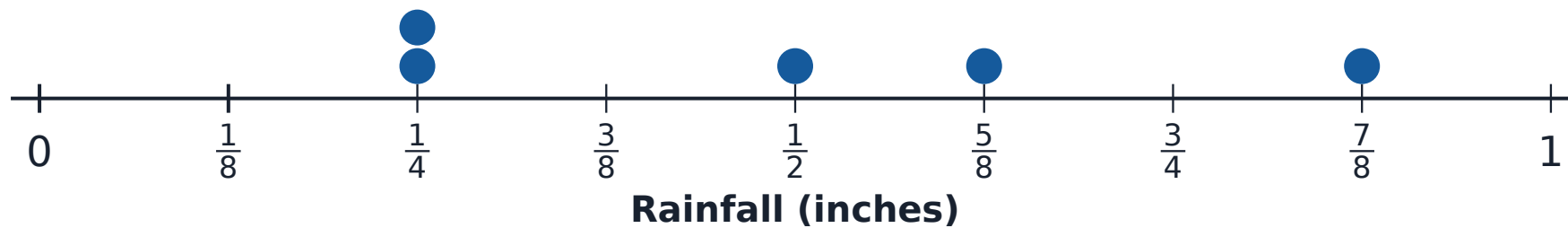
The two days with $\frac{1}{4}$ inch of rain had $\frac{1}{2}$ inch in all.



A. True

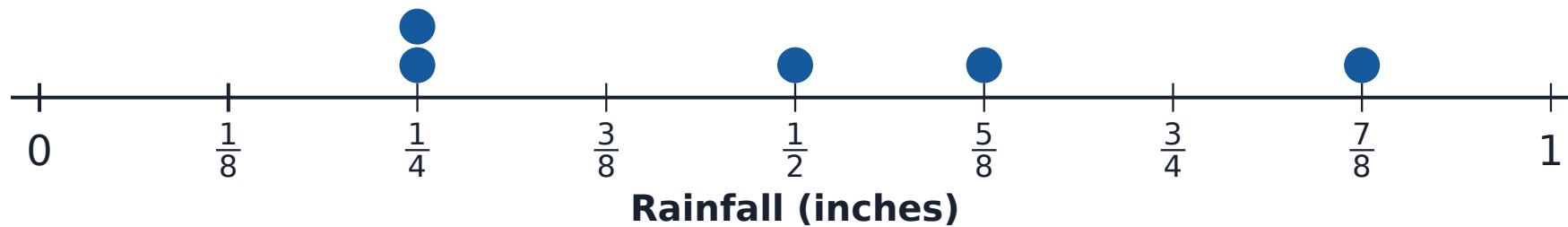
B. False

What is the difference between the greatest and the least rainfall?



- A.** $\frac{3}{8}$ inch **B.** $\frac{1}{2}$ inch **C.** $\frac{5}{8}$ inch **D.** $\frac{7}{8}$ inch

If the total rainfall had been spread equally over the 5 days, how much would have fallen each day?



Each day: _____

True or false?

The total of $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ is $\frac{3}{14}$.

A. True

B. False

Draw a scaled bar graph to represent data

MAIN	Grade 3 main question	3.MD.B.3
BACKWARD 1	Draw and read a bar graph with a single-unit scale	2.MD.D.10
BACKWARD 2	Read whole numbers on a number line scale	2.MD.B.6
FORWARD 1	Use a bar graph to solve multiplicative comparison problems — ne...	4.OA.A.2
FORWARD 2	Graph data as points in the first quadrant — nearest related (no di...	5.G.A.2

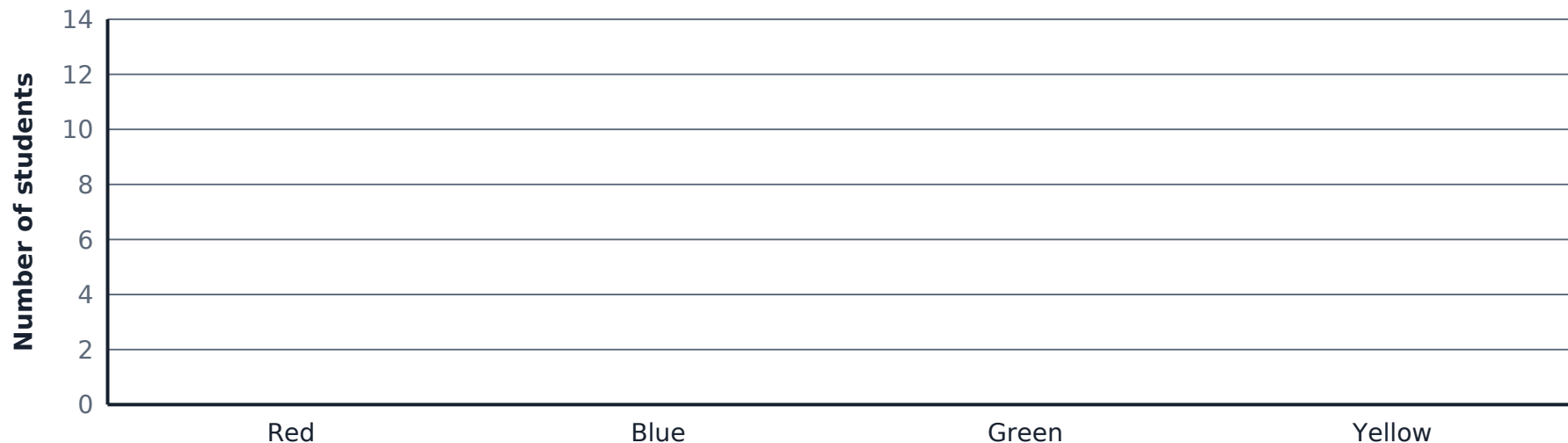
SET 44 • 3.MD.B.3

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.3

Make a bar graph of the data. The scale counts by 2s.

Red: 8, Blue: 12, Green: 6, Yellow: 10



A bar graph has a scale that counts by 2s. A bar ends halfway between 6 and 8.

What number does the bar show?

A. 14

B. 7

C. 8

D. 6

True or false?

On a bar graph with a scale that counts by 10s, a bar for 40 ends on the fourth line above 0.

- A.** True
- B.** False

Which scale is better for a bar graph of 10, 25, 40, and 35: counting by 1s or counting by 5s? Explain.

Scale: _____

Explanation: _____

True or false?

On a bar graph with a scale that counts by 2s, a bar for 9 ends exactly on a labeled line.

- A. True
- B. False

SET 44 • 3.MD.B.3

BACKWARD 1

Draw and read a bar graph with a single-unit scale • Grade 2 • Measurement & Data • 2.MD.D.10

Make a bar graph.

Apples: 4, Pears: 2, Plums: 5



True or false?

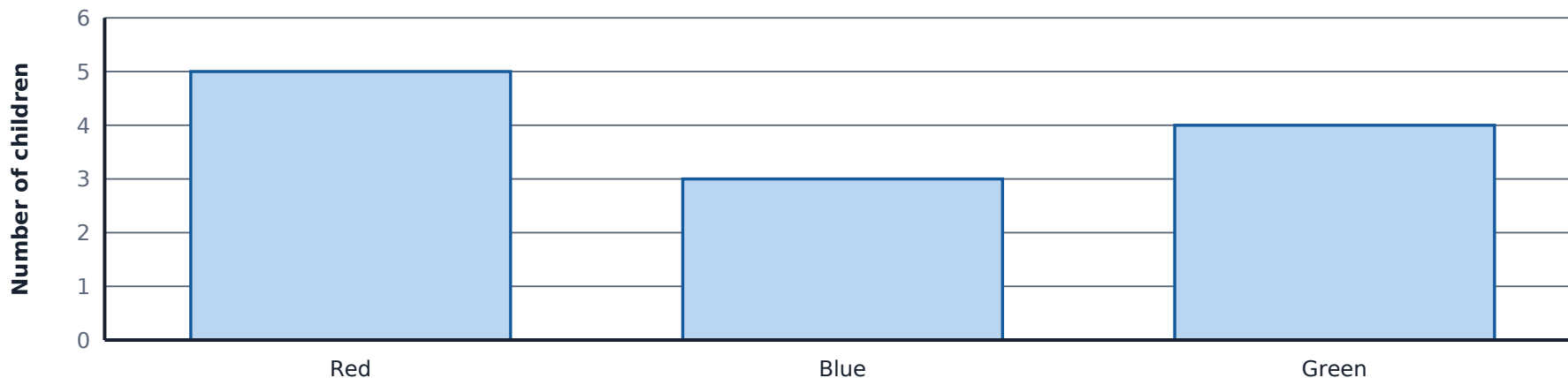
In a bar graph, a taller bar shows a greater number.

A. True

B. False

Use the bar graph.

How many children chose red?



A. 4

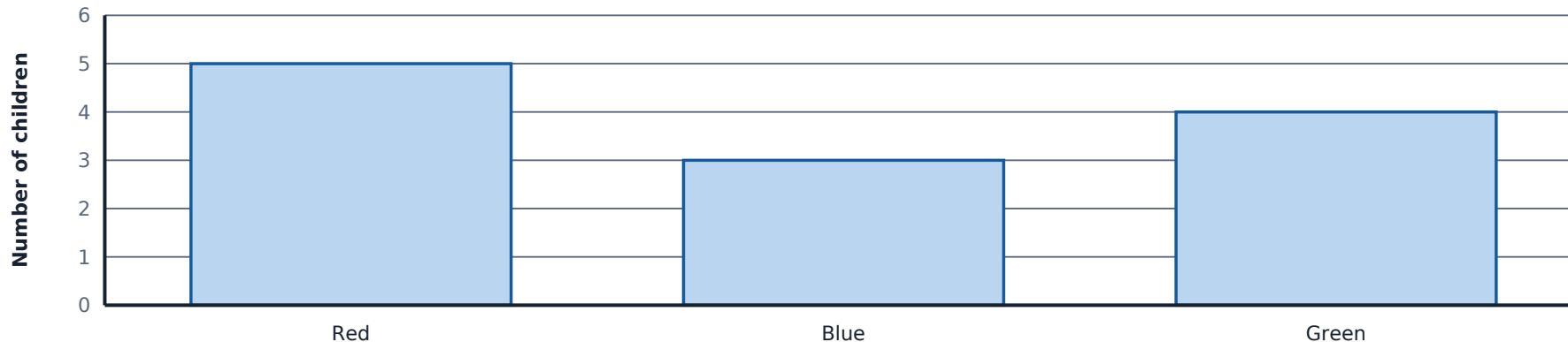
B. 5

C. 3

D. 6

Use the bar graph.

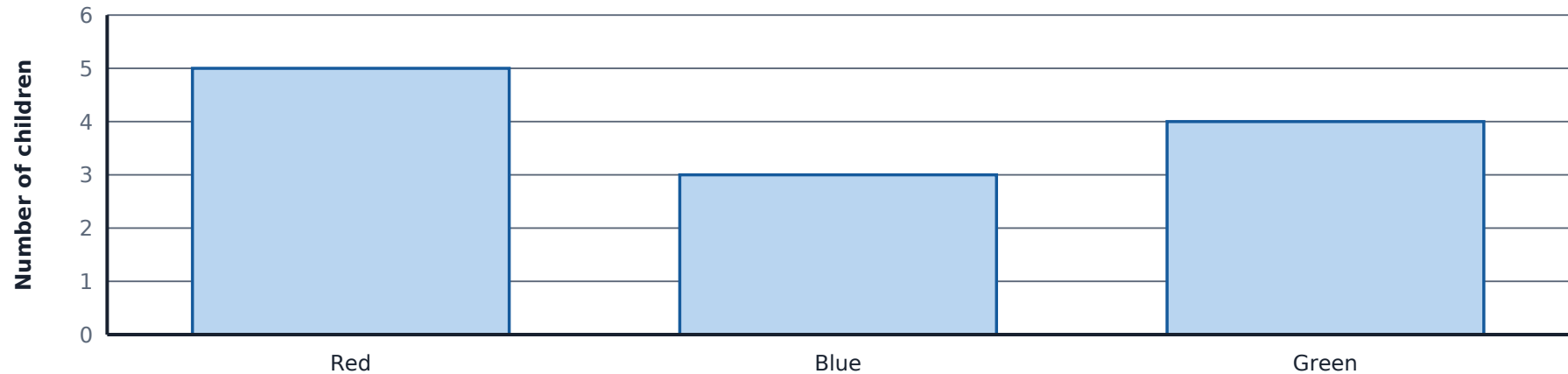
How many more children chose red than blue?



More: _____

True or false?

In the bar graph, green has the most votes.



A. True

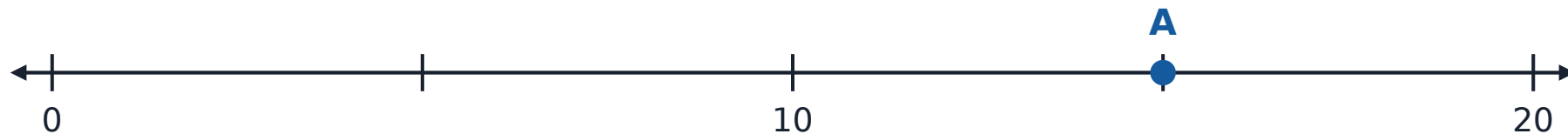
B. False

SET 44 • 3.MD.B.3

BACKWARD 2

Read whole numbers on a number line scale • Grade 2 • Measurement & Data
• 2.MD.B.6

What number is at point A?



A = _____

True or false?

On a number line that counts by 10s, 30 comes right after 20.

A. True

B. False

A number line counts by 2s: 0, 2, 4, 6, 8.

What number is halfway between 4 and 6?

- A.** 5
- B.** 10
- C.** 7
- D.** 3

What number is at point B?



B = _____

True or false?

On a number line that counts by 5s, the mark after 25 is 35.

- A. True
- B. False

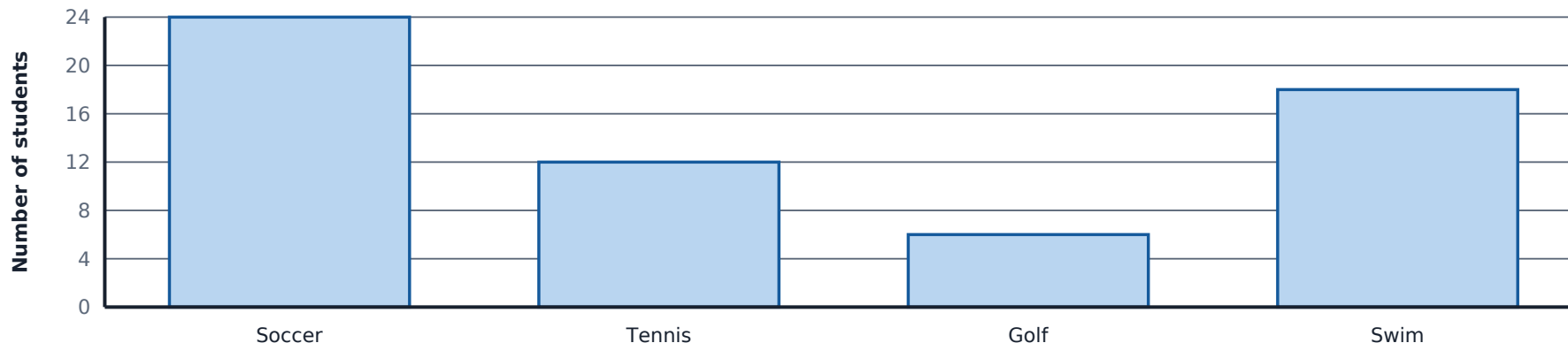
FORWARD 1

Use a bar graph to solve multiplicative comparison problems • Grade 4 •
Operations & Algebraic Thinking • 4.OA.A.2

► No Grade 4 standard directly continues this skill. This branch uses the nearest related Grade 4 standard.

Use the bar graph.

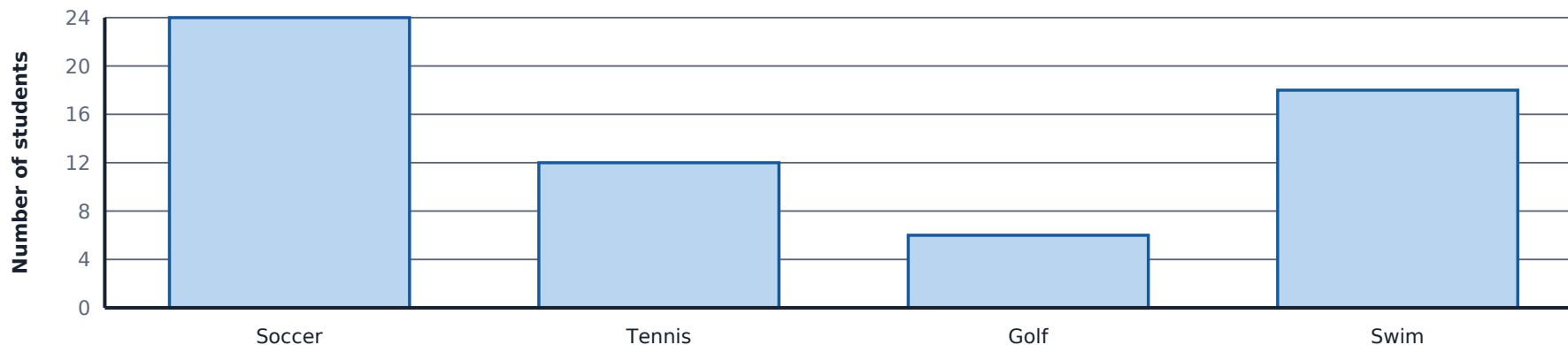
How many times as many students chose soccer as chose golf?



Times as many: _____

True or false?

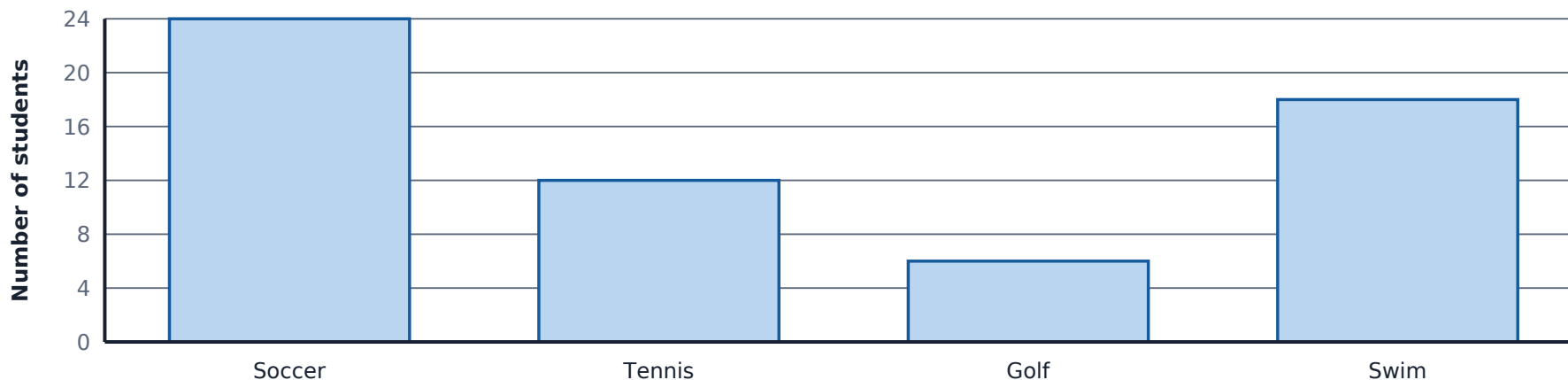
Use the bar graph. Twice as many students chose swim as chose golf.



- A. True
- B. False

Use the bar graph.

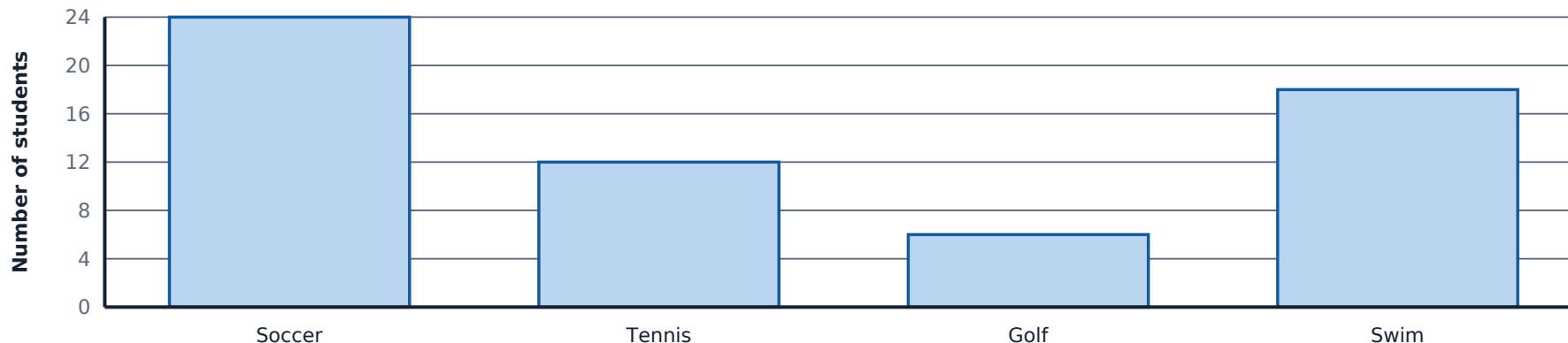
Which sport was chosen by 3 times as many students as golf?



- A.** Swim **B.** Golf **C.** Soccer **D.** Tennis

Use the bar graph.

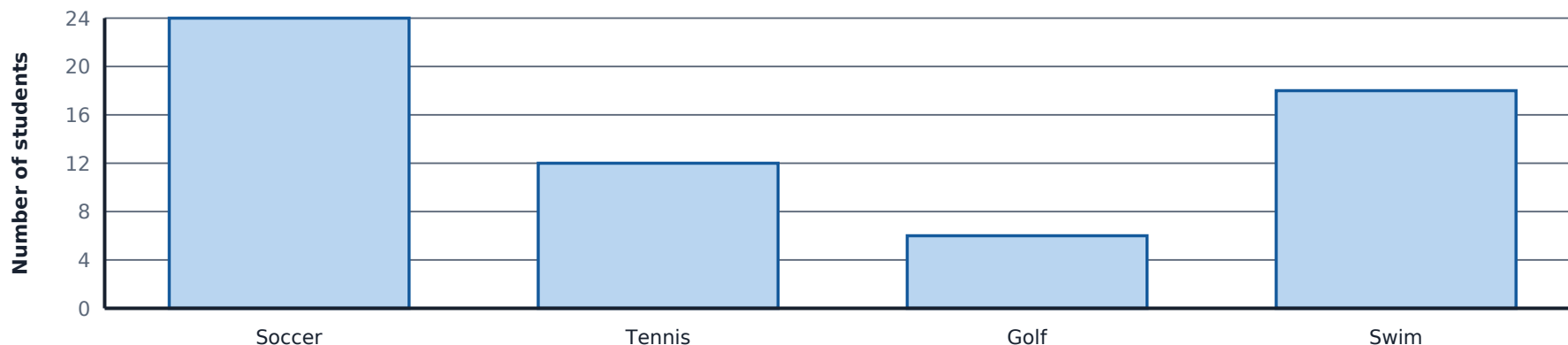
How many times as many students chose soccer as chose tennis?



Times as many: _____

True or false?

Use the bar graph. Soccer was chosen by 4 times as many students as golf.



A. True

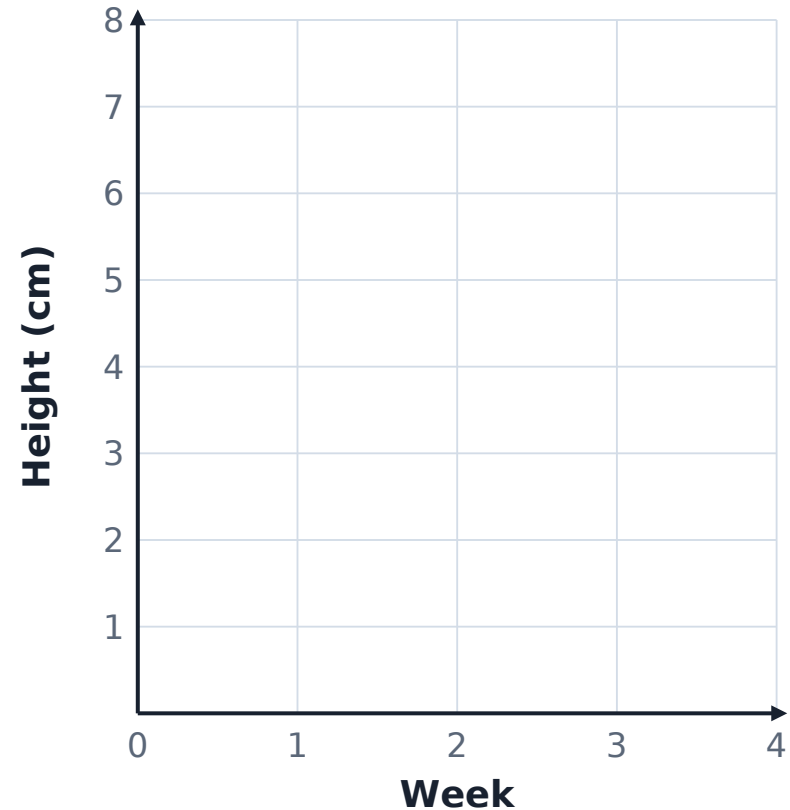
B. False

FORWARD 2

Graph data as points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A plant grows 2 cm each week. Graph the points (week, height): (0, 0), (1, 2), (2, 4), and (3, 6).

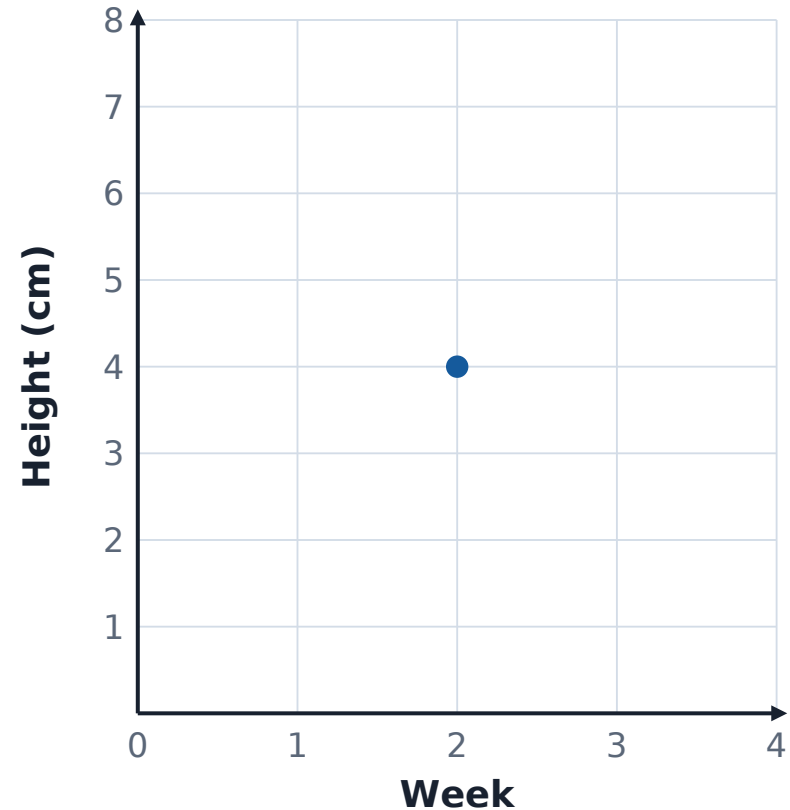


True or false?

On the graph, the point (2, 4) shows a height of 4 cm in week 2.

A. True

B. False



A store sells 5 hats each day.

Which point (days, hats) shows the total sold after 3 days?

A. (3, 5)

B. (15, 3)

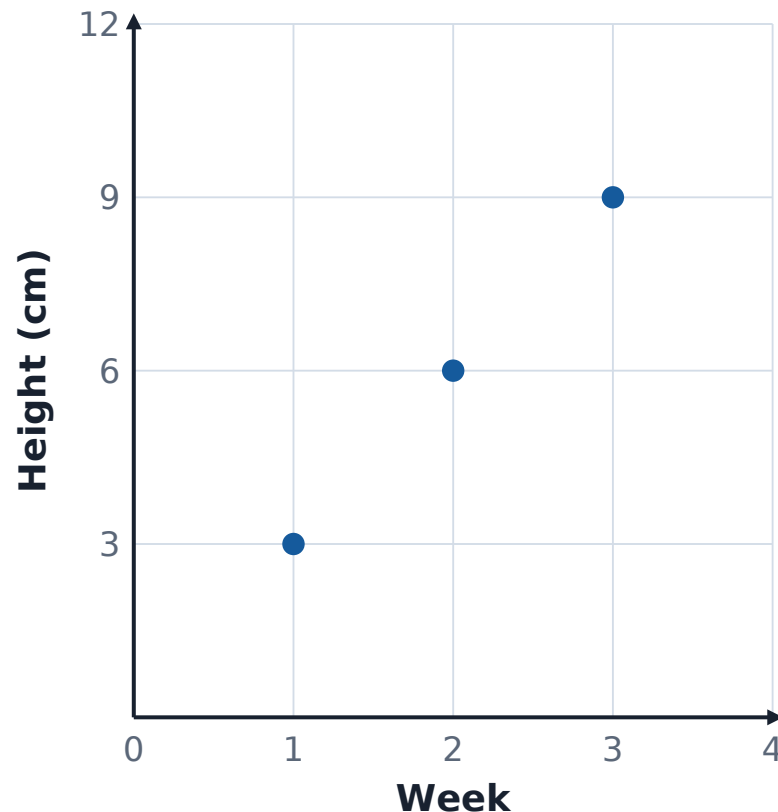
C. (3, 15)

D. (5, 3)

Use the graph.

What is the height in week 3?

Height: _____



True or false?

The point $(4, 0)$ is on the y -axis.

A. True

B. False

Solve "how many more" and "how many less" problems using scaled bar graphs and picture graphs

MAIN	Grade 3 main question	3.MD.B.3
BACKWARD 1	Solve compare problems with a single-unit bar graph	2.MD.D.10
BACKWARD 2	Solve compare word problems within 100	2.OA.A.1
FORWARD 1	Solve problems with data in a line plot of fractional measurements	4.MD.B.4
FORWARD 2	Find totals, differences, and equal shares of line plot data	5.MD.B.2

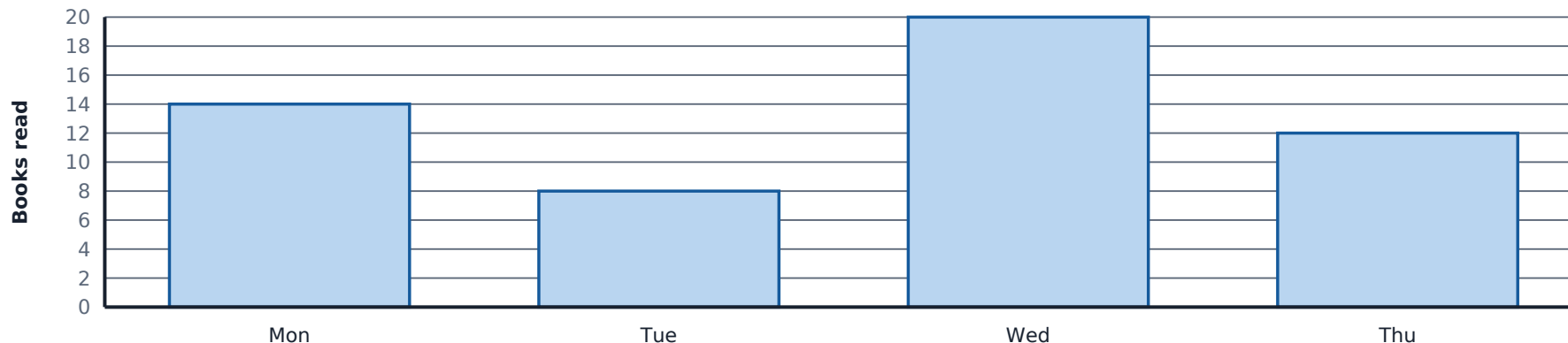
SET 45 • 3.MD.B.3

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.3

Use the bar graph.

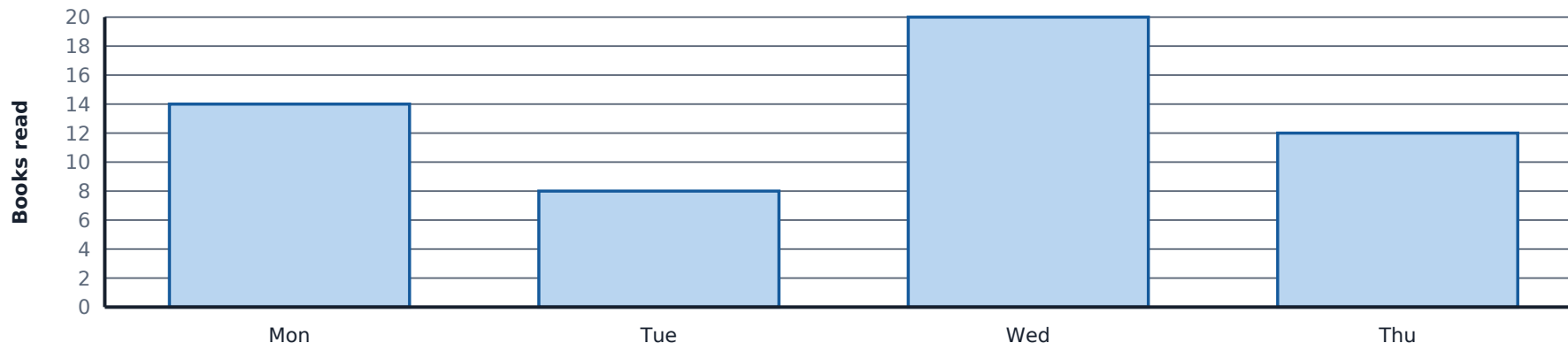
How many more books were read on Wednesday than on Tuesday?



More: _____

Use the bar graph.

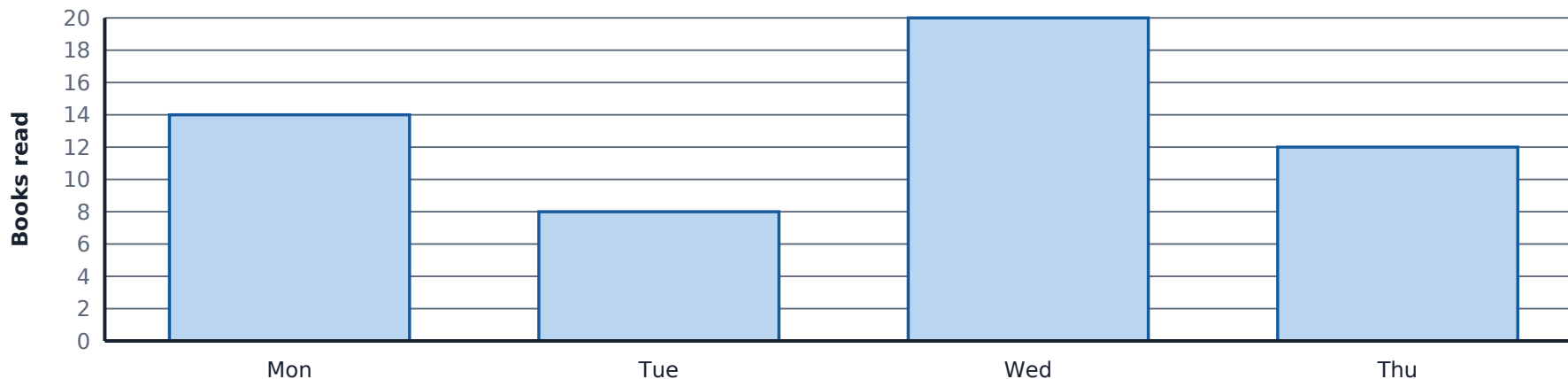
How many books were read on Monday and Thursday together?



Books: _____

Use the bar graph.

How many fewer books were read on Thursday than on Wednesday?



A. 32

B. 12

C. 8

D. 6

True or false?

Use the picture graph. Ana picked 6 more apples than Ben.

Apples picked	
Ana	● ● ● ● ●
Ben	● ● ●
Cy	● ● ● ●

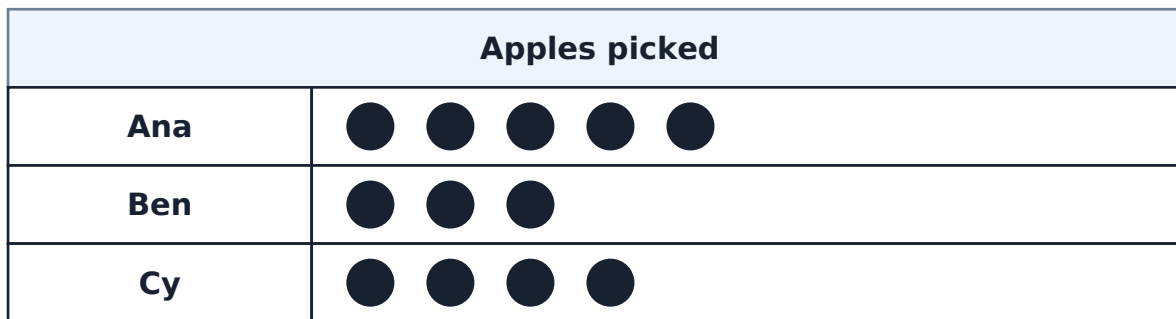
Key: ● = 3 apples

A. True

B. False

Use the picture graph.

How many apples did the three children pick in all?



Key: ● = 3 apples

Apples: _____

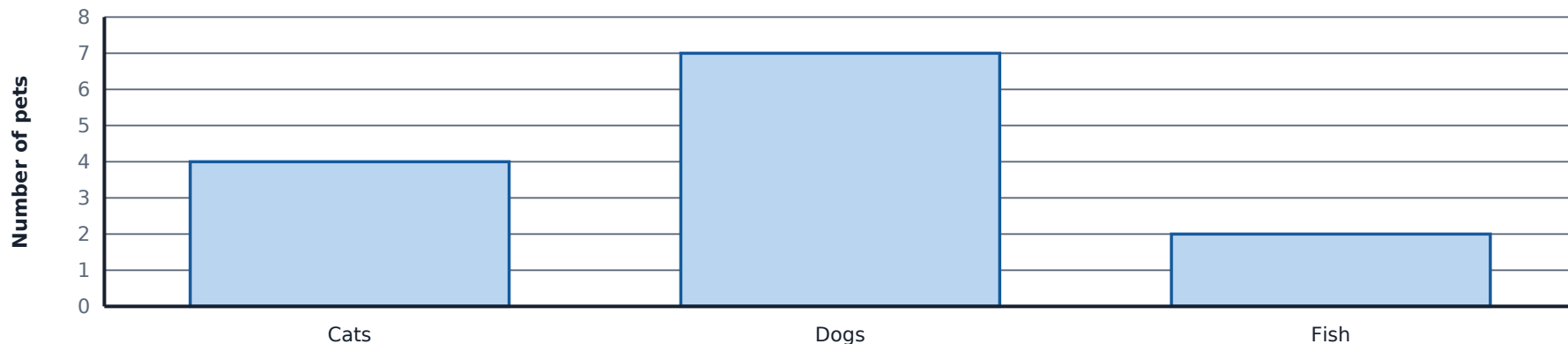
SET 45 • 3.MD.B.3

BACKWARD 1

Solve compare problems with a single-unit bar graph • Grade 2 • Measurement & Data • 2.MD.D.10

Use the bar graph.

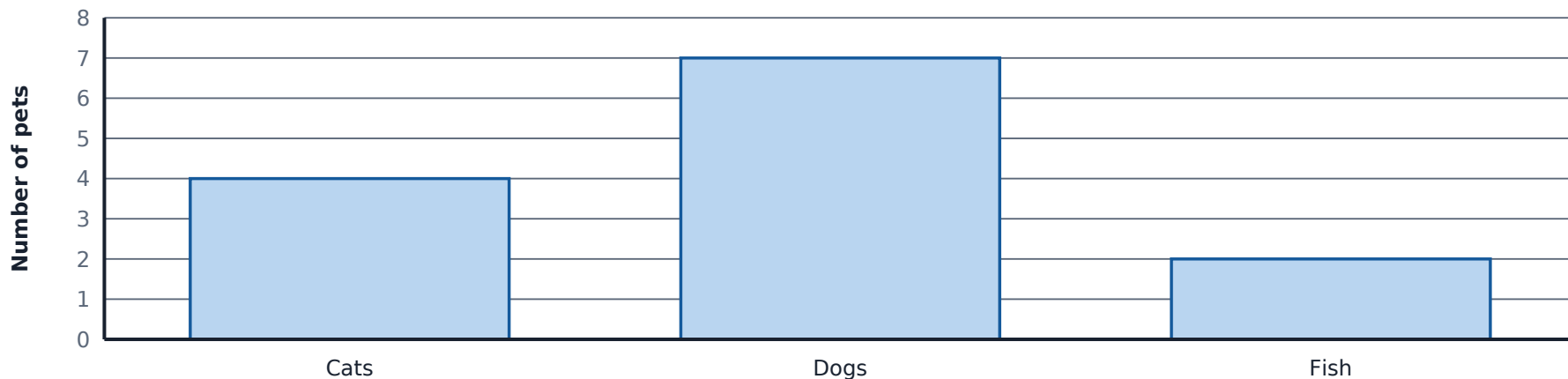
How many more dogs are there than cats?



More: _____

True or false?

Use the bar graph. There are 13 pets in all.

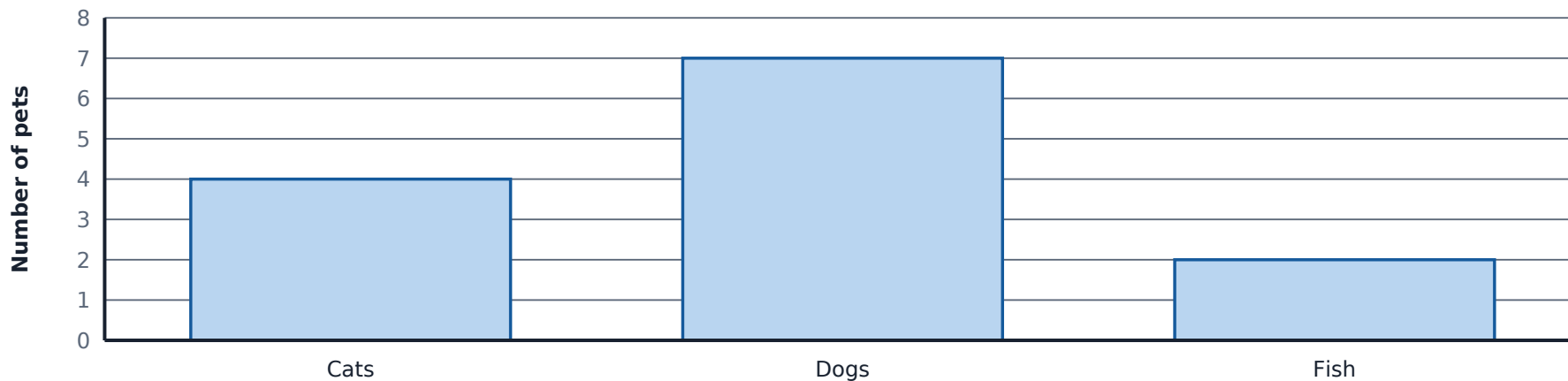


A. True

B. False

Use the bar graph.

How many fewer fish are there than dogs?



A. 9

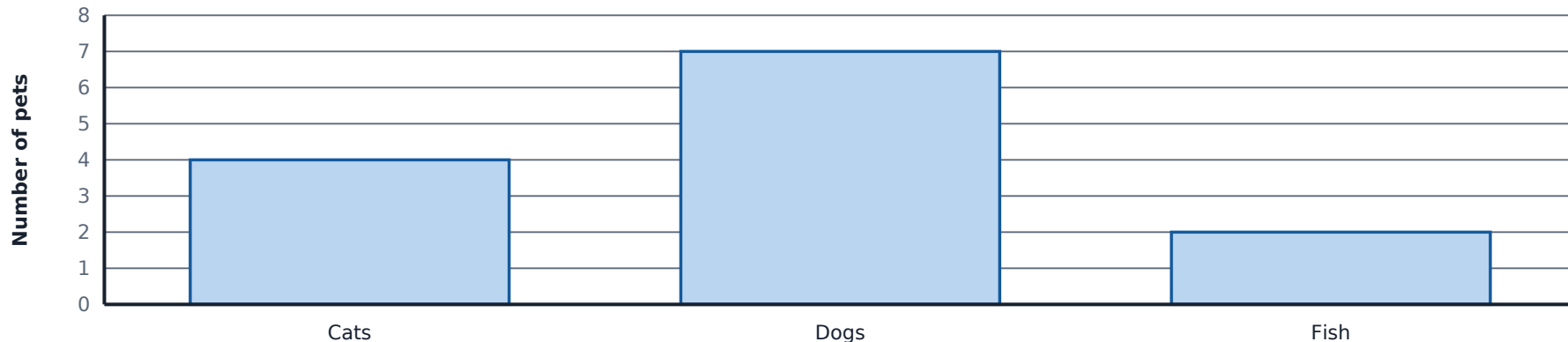
B. 2

C. 7

D. 5

Use the bar graph.

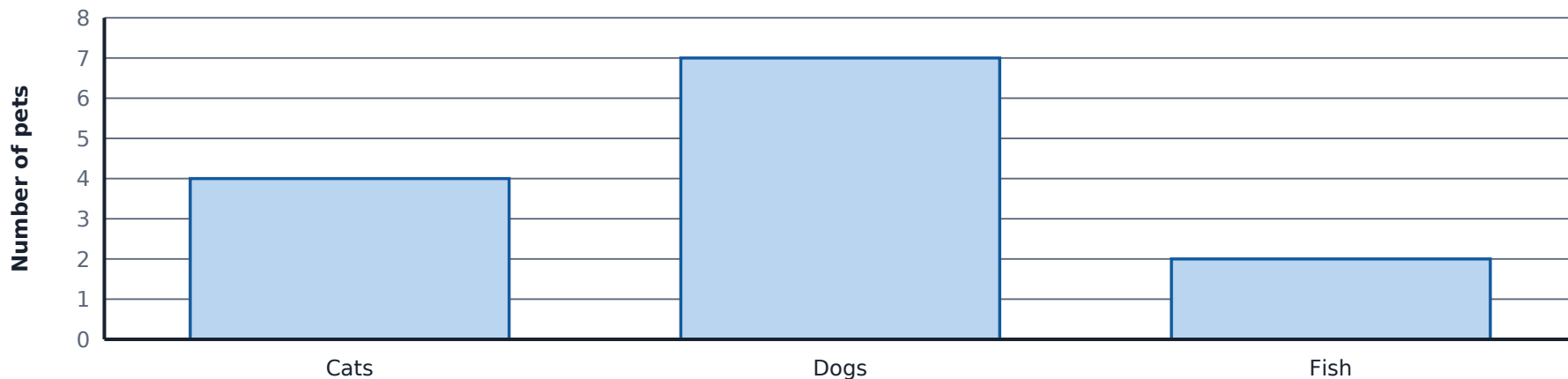
How many cats and fish are there together?



Pets: _____

True or false?

Use the bar graph. There are 3 more fish than cats.



A. True

B. False

SET 45 • 3.MD.B.3

BACKWARD 2

Solve compare word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Ty has 45 cards. Al has 28 cards.

How many more cards does Ty have?

More: _____

True or false?

A red ribbon is 50 cm long and a blue ribbon is 35 cm long. The red ribbon is 15 cm longer.

A. True

B. False

A school has 62 boys and 48 girls.

How many fewer girls than boys are there?

- A.** 24
- B.** 16
- C.** 14
- D.** 110

Kim read 33 pages. Jo read 19 more pages than Kim.
How many pages did Jo read?

Pages: _____

True or false?

Ann has 70 stamps. Bo has 25 fewer stamps than Ann. Bo has 95 stamps.

- A.** True
- B.** False

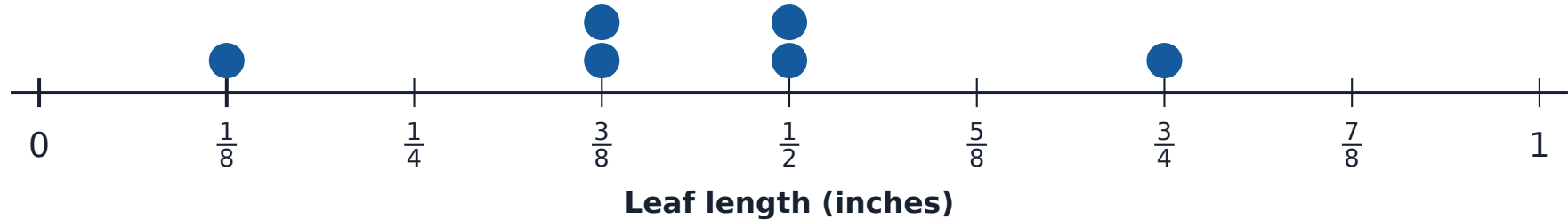
SET 45 • 3.MD.B.3

FORWARD 1

Solve problems with data in a line plot of fractional measurements • Grade 4 •
Measurement & Data • 4.MD.B.4

The line plot shows leaf lengths.

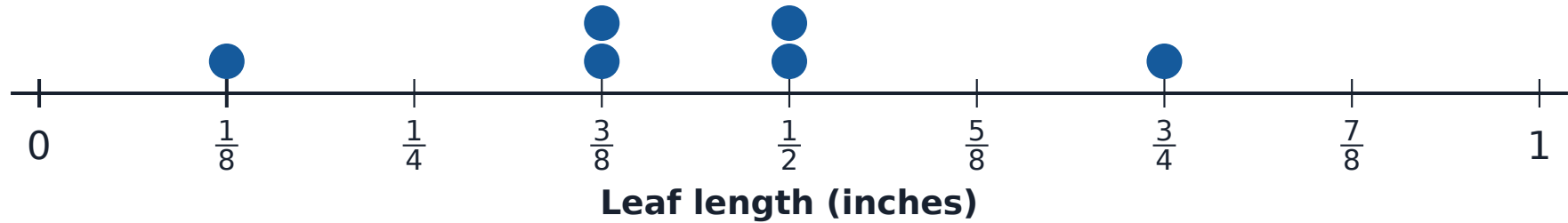
What is the difference between the longest and the shortest leaf?



Difference: _____

True or false?

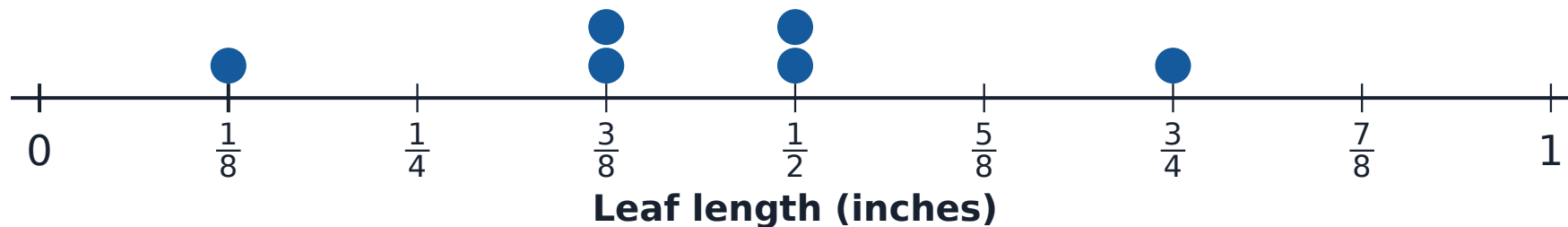
The two $\frac{3}{8}$ -inch leaves have a total length of $\frac{6}{8}$ inch.



A. True

B. False

How many leaves are longer than $\frac{3}{8}$ inch?



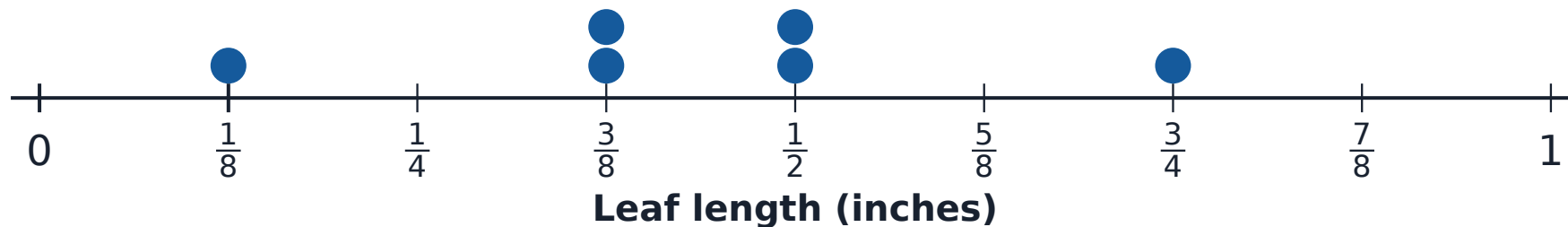
A. 4

B. 2

C. 1

D. 3

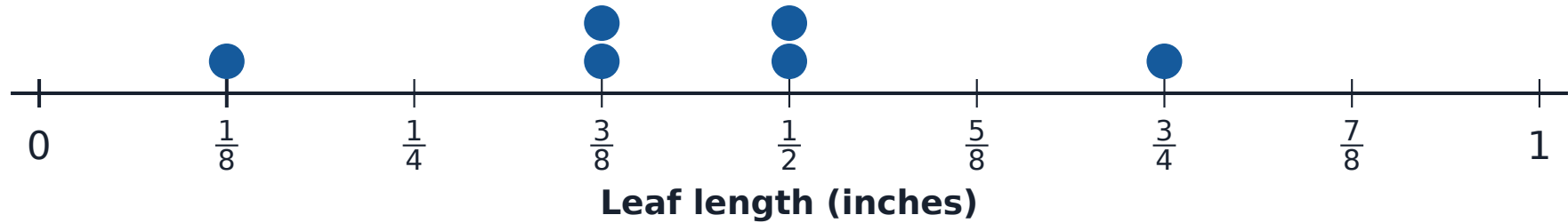
What is the total length of the two $\frac{1}{2}$ -inch leaves?



Total: _____

True or false?

The shortest leaf is $\frac{1}{4}$ inch long.



A. True

B. False

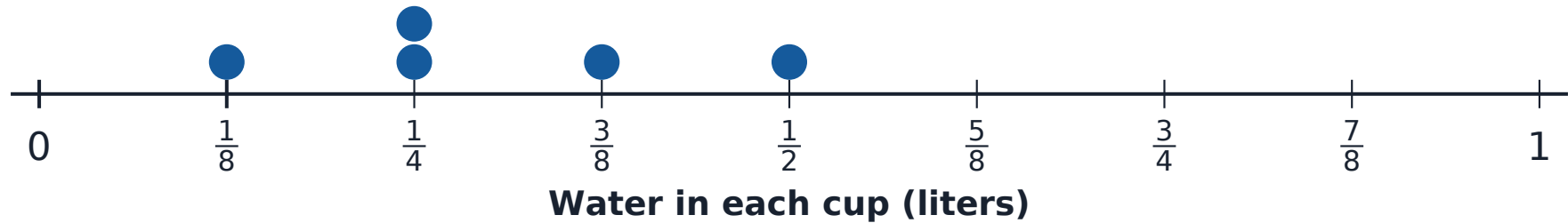
SET 45 • 3.MD.B.3

FORWARD 2

Find totals, differences, and equal shares of line plot data • Grade 5 •
Measurement & Data • 5.MD.B.2

The line plot shows the water in 5 cups.

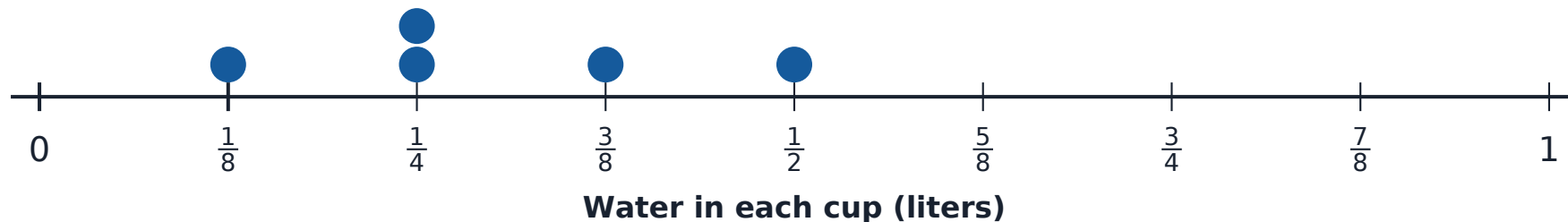
What is the total amount of water?



Total: _____

True or false?

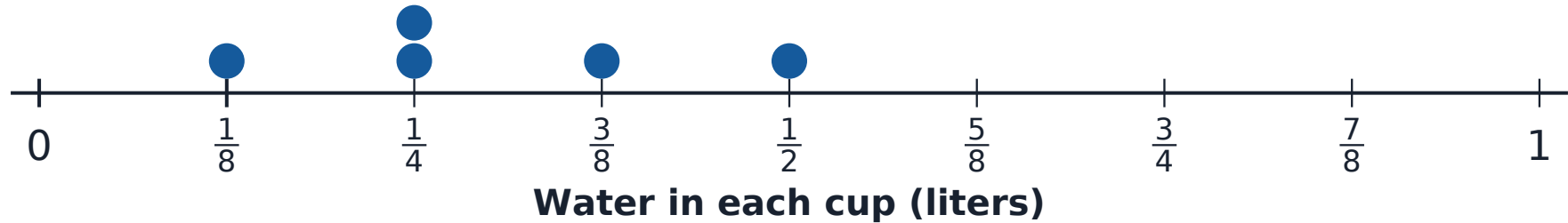
If all the water were poured into one jug, the jug would hold $1\frac{1}{2}$ liters.



A. True

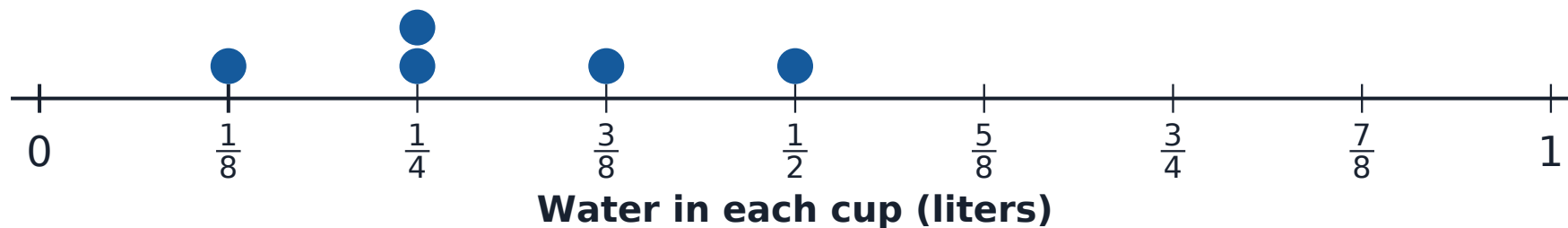
B. False

If the water were shared equally among the 5 cups, how much would each cup hold?



- A.** $\frac{3}{10}$ liter **B.** $1\frac{1}{2}$ liters **C.** $\frac{1}{4}$ liter **D.** $\frac{1}{5}$ liter

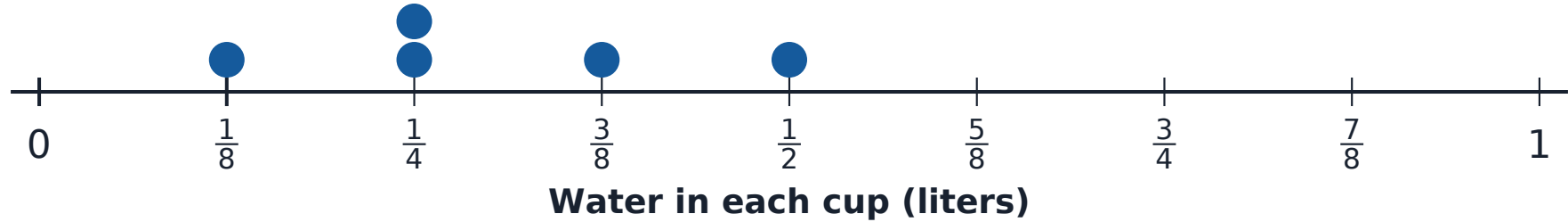
How much more water is in the fullest cup than in the cup with the least water?



More: _____

True or false?

The two cups with $\frac{1}{4}$ liter hold $\frac{1}{8}$ liter in all.



A. True

B. False

Measure lengths to the nearest half inch and quarter inch

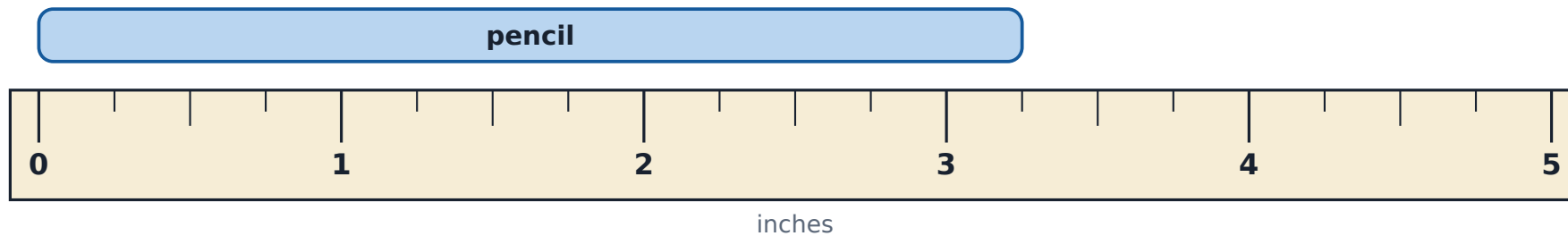
MAIN	Grade 3 main question	3.MD.B.4
BACKWARD 1	Measure lengths in whole inches and centimeters	2.MD.A.1
BACKWARD 2	Measure length by laying same-size units end to end	1.MD.A.2
FORWARD 1	Add mixed-number lengths in word problems	4.NF.B.3.d
FORWARD 2	Convert inches, feet, and yards with fractions	5.MD.A.1

SET 46 • 3.MD.B.4

MAIN

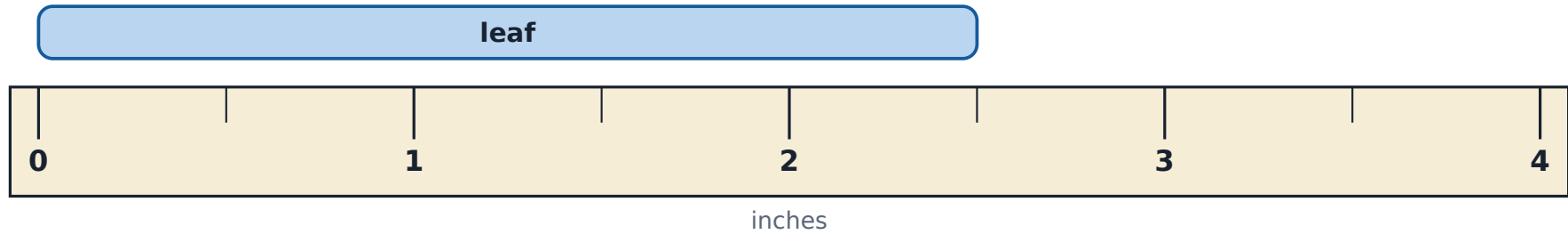
Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.4

How long is the pencil, to the nearest quarter inch?



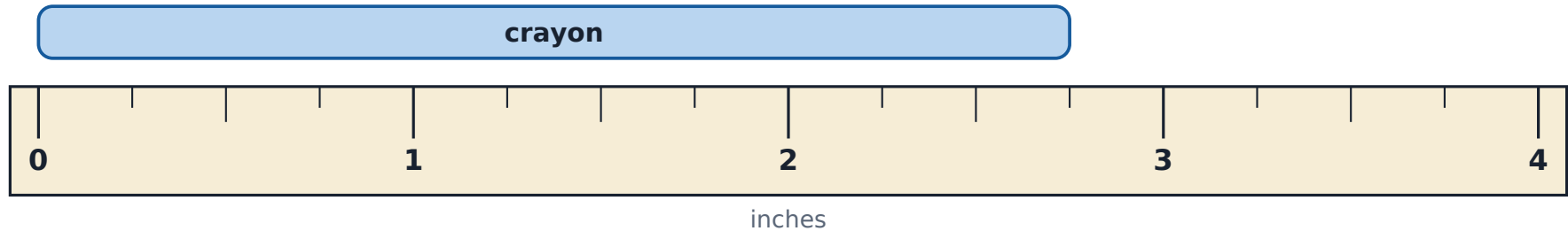
Length: _____

How long is the leaf, to the nearest half inch?



Length: _____

How long is the crayon, to the nearest quarter inch?



A. $3\frac{3}{4}$ inches

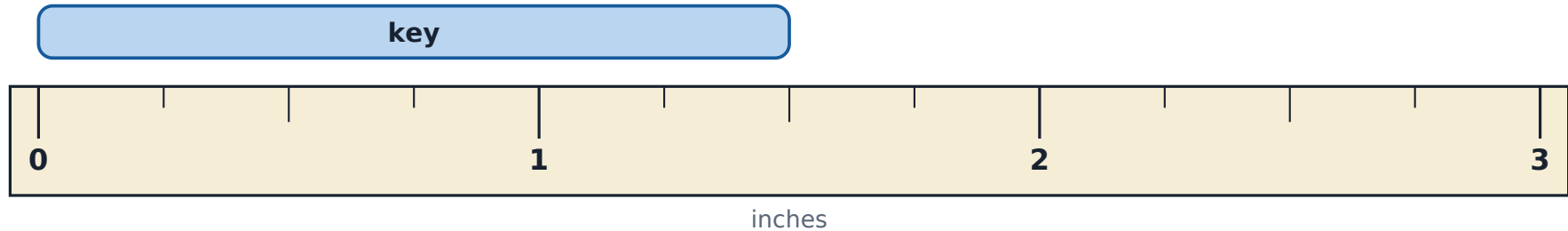
B. 3 inches

C. $2\frac{1}{2}$ inches

D. $2\frac{3}{4}$ inches

True or false?

The key is $1\frac{1}{2}$ inches long.

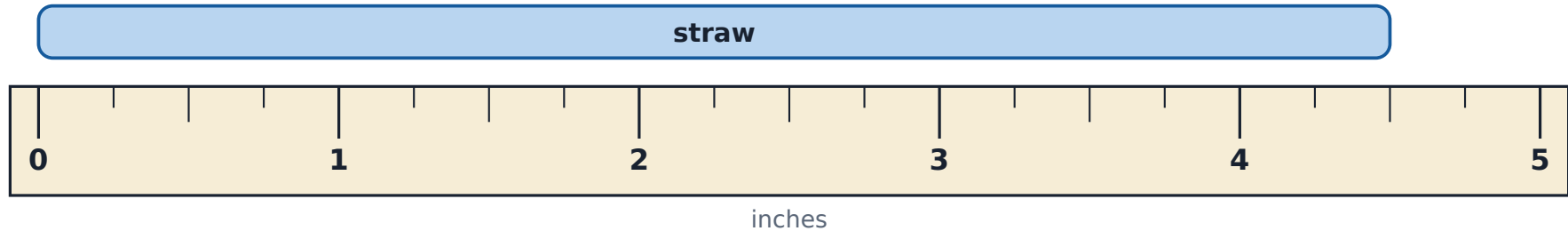


A. True

B. False

True or false?

The straw is $4\frac{1}{4}$ inches long.



A. True

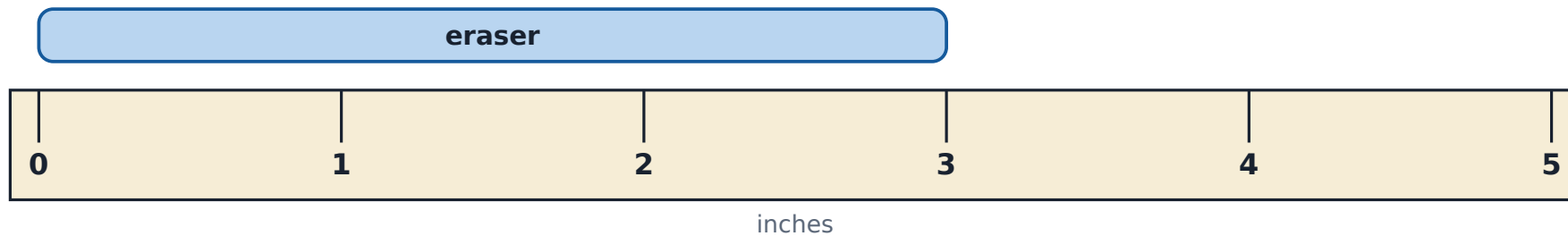
B. False

SET 46 • 3.MD.B.4

BACKWARD 1

Measure lengths in whole inches and centimeters • Grade 2 • Measurement & Data • 2.MD.A.1

How long is the eraser?



Length: _____

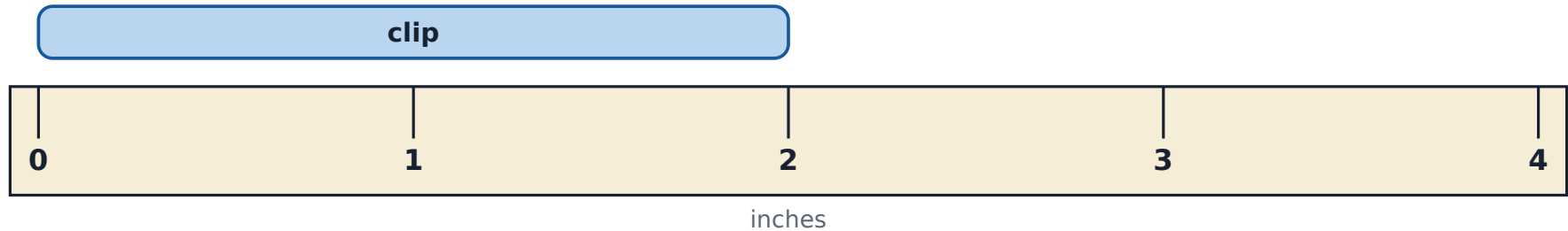
True or false?

A ruler can show inches or centimeters.

A. True

B. False

How long is the clip?



A. 2 inches

B. 1 inch

C. 4 inches

D. 3 inches

A string goes from 0 to 6 on a centimeter ruler.

How long is it?

Length: _____

True or false?

A nail that reaches from 0 to 5 on an inch ruler is 4 inches long.

- A.** True
- B.** False

SET 46 • 3.MD.B.4

BACKWARD 2

Measure length by laying same-size units end to end • Grade 1 • Measurement & Data • 1.MD.A.2

A pencil is as long as 5 paper clips laid end to end.
How many paper clips long is it?

Paper clips: _____

True or false?

When you measure with paper clips, there should be no gaps or overlaps.

- A. True
- B. False

A book is 8 cubes long. A box is 2 cubes longer.

How many cubes long is the box?

- A.** 6 cubes
- B.** 16 cubes
- C.** 8 cubes
- D.** 10 cubes

A rug is 9 shoes long. A mat is 3 shoes shorter.
How many shoes long is the mat?

Shoes: _____

True or false?

If the paper clips overlap when you measure, the count is still correct.

- A. True
- B. False

SET 46 • 3.MD.B.4

FORWARD 1

Add mixed-number lengths in word problems • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

A pencil is $3\frac{1}{4}$ inches long and a crayon is $2\frac{3}{4}$ inches long.

How long are they laid end to end?

Length: _____

True or false?

Ribbons of $2\frac{1}{2}$ inches and $1\frac{3}{4}$ inches laid end to end are $4\frac{1}{4}$ inches long.

A. True

B. False

A snail crawls $1\frac{1}{4}$ inches and then $2\frac{2}{4}$ inches.

How far does it crawl in all?

A. $3\frac{3}{4}$ inches

B. 4 inches

C. $3\frac{1}{4}$ inches

D. $1\frac{1}{4}$ inches

Three sticks are $\frac{1}{2}$ inch, $\frac{3}{4}$ inch, and $\frac{1}{4}$ inch long.

What is their total length?

Length: _____

True or false?

Three beads that are each $\frac{3}{4}$ inch long, in a row, are 2 inches long.

- A. True
- B. False

SET 46 • 3.MD.B.4

FORWARD 2

Convert inches, feet, and yards with fractions • Grade 5 • Measurement & Data
• 5.MD.A.1

How many feet is 30 inches?

Feet: _____

True or false?

18 inches = $1\frac{1}{2}$ feet

A. True

B. False

How many inches are in $2\frac{1}{4}$ feet?

A. 27

B. 24

C. 25

D. 29

A board is $1\frac{3}{4}$ yards long.

How many feet is that?

Feet: _____

True or false?

9 inches = $\frac{3}{4}$ yard

A. True

B. False

Make a line plot of measurement data in whole numbers, halves, or quarters

MAIN	Grade 3 main question	3.MD.B.4
BACKWARD 1	Make a line plot of whole-number lengths	2.MD.D.9
BACKWARD 2	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Make a line plot of measurements in halves, fourths, and eighths	4.MD.B.4
FORWARD 2	Use operations on fractions to solve problems with line plot data	5.MD.B.2

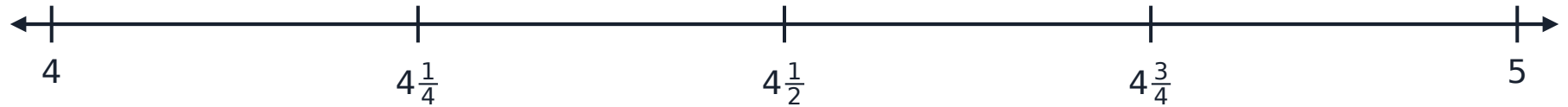
SET 47 • 3.MD.B.4

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.B.4

Make a line plot of the data.

Pencil lengths (inches): 4, $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $4\frac{1}{4}$, $4\frac{1}{2}$



Shell lengths (inches): 2 , $2\frac{1}{2}$, $2\frac{1}{2}$, 3 , $2\frac{1}{2}$

In a line plot of the data, how many X's go above $2\frac{1}{2}$?

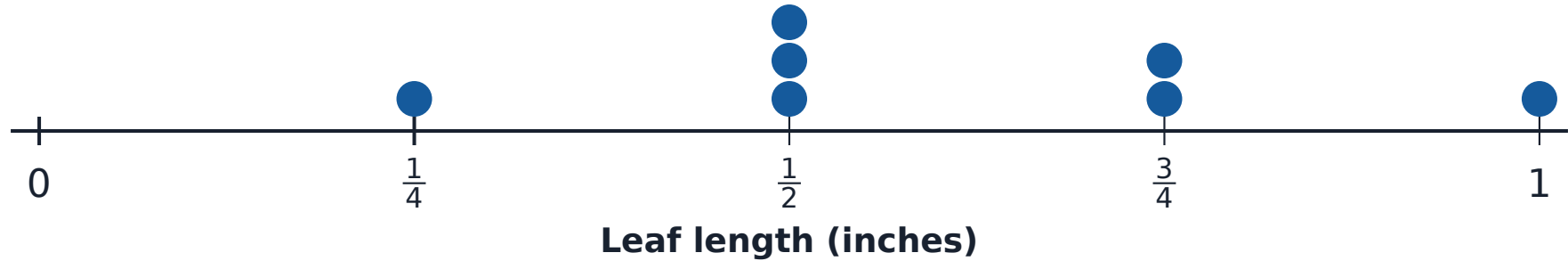
X's: _____

Which number line is best for a line plot of $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, and 2 inches?

- A.** From 1 to 2, marked in halves
- B.** From 0 to 10, marked in ones
- C.** From 1 to 2, marked in quarters
- D.** From 0 to 1, marked in halves

True or false?

The line plot shows that 2 leaves are $\frac{3}{4}$ inch long.

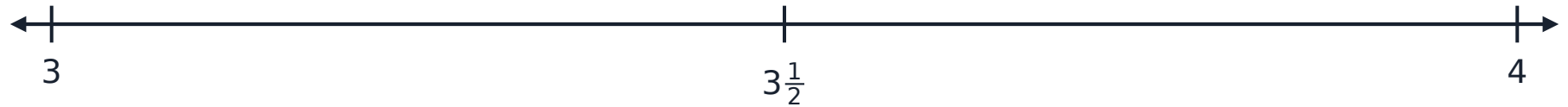


A. True

B. False

Make a line plot of the data.

Ribbon lengths (inches): 3, $3\frac{1}{2}$, 4, 3, $3\frac{1}{2}$, 3



SET 47 • 3.MD.B.4

BACKWARD 1

Make a line plot of whole-number lengths • Grade 2 • Measurement & Data •
2.MD.D.9

Make a line plot of the data.

Lengths (inches): 5, 6, 6, 8, 6



True or false?

A line plot shows each measurement as an X above a number line.

- A.** True
- B.** False

Which number line fits a line plot of 3, 4, 4, and 7?

- A.** A number line from 10 to 20
- B.** A number line from 5 to 9
- C.** A number line from 0 to 10
- D.** A number line from 0 to 4

Lengths (cm): 9, 10, 10, 12

In a line plot of the data, how many X's go above 10?

X's: _____

True or false?

A line plot of 2, 2, 2, and 3 has 2 X's above 2.

- A. True
- B. False

SET 47 • 3.MD.B.4

BACKWARD 2

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

True or false?

On a number line, numbers get greater to the right.

- A.** True
- B.** False

Which number is between 4 and 6 on a number line?

- A.** 7
- B.** 3
- C.** 10
- D.** 5

What number is at point B?



B = _____

True or false?

On a number line, 9 is to the left of 6.

A. True

B. False

SET 47 • 3.MD.B.4

FORWARD 1

Make a line plot of measurements in halves, fourths, and eighths • Grade 4 •
Measurement & Data • 4.MD.B.4

Make a line plot of the data.

Screw lengths (inches): $\frac{3}{8}$, $\frac{1}{2}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{4}$, $\frac{3}{8}$



True or false?

$\frac{1}{2}$ inch and $\frac{4}{8}$ inch go above the same mark on a line plot marked in eighths.

- A.** True
- B.** False

Lengths (inches): $\frac{1}{4}$, $\frac{2}{8}$, $\frac{3}{8}$, $\frac{1}{4}$

In a line plot marked in eighths, how many X's go above $\frac{2}{8}$?

A. 4

B. 1

C. 3

D. 2

Make a line plot of the data.

Bead sizes (inches): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$



True or false?

A line plot marked in eighths has a mark for $\frac{1}{3}$ inch.

- A.** True
- B.** False

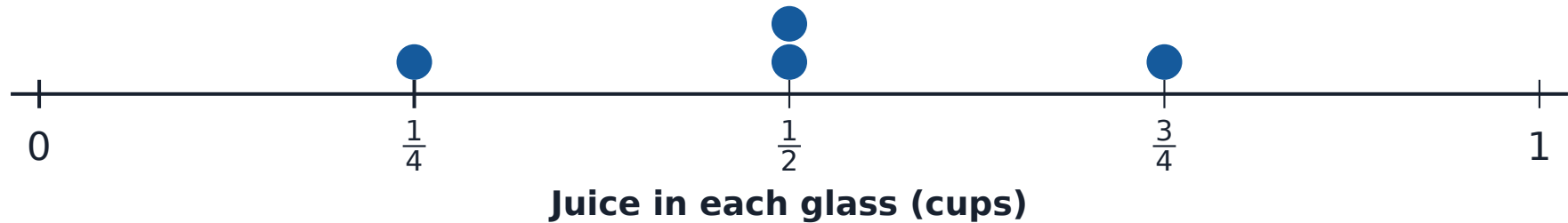
SET 47 • 3.MD.B.4

FORWARD 2

Use operations on fractions to solve problems with line plot data • Grade 5 •
Measurement & Data • 5.MD.B.2

The line plot shows the juice in 4 glasses.

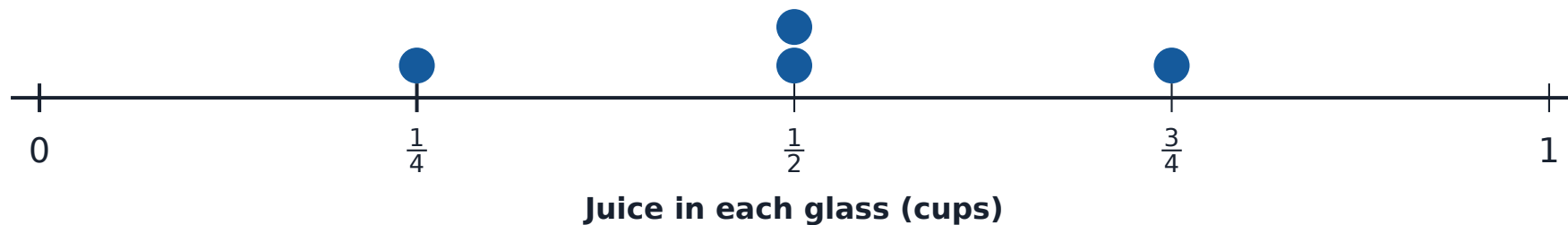
What is the total amount of juice?



Total: _____

True or false?

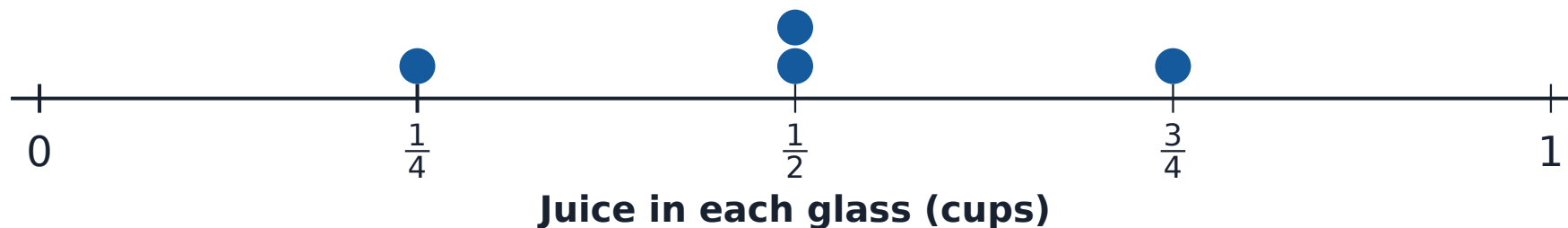
If the juice were shared equally among the 4 glasses, each glass would hold $\frac{1}{2}$ cup.



A. True

B. False

How much more juice is in the fullest glass than in the emptiest glass?



A. $\frac{3}{4}$ cup

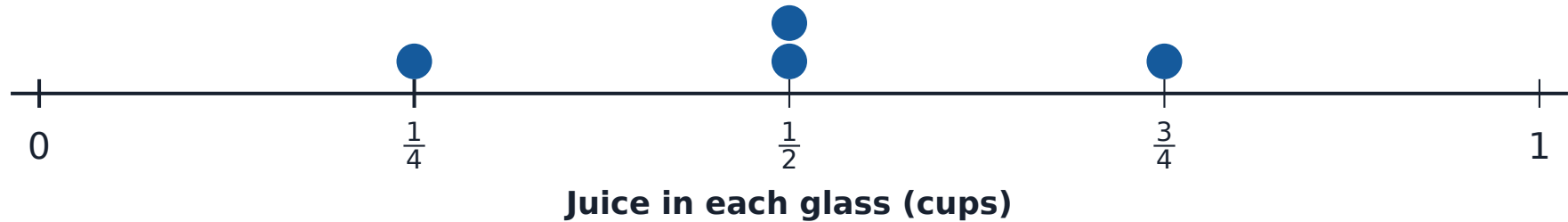
B. 1 cup

C. $\frac{1}{2}$ cup

D. $\frac{1}{4}$ cup

The juice in the two $\frac{1}{2}$ -cup glasses is poured together.

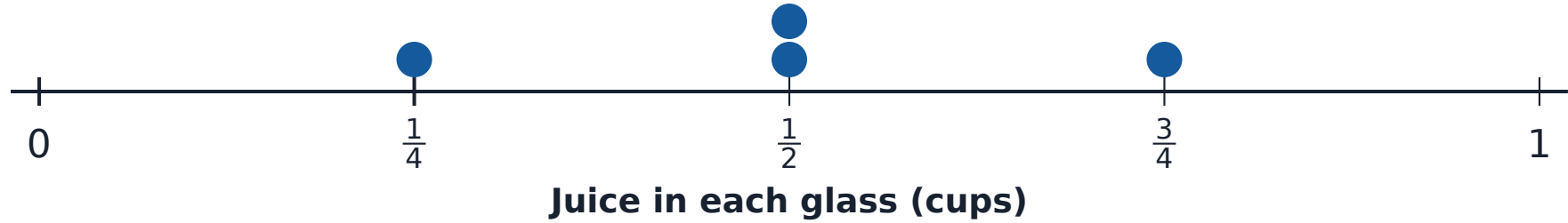
How much juice is that?



Juice: _____

True or false?

The glasses with less than $\frac{1}{2}$ cup hold $\frac{3}{4}$ cup in all.



A. True

B. False

Understand a unit square and that it is used to measure area

MAIN	Grade 3 main question	3.MD.C.5.a
BACKWARD 1	Count the same-size squares in a rectangle	2.G.A.2
BACKWARD 2	Defining attributes of a square	1.G.A.1
FORWARD 1	Find the area of a rectangle with the area formula	4.MD.A.3
FORWARD 2	Understand a unit cube and that it is used to measure volume —...	5.MD.C.3.a

SET 48 • 3.MD.C.5.a

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides 1 unit long is a unit square, and its area is 1 square unit.

A. True

B. False

Which shape is a unit square?

- A.** A rectangle 1 cm by 2 cm
- B.** A square with sides 1 cm long
- C.** A circle 1 cm across
- D.** A square with sides 2 cm long

What is the area of a square with sides 1 inch long?

Area: _____

Why can unit squares be used to measure area?

Explanation: _____

True or false?

A square with sides 1 meter long has an area of 1 square centimeter.

- A. True
- B. False

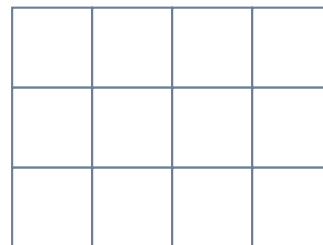
SET 48 • 3.MD.C.5.a

BACKWARD 1

Count the same-size squares in a rectangle • Grade 2 • Geometry • 2.G.A.2

How many same-size squares
are in the rectangle?

Squares: _____



True or false?

A rectangle split into 2 rows of 3 same-size squares has 6 squares.

- A.** True
- B.** False

A rectangle has 4 rows of 2 same-size squares.

How many squares are there?

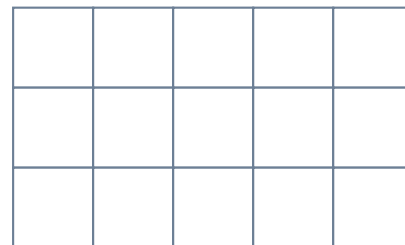
A. 2

B. 6

C. 8

D. 4

How many same-size squares
are in each column of the
rectangle?



Squares in each column: ____

True or false?

A rectangle split into 3 rows of 3 same-size squares has 6 squares.

- A. True
- B. False

SET 48 • 3.MD.C.5.a

BACKWARD 2

Defining attributes of a square • Grade 1 • Geometry • 1.G.A.1

True or false?

Every square has 4 sides of the same length.

A. True

B. False

Which is a defining attribute of a square?

- A.** It is big.
- B.** 4 equal sides and 4 square corners
- C.** It is turned on a corner.
- D.** It is blue.

How many sides does a square have?

Sides: _____

Is a very small square still a square?

Answer: _____

True or false?

A shape with 3 sides can be a square.

A. True

B. False

SET 48 • 3.MD.C.5.a

FORWARD 1

Find the area of a rectangle with the area formula • Grade 4 • Measurement & Data • 4.MD.A.3

Find the area of a rectangle that is 12 m by 7 m.

Area: _____

True or false?

A 9-by-11 rectangle has an area of 99 square units.

A. True

B. False

A rug is 15 feet by 6 feet.

What is its area?

- A.** 21 square feet
- B.** 42 square feet
- C.** 90 square feet
- D.** 96 square feet

A square has sides that are 13 cm long.

What is its area?

Area: _____

True or false?

A rectangle that is 20 in. by 5 in. has an area of 50 square inches.

- A.** True
- B.** False

FORWARD 2

Understand a unit cube and that it is used to measure volume • Grade 5 •
Measurement & Data • 5.MD.C.3.a

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

True or false?

A cube with edges 1 unit long is a unit cube, and its volume is 1 cubic unit.

- A.** True
- B.** False

Which can be used to measure volume?

- A.** A clock
- B.** A ruler
- C.** A unit square
- D.** A unit cube

What is the volume of a cube with edges 1 cm long?

Volume: _____

How is a unit cube like a unit square?

Answer: _____

True or false?

A unit cube has a volume of 6 cubic units because it has 6 faces.

A. True

B. False

A figure covered by n unit squares without gaps or overlaps has an area of n square units

MAIN	Grade 3 main question	3.MD.C.5.b
BACKWARD 1	Measure length with same-size units and no gaps or overlaps	1.MD.A.2
BACKWARD 2	Count the same-size squares that fill a rectangle	2.G.A.2
FORWARD 1	Find an unknown side length of a rectangle from its area	4.MD.A.3
FORWARD 2	A solid packed with n unit cubes has a volume of n cubic units —...	5.MD.C.3.b

SET 49 • 3.MD.C.5.b

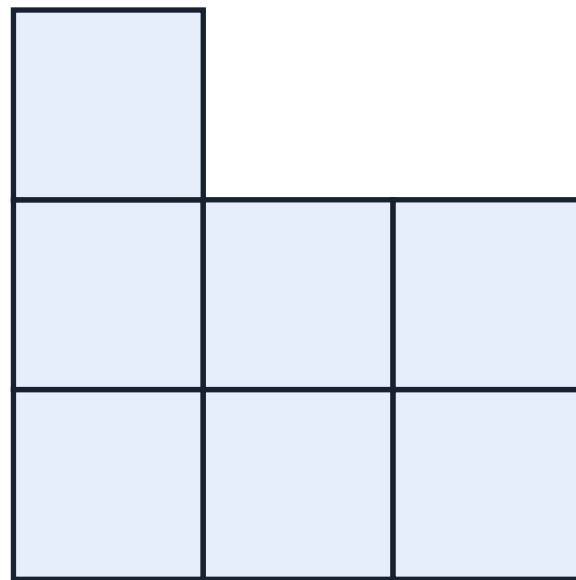
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.5.b

Each small square is 1 square unit.

What is the area of the shape?

Area: _____



True or false?

A shape covered by 15 unit squares with no gaps or overlaps has an area of 15 square units.

- A. True
- B. False

A rectangle is covered by 12 unit squares with no gaps or overlaps.

What is its area?

- A.** 12 square units
- B.** 6 square units
- C.** 24 square units
- D.** 12 units

Ben covers a shape with unit squares, but he leaves gaps between them. He counts 10 squares.

Is the area exactly 10 square units? Explain.

Answer: _____

Explanation: _____

True or false?

If unit squares overlap, counting them gives the exact area.

- A. True
- B. False

SET 49 • 3.MD.C.5.b

BACKWARD 1

Measure length with same-size units and no gaps or overlaps • Grade 1 •
Measurement & Data • 1.MD.A.2

A crayon is as long as 6 cubes laid end to end with no gaps.

How many cubes long is it?

Cubes: _____

True or false?

Length units laid end to end must have no gaps or overlaps.

A. True

B. False

Which shows the correct way to measure with paper clips?

- A.** Clips in a pile
- B.** Clips that overlap
- C.** Clips with spaces between them
- D.** Clips end to end with no gaps

A desk is 9 hands long. A table is 4 hands longer.
How many hands long is the table?

Hands: _____

True or false?

If the cubes overlap when you measure, the count is still correct.

A. True

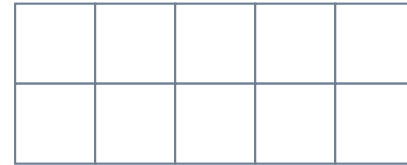
B. False

SET 49 • 3.MD.C.5.b

BACKWARD 2

Count the same-size squares that fill a rectangle • Grade 2 • Geometry •
2.G.A.2

How many same-size squares
fill the rectangle?



Squares: _____

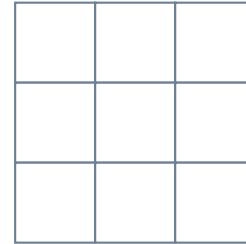
True or false?

A rectangle split into 3 rows of 4 same-size squares has 12 squares.

- A.** True
- B.** False

How many same-size squares
fill the rectangle?

- A.** 6
- B.** 9
- C.** 12
- D.** 3



A rectangle has 5 rows of 2 same-size squares.
How many squares fill it?

Squares: _____

True or false?

A rectangle with 2 rows of 4 same-size squares has 6 squares.

A. True

B. False

SET 49 • 3.MD.C.5.b

FORWARD 1

Find an unknown side length of a rectangle from its area • Grade 4 •
Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a length of 8 feet.

What is its width?

Width: _____

True or false?

A rectangle with an area of 72 square cm and a width of 6 cm has a length of 12 cm.

A. True

B. False

A garden has an area of 90 square meters. One side is 9 meters long.

How long is the other side?

A. 81 m

B. 18 m

C. 10 m

D. 99 m

A square has an area of 64 square inches.
How long is each side?

Side: _____

True or false?

A rectangle with an area of 56 square units and a length of 7 units has a width of 9 units.

- A. True
- B. False

FORWARD 2

A solid packed with n unit cubes has a volume of n cubic units • Grade 5 •
Measurement & Data • 5.MD.C.3.b

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

A solid built from 18 unit cubes has a volume of 18 cubic units.

- A. True
- B. False

A prism has 4 layers with 6 unit cubes in each layer.

What is its volume?

- A.** 46 cubic units
- B.** 10 cubic units
- C.** 24 square units
- D.** 24 cubic units

How many unit cubes fill a box with a volume of 36 cubic units?

Unit cubes: _____

True or false?

If the cubes in a box have gaps between them, counting them gives the exact volume.

- A. True
- B. False

Measure areas by counting unit squares in square centimeters, meters, inches, feet, and other units

MAIN	Grade 3 main question	3.MD.C.6
BACKWARD 1	Count objects to tell how many	K.CC.B.5
BACKWARD 2	Estimate lengths in inches, feet, centimeters, and meters	2.MD.A.3
FORWARD 1	Find areas of rectangles in real-world problems with the area for...	4.MD.A.3
FORWARD 2	Measure volume by counting unit cubes in cubic cm, cubic in., and...	5.MD.C.4

SET 50 • 3.MD.C.6

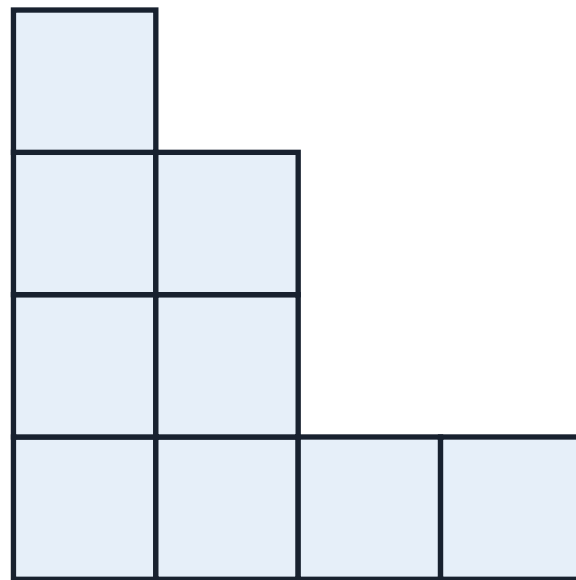
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.6

Each square is 1 square centimeter.

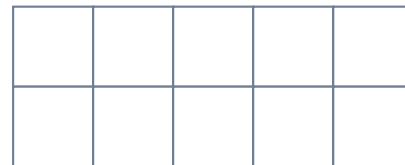
What is the area of the shape?

Area: _____



Each square is 1 square foot.

What is the area of the rug?

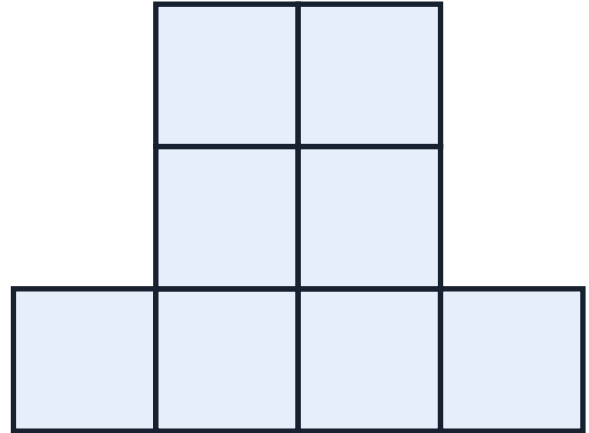


Area: _____

Each square is 1 square inch.

What is the area of the shape?

- A.** 6 square inches **B.** 8 inches
- C.** 12 square inches **D.** 8 square inches

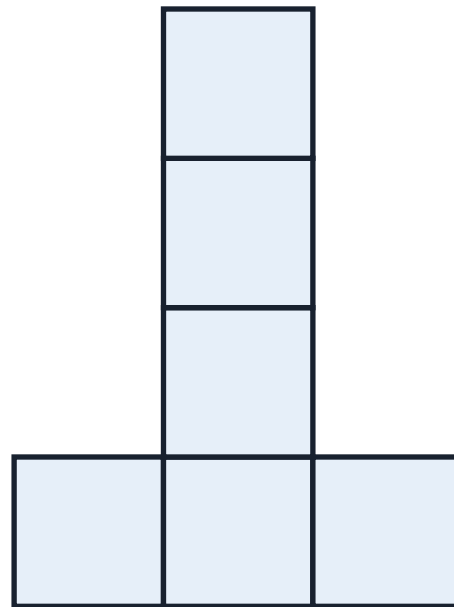


True or false?

Each square is 1 square meter. The shape has an area of 6 square meters.

A. True

B. False

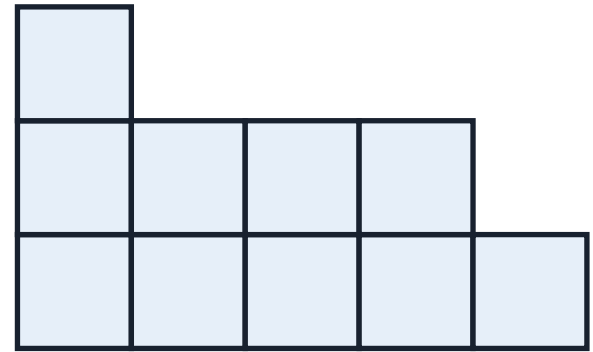


True or false?

Each square is 1 square unit. The shape has an area of 11 square units.

A. True

B. False

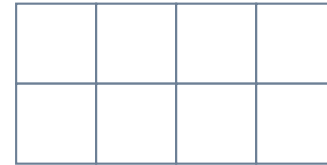


SET 50 • 3.MD.C.6

BACKWARD 1

Count objects to tell how many • Grade K • Counting & Cardinality • K.CC.B.5

How many squares are in the grid?



Squares: _____

True or false?

Counting each square once, a row of 6 squares has 6 squares.

- A.** True
- B.** False

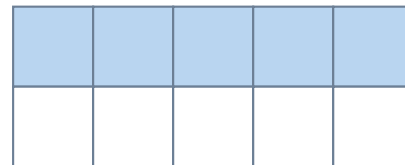
How many squares are shaded?

A. 4

B. 6

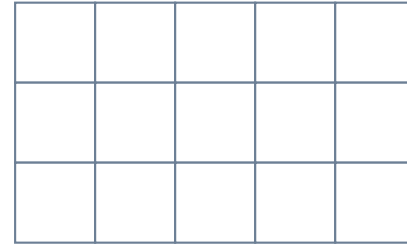
C. 10

D. 5



How many squares are in the grid?

Squares: _____



True or false?

When you count objects, you may count one object twice.

A. True

B. False

SET 50 • 3.MD.C.6

BACKWARD 2

Estimate lengths in inches, feet, centimeters, and meters • Grade 2 •
Measurement & Data • 2.MD.A.3

Would you measure the length of a classroom in meters or in centimeters?

Unit: _____

True or false?

A centimeter is shorter than a meter.

A. True

B. False

Which unit is best for the length of a bug?

- A.** Meters
- B.** Feet
- C.** Yards
- D.** Centimeters

Would you measure a book in inches or in feet?

Unit: _____

True or false?

A foot is shorter than an inch.

A. True

B. False

SET 50 • 3.MD.C.6

FORWARD 1

Find areas of rectangles in real-world problems with the area formula • Grade 4
• Measurement & Data • 4.MD.A.3

A classroom floor is 9 m by 8 m.

What is its area?

Area: _____

True or false?

A 14-ft by 10-ft deck has an area of 140 square feet.

A. True

B. False

A poster is 24 in. by 18 in.

What is its area?

- A.** 84 square inches
- B.** 422 square inches
- C.** 42 square inches
- D.** 432 square inches

A garden is 25 ft by 4 ft.

What is its area?

Area: _____

True or false?

A 16-cm by 5-cm card has an area of 42 square cm.

A. True

B. False

FORWARD 2

Measure volume by counting unit cubes in cubic cm, cubic in., and cubic ft •

Grade 5 • Measurement & Data • 5.MD.C.4

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box holds 2 layers of 6 cubes. Each cube is 1 cubic centimeter.

What is the volume of the box?

Volume: _____

True or false?

A stack of 3 layers of 4 one-inch cubes has a volume of 12 cubic inches.

- A. True
- B. False

A prism is made of 20 cubes that are each 1 cubic foot.
What is its volume?

- A.** 20 square feet
- B.** 20 cubic feet
- C.** 40 cubic feet
- D.** 20 feet

A figure is built from 15 unit cubes.

What is its volume?

Volume: _____

True or false?

A box that holds 2 layers of 9 one-centimeter cubes has a volume of 11 cubic centimeters.

- A. True
- B. False

Find the area of a rectangle by tiling, and show that it equals the product of the side lengths

MAIN	Grade 3 main question	3.MD.C.7.a
BACKWARD 1	Find the total of an array by adding equal addends	2.OA.C.4
BACKWARD 2	Split a rectangle into rows and columns of same-size squares	2.G.A.2
FORWARD 1	Write and apply the area formula $A = l \times w$	4.MD.A.3
FORWARD 2	Find the volume of a prism by packing it with unit cubes and by...	5.MD.C.5.a

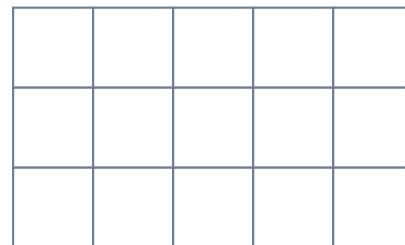
SET 51 • 3.MD.C.7.a

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.a

The rectangle is tiled with unit squares.

Count the squares. Then multiply the side lengths. Do you get the same area?



Count: _____

Multiply: _____

Same? _____

A rectangle is tiled with 4 rows of 6 unit squares.
Which equation gives its area?

A. $4 + 6 = 10$

B. $6 - 4 = 2$

C. $4 \times 6 = 24$

D. $4 + 4 + 6 + 6 = 20$

True or false?

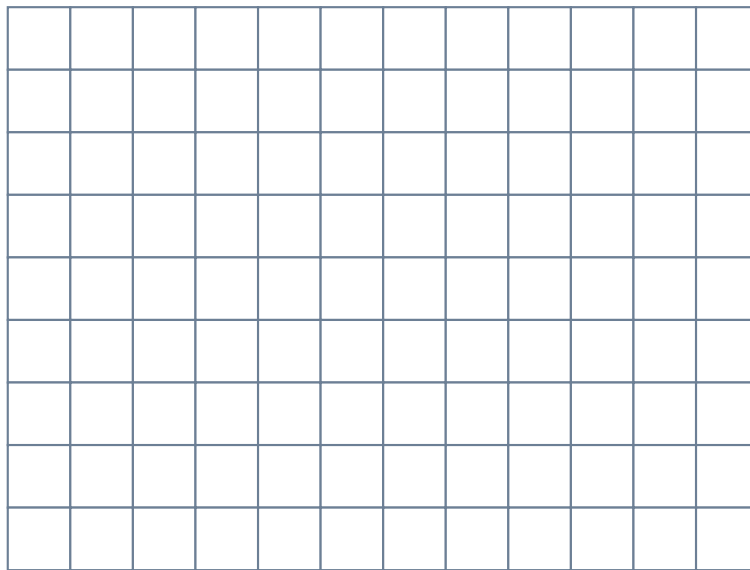
A rectangle 5 units long and 2 units wide can be tiled with 10 unit squares.

- A.** True
- B.** False

Draw a rectangle 4 units long and 3 units wide on the grid, and show its unit squares.

Write a multiplication equation for its area.

Equation: _____



True or false?

Tiling a 6-by-3 rectangle takes 9 unit squares.

A. True

B. False

SET 51 • 3.MD.C.7.a

BACKWARD 1

Find the total of an array by adding equal addends • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

How many squares are in 3 rows of 5 squares? Write an addition equation.

Equation: _____

True or false?

An array of 4 rows of 4 has 16 objects.

A. True

B. False

Which addition matches 2 rows of 5 squares?

A. $2 + 2$

B. $5 + 2 + 5$

C. $5 + 5$

D. $2 + 5$

How many squares are in 5 rows of 3 squares?

Squares: _____

True or false?

$3 + 3 + 3 + 3$ matches 3 rows of 3 squares.

A. True

B. False

SET 51 • 3.MD.C.7.a

BACKWARD 2

Split a rectangle into rows and columns of same-size squares • Grade 2 •
Geometry • 2.G.A.2

Draw lines to split the rectangle into 2 rows of 4 same-size squares.



True or false?

A rectangle split into 3 rows and 2 columns has 6 same-size squares.

A. True

B. False

A rectangle is split into 2 rows and 3 columns of same-size squares.

How many squares are there?

A. 4

B. 9

C. 6

D. 5

A rectangle is split into 4 rows and 4 columns of same-size squares.

How many squares are there?

Squares: _____

True or false?

A rectangle split into 5 rows and 1 column has 6 squares.

A. True

B. False

SET 51 • 3.MD.C.7.a

FORWARD 1

Write and apply the area formula $A = l \times w$ • Grade 4 • Measurement & Data •
4.MD.A.3

Write a formula for the area A of a rectangle with length l and width w .

Formula: _____

True or false?

A rectangle with $l = 13$ and $w = 6$ has $A = 78$.

A. True

B. False

Use $A = l \times w$.

What is the area when $l = 25$ cm and $w = 4$ cm?

- A.** 104 square cm
- B.** 58 square cm
- C.** 29 square cm
- D.** 100 square cm

Find the area of a rectangle with $l = 18$ ft and $w = 5$ ft.

Area: _____

True or false?

A rectangle with $l = 11$ and $w = 9$ has $A = 40$.

A. True

B. False

FORWARD 2

Find the volume of a prism by packing it with unit cubes and by multiplying •
Grade 5 • Measurement & Data • 5.MD.C.5.a

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box is packed with unit cubes: 3 cubes long, 2 cubes wide, and 4 cubes tall.

Count the cubes and then multiply. Do you get the same volume?

Count: _____

Multiply: _____

Same? _____

True or false?

A prism 5 units by 2 units by 3 units can be packed with 30 unit cubes.

- A.** True
- B.** False

Which equation gives the volume of a prism 4 by 3 by 2?

A. $2 \times 4 + 3 = 11$

B. $4 + 3 + 2 = 9$

C. $4 \times 3 = 12$

D. $4 \times 3 \times 2 = 24$

A prism has a base area of 6 square units and a height of 5 units.

What is its volume?

Volume: _____

True or false?

A 2-by-2-by-2 cube can be packed with 6 unit cubes.

A. True

B. False

Multiply side lengths to find the area of a rectangle in real-world problems

MAIN	Grade 3 main question	3.MD.C.7.b
BACKWARD 1	Measure side lengths with a ruler	2.MD.A.1
BACKWARD 2	Find the total of equal rows by repeated addition	2.OA.C.4
FORWARD 1	Use the area formula to solve real-world problems with larger num...	4.MD.A.3
FORWARD 2	Find the area of a rectangle with fractional side lengths	5.NF.B.4.b

SET 52 • 3.MD.C.7.b

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rug is 6 feet long and 4 feet wide.

What is its area?

Area: _____

Find the area of the rectangle.

Area: _____



A garden is 8 yards long and 9 yards wide.

What is its area?

- A.** 34 square yards
- B.** 72 yards
- C.** 72 square yards
- D.** 17 square yards

True or false?

A poster 3 feet by 2 feet has an area of 5 square feet.

- A. True
- B. False

True or false?

A tile 8 inches by 8 inches has an area of 16 square inches.

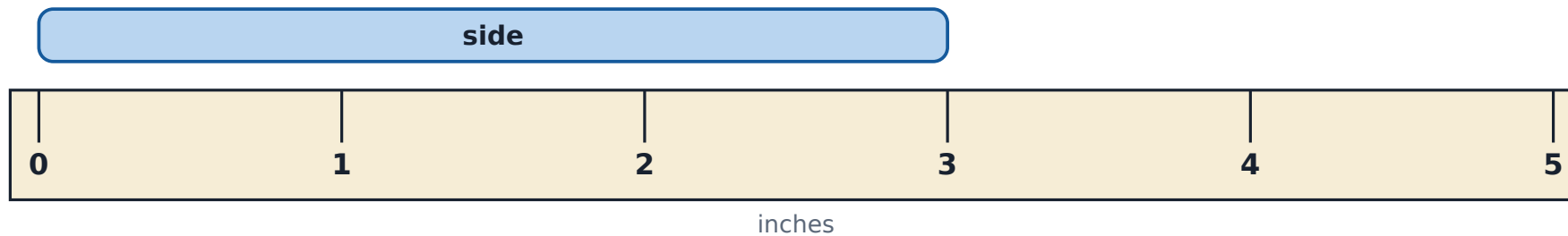
- A. True
- B. False

SET 52 • 3.MD.C.7.b

BACKWARD 1

Measure side lengths with a ruler • Grade 2 • Measurement & Data • 2.MD.A.1

How long is the side?



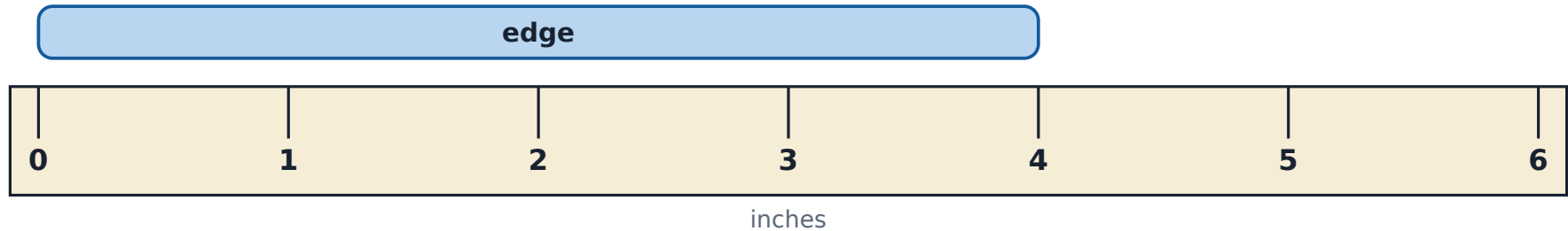
Length: _____

True or false?

To measure with a ruler, line up one end of the side with 0.

- A.** True
- B.** False

How long is the edge?



A. 4 inches

B. 3 inches

C. 6 inches

D. 5 inches

A side reaches from 0 to 7 on a centimeter ruler.

How long is it?

Length: _____

True or false?

A side that reaches from 0 to 2 on an inch ruler is 3 inches long.

- A. True
- B. False

SET 52 • 3.MD.C.7.b

BACKWARD 2

Find the total of equal rows by repeated addition • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4